On the equation $n'' = n$ and a problem of Erdős and Barbeau on products of prime–reciprocal sums

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Abstract

Erdős Problem #307 (posed by Barbeau, 1976; recorded in Erdős–Graham, 1980) asks whether there exist two finite sets P, Q of primes with $\left(\sum_{p\in P}\frac{1}{p}\right)\left(\sum_{q\in Q}\frac{1}{q}\right) = 1$. Writing $a = \prod P$, $b = \prod Q$, a solution exists if and only if the *arithmetic derivative* has a two–cycle $a' = b$, $b' = a$ with $a \neq b$; equivalently iff $n'' = n$ has a solution that is not a fixed point p^p. The two–cycle reformulation is due to Seva (2019) and, independently, Bado (2026); our starting point is to identify the map with the arithmetic derivative, which ties #307 to the Ufnarovski–Åhlander dynamics: the conjecture that every $n'' = n$–cycle has period one would force #307 to have answer NO, while a YES would refute it. Exploiting the equivalence we prove a *rigidity lemma* (each prime–reciprocal sum is automatically in lowest terms, so $N_P = D_Q$, $N_Q = D_P$, and $P \cap Q = \varnothing$) and a quantitative *barrier*: building on Rosen's bound $|P \cup Q| \geq 59$, any solution has $\min(\prod P, \prod Q) \geq 2.09 \times 10^{56}$, improving the best published lower bound for $n'' = n$ solutions by about 47 orders of magnitude. A finite verification over the complete list of 49,961 admissible 59–prime supports then closes the 59–prime level: any solution in fact has $|P \cup Q| \geq 60$ and $\min(\prod P, \prod Q) > 3.50 \times 10^{57}$; a quadratic–reciprocity certificate, requiring no factorisation, then proves the canonical tail family (the first 59 primes plus any q) empty, and together with a factoring pass 46,483 of the 49,961 one–new–prime families at level 60 (93%; the residual is factoring–hard). We add a parity dichotomy (a hypothetical all–odd cycle would need ≥ 1412 primes and have $\min(\prod P, \prod Q) > 10^{2519}$), a proof that longer cycles are strictly harder ($k = 3$ already forces 361,139 primes), and a function–field theorem: over $\mathbb{F}_q[t]$ the derivative has *no* cycles, so the obstruction over $\mathbb{Z}$ is purely archimedean. A local congruence anatomy of the cycle equation, the Erdős–Wintner fact that the abundance density 0.0420 is a genuine constant, and a Thābit/Euler–type constructive calculus (the *Frame Rule* and a divisor–trick) that faithfully parametrises solutions but, by an exact identity $\Delta_0 = M_0 N(1 - s_{M_0} s_N)$, provably does *not* reduce the difficulty below #307 itself (the natural "make Δ_0 small" objective is #307 in disguise). Recast as a bilinear breeder, the rule yields the maximal honest consequence (a conditional reduction (H) $\Rightarrow$ YES to an explicit prime–density hypothesis) which a quantitative ledger shows is *not* a feasible finite search. The rigidity result and the barrier are fully formalised in Lean 4 (mathlib), the smallest–primes extremality being proved rather than assumed; the only non–logical input is a verified evaluation of the first 59 primes. The problem remains open; the heuristics weakly favour existence.

1 Introduction

Let P, Q be finite, nonempty sets of (distinct) primes. Erdős Problem #307 [10], attributed to Barbeau's 1976 "Computer challenge corner" problem #477 [2] and listed in Erdős–Graham [3], asks:

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do there exist finite sets of primes P, Q with $\Big(\sum_{p\in P}\frac{1}{p}\Big)\Big(\sum_{q\in Q}\frac{1}{q}\Big) = 1$?

The problem is verifiable in the sense that a single example would settle it. No example is known; the only published partial fact (a remark of J. Rosen on MathOverflow [9], paraphrased on the problem page) is that any solution needs at least 59 primes in $P \cup Q$. The known "solutions" of Cambie and others solve only the *weakened* version in which 1 is permitted as an element, e.g. $(1+\frac{1}{5})(\frac{1}{2}+\frac{1}{3}) = 1$; these do not bear on the prime version.

Expanding the product exhibits #307 as a rigidly shaped Egyptian fraction for 1: $1 = \sum_{p\in P}\sum_{q\in Q} 1/(pq)$, whose denominators are the *rectangular grid* $P \times Q$ of semiprimes; a *rank–one*, multiplicatively separable representation. The unrestricted companion (representing 1 by reciprocals of distinct two–prime–factor numbers, with no product structure imposed) is by contrast *solved*: Johnson (1978) gave a 48–term decomposition, Watanabe [5] 47–term ones, and Graham's survey [4] records exactly this contrast, noting after Barbeau that the product form "is not known." Thus #307 is precisely the rank–one restriction of a solved problem. Nor is the density–zero nature of the prime denominators the obstruction in itself: Bloom [6] represents every positive rational by distinct unit fractions drawn from the (equally sparse) shifted primes $\{p-1\}$. What resists is the rank–one rigidity (the demand that the representation *factor* as a product of two prime–reciprocal sums) which the barrier (Theorem 4.3) shows costs at least 59 primes, hence of order 870 semiprime terms against Johnson's 47.

Our starting point is an identification of #307 with a question about the *arithmetic derivative* $n \mapsto n'$, the unique map with $0' = 1' = 0$, $p' = 1$ for primes p, and the Leibniz rule $(ab)' = a'b + ab'$. (The map goes back to J. Mingot Shelly (1911) and appeared as a 1950 Putnam problem; Barbeau [1] gave the first systematic study, with the modern account in Ufnarovski–Åhlander [7] and OEIS [11].) For $n = \prod_i p_i^{e_i}$ one has $n' = n\sum_i e_i/p_i$. We show (Section 3) that #307 is *equivalent* to the existence of a two–cycle of $n \mapsto n'$ that is not a fixed point. The two–cycle reformulation itself is not new: it was stated "in disguise" by Seva [19] in 2019 (via the map $Q \mapsto Q\sum_{p|Q} 1/p$, which agrees with the arithmetic derivative on squarefree integers, where the question lives), and obtained independently by Bado [20]; see the note in Section 12. Neither, however, identifies the map with the *arithmetic derivative* of Barbeau [1] and Ufnarovski–Åhlander [7]. That identification (and the resulting two–way bridge to the $n'' = n$ literature (importing the Ufnarovski–Åhlander conjecture, exporting our bounds)) is the starting contribution of this note; remarkably, Barbeau authored foundational papers on both sides (1961 and 1976 [2]).

The bridge is productive in both directions.

- *Importing* from the derivative literature: the Ufnarovski–Åhlander theorem that any two–cycle has squarefree members with disjoint supports [7] makes the "finite sets of distinct primes" hypothesis of #307 automatic, and links #307 to the Ufnarovski–Åhlander conjecture that all derivative cycles have period one.

- *Exporting* to that literature: our barrier ($\min(\prod P, \prod Q) \geq 2.09 \times 10^{56}$, Theorem 4.3) improves Kovič's lower bound $\approx 3.2 \times 10^9$ [8] on the odd member of an $n'' = n$ solution by roughly 47 orders of magnitude.

Results and organisation. Section 2 proves the Rigidity Lemma (Theorem 2.1): the fraction $\sum_{p\in S} 1/p$ is automatically in lowest terms, which forces, for any solution, the exact identities $N_P = D_Q$, $N_Q = D_P$, disjointness $P \cap Q = \varnothing$, and $D_Q = s\,D_P$ where $s = \sum_{p\in P} 1/p$. The disjointness core was anticipated by R. Israel [9]; the full rigidity is what powers everything later. Section 3 establishes the bridge. Section 4 proves the barrier, the parity dichotomy, and that longer cycles are strictly harder (so two–cycles are extremal). Section 5 gives a local congruence anatomy, a heuristic model, and a proof (via Erdős–Wintner) that the relevant abundance density is a genuine constant. Section 6 adds a function–field lens: over $\mathbb{F}_q[t]$ the derivative has *no* cycles, so the obstruction in $\mathbb{Z}$ is purely archimedean. Section 7 develops the Frame Rule and divisor–trick, and, crucially, records the exact identity that renders the natural constructive objective circular. Section 8 recasts the rule as a bilinear breeder, records a conditional reduction (H) $\Rightarrow$ YES to an explicit prime–density hypothesis, and shows quantitatively why (H) does not amount to a feasible finite search. Section 9 documents the computations, and Section 10 the Lean 4 formalisation. We are explicit throughout about what is proved, what is heuristic, and what is borrowed; the problem remains open.

Notation. For a finite set S of distinct primes put

$$D_S := \prod_{p\in S} p, \qquad N_S := \sum_{p\in S} \frac{D_S}{p}, \qquad \text{so} \qquad \sum_{p\in S}\frac{1}{p} = \frac{N_S}{D_S}.$$

We call N_S the *cofactor sum*. For a solution we write $a = D_P$, $b = D_Q$, $s = N_P/D_P = \sum_{p\in P} 1/p$ and $t = N_Q/D_Q = \sum_{q\in Q} 1/q$, so $st = 1$, and (without loss of generality) $s > 1 > t$. We write $\omega(n)$ for the number of distinct primes dividing n.

2 The Rigidity Lemma

Lemma 2.1 (Rigidity / lowest terms)**.** *For a finite set S of distinct primes,* $\gcd(N_S, D_S) = 1$. *Equivalently, $\sum_{p\in S} 1/p$ is already in lowest terms, and $v_p\big(\sum_{r\in S} 1/r\big) = -1$ for every $p \in S$.*

Proof. The prime divisors of $D_S = \prod_{r\in S} r$ are exactly the elements of S, so it suffices to show $p \nmid N_S$ for each $p \in S$. Fix $p \in S$ and reduce $N_S = \sum_{r\in S} D_S/r$ modulo p. For $r \neq p$ the term $D_S/r = \prod_{u\in S\setminus\{r\}} u$ still contains the factor p, so $p \mid D_S/r$. The single surviving term is $D_S/p = \prod_{u\in S\setminus\{p\}} u$, whence

$$N_S \equiv \prod_{u\in S\setminus\{p\}} u \pmod p.$$

Each u in this product is a prime distinct from p, so $p \nmid u$; as p is prime it does not divide the product. Thus $N_S \not\equiv 0 \pmod p$, i.e. $p \nmid N_S$. As p ranged over all prime divisors of D_S, $\gcd(N_S, D_S) = 1$. The valuation statement is the same computation: $v_p(N_S/D_S) = v_p(N_S) - v_p(D_S) = 0 - 1 = -1$. ☐

Remark 2.2. Distinctness and primality are both essential: v_p of the cofactor of a repeated prime need not be -1, and the final step uses Euclid's lemma. The case $|S| = 1$ is consistent ($N_{\{p\}} = 1$).

Theorem 2.3 (Solution structure)**.** *Suppose P, Q are finite sets of distinct primes with $\dfrac{N_P}{D_P}\cdot\dfrac{N_Q}{D_Q} = 1$. Then*

$$N_P = D_Q, \qquad N_Q = D_P, \qquad P \cap Q = \varnothing,$$

N_P is squarefree with prime support exactly Q (and N_Q squarefree with support P), and $D_Q = s\, D_P$ where $s = N_P/D_P$.

Proof. Cross-multiplying, $N_P N_Q = D_P D_Q$. By Lemma 2.1 both N_P/D_P and N_Q/D_Q are in lowest terms. Since $N_Q/D_Q = D_P/N_P$ and the lowest–terms representative of a positive rational is unique, comparing N_Q/D_Q with D_P/N_P gives $N_Q = D_P$ and $D_Q = N_P$.

Disjointness (the Israel mechanism [9]). If $r \in P \cap Q$ then $r \mid D_P$ and, by $N_P = D_Q = \prod Q$, also $r \mid N_P$, but $\gcd(N_P, D_P) = 1$ forces $r \nmid N_P$, a contradiction. (Equivalently $v_r(s) = v_r(t) = -1$, so $v_r(st) = -2 \neq 0 = v_r(1)$.) Hence $P \cap Q = \varnothing$.

Finally $N_P = D_Q = \prod_{q \in Q} q$ is a product of distinct primes, hence squarefree with prime support Q; symmetrically for N_Q. And $s\,D_P = (N_P/D_P)D_P = N_P = D_Q$. □

The disjointness in Theorem 2.3 means a solution is genuinely a partition of $U \coloneqq P \cup Q$, and $D_P D_Q = \prod_{p \in U} p$. The valuation form $v_p(\sum 1/p_i) = -1$ underlying disjointness is due to R. Israel [9], and appears also in Seva [19]; the cross–matching form ($N_P = D_Q$, $N_Q = D_P$) was obtained independently and concurrently by Bado [20, Thm. A] (Section 12).

3 The bridge to the arithmetic derivative

Lemma 3.1. *For squarefree m, the arithmetic derivative equals the cofactor sum: $m' = \sum_{p \mid m} m/p = N_{\{p:\, p \mid m\}}$. Moreover $\gcd(m, m') = 1$.*

Proof. For $m = \prod_{p \mid m} p$ squarefree, $m' = m \sum_{p \mid m} 1/p = \sum_{p \mid m} m/p$, which is exactly the cofactor sum of the prime set of m. Coprimality is Lemma 2.1. □

Theorem 3.2 (Bridge)**.** *The following are equivalent.*

(1) *Erdős #307 has a solution: finite prime sets P, Q with $\left(\sum_{p \in P} 1/p\right)\left(\sum_{q \in Q} 1/q\right) = 1$.*

(2) *The arithmetic derivative has a two–cycle: integers $a \neq b$ with $a' = b$ and $b' = a$.*

(3) *The equation $n'' = n$ has a solution n that is not a fixed point (i.e. $n' \neq n$, so n is not of the form p^p).*

(*The implication* (2) ⇒ (1) *uses the Ufnarovski–Åhlander squarefreeness theorem for cycles; all other implications, and everything used in the barrier of Section* 4*, are unconditional; see Remark* 3.3*.*)

Proof. (1) ⇒ (2). Let $a = D_P, b = D_Q$. By Lemma 3.1, $a' = N_P$ and $b' = N_Q$, and by Theorem 2.3, $N_P = D_Q = b$ and $N_Q = D_P = a$. Thus $a' = b$, $b' = a$, and $a \neq b$ since $s \neq 1$ (a prime–reciprocal sum is never 1, being N/D in lowest terms with $D > 1$).

(2) ⇒ (1). Let $a' = b, b' = a$ with $a \neq b$. By the Ufnarovski–Åhlander analysis of two–cycles (cf. [7, 8]), both a and b are squarefree; their prime supports P, Q are then disjoint since $\gcd(a, b) = \gcd(a, a') = 1$. By Lemma 3.1, $a' = N_P$ and $b' = N_Q$; the cycle gives $N_P = b = D_Q$ and $N_Q = a = D_P$, so $\left(\sum_{p \in P} 1/p\right)\left(\sum_{q \in Q} 1/q\right) = \frac{N_P}{D_P}\frac{N_Q}{D_Q} = \frac{D_Q}{D_P}\frac{D_P}{D_Q} = 1$.

(2) ⇔ (3). A two–cycle $a \neq b$ gives $a'' = a$ with $a' = b \neq a$, so a is a non–fixed solution of $n'' = n$. Conversely a solution of $n'' = n$ with $n' \neq n$ yields the two–cycle (n, n'). The fixed points of $n \mapsto n'$ are exactly the numbers p^p [1, 7]. □

Remark 3.3 (Status of the cited input)**.** Part (2) ⇒ (1) uses that a derivative two–cycle has squarefree, support–disjoint members. Squarefreeness is the content of the Ufnarovski–Åhlander analysis of cycles (a non–squarefree n has $p \mid \gcd(n, n')$ for some p, and along $n \mapsto n'$ such common factors propagate, precluding return); disjointness then follows from $\gcd(a, a') = 1$. We use it as a cited theorem. Even *without* it, the implication (1) ⇒ (2) and the entire barrier below are unconditional, because Rigidity already delivers squarefree, disjoint P, Q from a #307 solution.

Corollary 3.4. *If the Ufnarovski–Åhlander conjecture holds in the weak form "every cycle of $n \mapsto n'$ has period one," then Erdős #307 has answer* NO*. Conversely, a* YES *to #307 exhibits a period–two cycle and refutes that statement.*

4 The barrier

Lemma 4.1 (Strict AM–GM)**.** *If $s, t > 0$ with $st = 1$ and $s \neq 1$, then $s + t > 2$.*

Proof. $s + t - 2 = s + 1/s - 2 = (s-1)^2/s > 0$ since $s \neq 1$. □

Lemma 4.2 (The number 59, after Rosen [9])**.** *Any finite set U of distinct primes with $\sum_{p\in U} 1/p > 2$ has $|U| \geq 59$. Moreover, among k–element prime sets, $\sum 1/p$ is maximised by the k smallest primes.*

Proof. Replacing any prime of U by a smaller prime not in U increases $\sum 1/p$, so the maximum over k–element sets is at the first k primes. The reciprocal sum of the first 58 primes is $1.99874\ldots < 2$ and of the first 59 primes is $2.00235\ldots > 2$, so a sum exceeding 2 needs at least 59 primes. □

Theorem 4.3 (Barrier)**.** *For any solution (P, Q) of #307, with $U = P \cup Q$:*

(a) $|U| \geq 59$ *and* $\displaystyle\prod_{p\in U} p \geq \Pi_{59} := \prod_{i=1}^{59} p_i \approx 8.77 \times 10^{112}$*;*

(b) $\min(\prod P, \prod Q) \;\geq\; \sqrt{\Pi_{59}/T_{59}} \;=\; 2.0929\ldots \times 10^{56}$*, where* $T_{59} = \sum_{i=1}^{59} 1/p_i = 2.00235\ldots$*. In particular both $\prod P$ and $\prod Q$ exceed* 2.09×10^{56}*.*

Consequently every $n'' = n$ solution has both members $> 2.09 \times 10^{56}$, improving Kovič's bound $\approx 3.2 \times 10^9$ [8] by about 47 orders of magnitude.

Proof. (a) By $st = 1$, $s \neq 1$ and Lemma 4.1, $\sum_{p\in U} 1/p = s + t > 2$, so $|U| \geq 59$ by Lemma 4.2; since P, Q are disjoint (Theorem 2.3), $\prod_{p\in U} p = D_P D_Q$, and a product of ≥ 59 distinct primes is at least the product of the 59 smallest, Π_{59}.

(b) Take $D_P \leq D_Q$ to be the minimal side. By Theorem 2.3, $D_Q = s\, D_P$ with $s > 1$, hence

$$s\, D_P^2 \;=\; D_P\, D_Q \;=\; \prod_{p\in U} p \;=: R, \qquad \text{so} \qquad D_P^2 = \frac{R}{s}.$$

Since $s \leq s + t = T(U) := \sum_{p\in U} 1/p$ (as $t > 0$),

$$D_P^2 = \frac{R}{s} \;\geq\; \frac{R}{T(U)}.$$

It remains to bound $R/T(U)$ below over all admissible U. Among k–element prime sets the quotient $\bigl(\prod U\bigr)/\bigl(\sum_{p\in U} 1/p\bigr)$ is minimised by the k smallest primes (smallest numerator and, by Lemma 4.2, largest denominator simultaneously), and passing from 59 to more primes only increases it (the product grows multiplicatively while the reciprocal sum grows by a single small term). Hence the global minimum over admissible U is at $U = \{p_1, \ldots, p_{59}\}$, giving $R/T(U) \geq \Pi_{59}/T_{59}$. Therefore

$$D_P^2 \;\geq\; \frac{\Pi_{59}}{T_{59}} \;=\; \frac{\Pi_{59}^2}{N_{59}} \;=\; 4.3806\ldots \times 10^{112}, \qquad D_P \;\geq\; 2.0929\ldots \times 10^{56},$$

where $N_{59} = T_{59}\, \Pi_{59}$ is the integer cofactor sum of the first 59 primes. □

Remark 4.4 (Why the *minimum* is bounded, not only the maximum)**.** The product identity $D_P D_Q \geq \Pi_{59}$ alone gives only $\max(\prod P, \prod Q) \geq \sqrt{\Pi_{59}} \approx 2.96 \times 10^{56}$. The stronger statement (b), a lower bound on the *smaller* side, genuinely uses rigidity: it is the identity $D_Q = s\, D_P$ (equivalently $D_P^2 = R/s$) together with $s \leq T(U)$ that converts the product bound into a bound on D_P. We verified numerically that $\min_U R(U)/T(U)$ is attained exactly at the first 59 primes (the minimising configuration is unique), so the constant 2.09×10^{56} is sharp for this method.

Proposition 4.5 (Closing the 59–prime level)**.** *No solution of #307 has* $|P \cup Q| = 59$*. Hence* $|P \cup Q| \geq 60$*,* $\prod_{p \in U} p \ \geq \ \Pi_{60} > 2.46 \times 10^{115}$*, and*

$$\min\left(\prod P, \prod Q\right) \ \geq \ \sqrt{\Pi_{60}/T_{60}} \ = \ 3.5053\ldots \times 10^{57}.$$

Proof (finite verification). Let $U = P \cup Q$ with $|U| = 59$; as in Theorem 4.3(a), $T(U) > 2$. The candidate list is finite and complete: (i) every prime $p \leq 167$ lies in U; omitting p caps $T(U)$ at $T_{60} - 1/p < 2$, since the 59 largest prime reciprocals avoiding p are those of the first 60 primes less p, and $1/p > T_{60} - 2 = 0.00590\ldots$ exactly for $p \leq 167$; (ii) every element $p' \geq 281$ satisfies $T_{58} + 1/p' > 2$ (the other 58 reciprocals sum to at most T_{58}), so $p' \leq 787$. Thus U consists of the 39 primes up to 167 plus twenty primes from $(167, 787]$, and exactly 49,961 such sets have $T(U) > 2$. For each, a solution supported on U forces both $N' + 2N = (a+b)^2$ and $N' - 2N = (a-b)^2$ to be perfect squares, where $N = \prod U$ and $N' = a^2 + b^2$ (rigidity; the $k = 0$ case of Proposition 12.2); one integer test that simultaneously excludes all 2^{58} splits of U. Exact–integer verification shows $N' + 2N$ is a non–square for *every* one of the 49,961 candidates, by two independent implementations with different enumeration orders and prunings (`code/close59.py` and `hunt/close59.rs` in the repository). Hence no 59–element support closes, and $|U| \geq 60$. The bound on the minimal side repeats the proof of Theorem 4.3(b) with the minimum of R/T now over $|U| \geq 60$, attained at the first 60 primes: $D_P^2 \geq \Pi_{60}/T_{60} = \Pi_{60}^2/N_{60}$, where N_{60} is the cofactor sum of the first 60 primes. □

Corollary 4.6 (The coprime variant inherits the closed level)**.** *Consider the weakened variant in which the elements of* P *and* Q *are merely pairwise coprime integers* ≥ 2 (*the open form of the variant,* $1 \notin P \cup Q$)*. Any solution has* $|P \cup Q| \geq 60$ *elements, and* $\min(\prod P, \prod Q) > 3.50 \times 10^{57}$*.*

Proof. Rigidity holds verbatim for pairwise–coprime families (the cofactor–sum argument uses only coprimality), so again $T(U) = s + t > 2$. The least prime factors $\ell(e)$ of the elements are distinct, and $\sum_e 1/e \leq \sum_e 1/\ell(e)$. Suppose $|U| = 59$ and some element e is not prime, so $e \geq \ell(e)^2$. If $\ell(e) \leq 277$, the loss is $1/\ell - 1/e \geq (\ell - 1)/\ell^2 \geq 276/277^2 = 0.00359\ldots$, exceeding the maximal available slack $T_{59} - 2 = 0.00235\ldots$; if $\ell(e) \geq 281$, then $e \geq 281^2$ while the other 58 reciprocals sum to at most T_{58}, giving $T(U) \leq T_{58} + 281^{-2} = 1.99875\ldots < 2$. Either way $T(U) < 2$, a contradiction: at 59 elements every element is prime, and Proposition 4.5 applies verbatim. The product bound transfers because $\prod_e e \geq \prod_e \ell(e)$ and $T(U) \leq \sum_e 1/\ell(e)$, so the extremal ratio bound at the first 60 primes still applies. □

Remark 4.7 (Verification tier, and why the ladder stops here)**.** Theorem 4.3 is machine–checked in Lean 4, and so is Proposition 4.5: the theorem `erdos307_sixty` formalises the entire argument; the forced–primes and element bounds, the Pythagorean plus–square, and a pruned depth–first search over the pool whose *completeness is proved, not assumed* (`dfs_sound`), the search itself running under `native_decide`; on top of the two independent unverified programs. The level cannot be climbed further by finite means: already at $|U| = 60$ the family $U = \{p_1, \ldots, p_{59}\} \cup \{q\}$ qualifies for *every* prime $q > 277$, and on it the two square conditions read $x^2 = Aq + \Pi_{59}$, $y^2 = Bq + \Pi_{59}$ with $A, B = N_{59} \pm 2\Pi_{59}$, i.e. the Pell–type system $Bx^2 - Ay^2 = -4\Pi_{59}^2$ with 113–digit coefficients. We verified that no accessible modulus obstructs it (mod 32, and all odd $\ell < 4000$: the conditions pin $q \equiv 5 \pmod 8$ and thin each $\ell \mid \Pi_{59}$ to about half of its residue classes and each $\ell \nmid \Pi_{59}$ to about a quarter, but never to zero), consistent with the local–solubility theme of Section 5. The obstruction, however, lives at the primes *of* A and B, and it is detectable *without factoring*: Proposition 4.8 below closes this family for *every* q by a single Jacobi–symbol computation. (An exact sweep to $q \leq 10^{12}$, `hunt/tail_sweep.rs`, 1.25×10^{11} candidates, none with $Aq + \Pi_{59}$ even a single square, which corroborates it independently.) The tail family also exposes the self–similarity of the ladder:

writing $a = qm$, so that $m\,b = \Pi_{59}$ partitions the first 59 primes, its two closing conditions collapse to the exact equation

$$D(m)\,D(b) + m^2 \;=\; \Pi_{59}$$

together with the divisibility $m \mid D(b)$ (then $q = D(b)/m$, required prime); verified exhaustively on a 12–prime toy universe. Equivalently, the deficit $1 - \sigma(m)\sigma(b)$ must equal m^2/Π_{59} *exactly*: the circularity $\Delta_0 = M_0 N(1 - s_{M_0} s_N)$ of Section 7 in its purest form, with the deviation forced to be the perfect square m^2. Closing even this one family is thus a 2^{58}–fold exact–coincidence search of the same type as #307 itself (expected count $\sim 2^{58}/\Pi_{59} \approx 10^{-95}$): each rung of the ladder past 60 contains the whole problem. The one case with *no* free large prime (a 60–set drawn entirely from small primes) is instead a finite search, and empty in the minimal range: a direct both–squares enumeration of all 60–prime sets with $T > 2$ supported in the first 70 primes (4,011,289 sets; `hunt/close60.rs`) finds none passing even the plus–square, so no level–60 solution has all its primes ≤ 349. (The first 68 primes give 1,258,448 sets, also empty, and reproduce independently.) By Corollary 12.9 this verdict is decisive rather than provisional: at these masses a support passing the plus– and minus–squares, with the parity of Proposition 12.5(2), *is* a two–cycle, so a survivor would have been a solution and not a candidate. The enumeration extends with compute.

Proposition 4.8 (The canonical tail family is empty: a reciprocity kill)**.** *No solution of #307 has support $U = \{p_1, \ldots, p_{59}\} \cup \{q\}$, for any prime q.*

Proof. A solution supported on U forces $x^2 = Aq + \Pi_{59}$ with $x = a + b$ and $A = N_{59} + 2\Pi_{59}$ (Remark 4.7). Every prime $\ell \mid A$ exceeds 277: for $\ell \leq 277$ one has $\ell \mid \Pi_{59}$, so $\ell \mid A$ would force $\ell \mid N_{59}$, contradicting $\gcd(N_{59}, \Pi_{59}) = 1$ (rigidity). Hence $\ell \nmid \Pi_{59}$, and reduction mod ℓ gives $x^2 \equiv \Pi_{59} \pmod{\ell}$, requiring the Legendre symbol $(\Pi_{59} \mid \ell) \neq -1$. But the Jacobi symbol satisfies

$$\left(\tfrac{\Pi_{59}}{A}\right) \;=\; -1$$

(a finite gcd–speed computation, script `code/tailkill.py`; no factorisation of the 114–digit A is needed), so some $\ell \mid A$ of odd multiplicity has $(\Pi_{59} \mid \ell) = -1$. The family is empty, for every q.

The companion symbol at $B = N_{59} - 2\Pi_{59}$ equals -1 as well; necessarily: for any admissible base S (writing $D = D_S$, $N = N_S$, $A_S, B_S = N \pm 2D$) one has $A_S \equiv B_S \pmod 8$ (as $4D \equiv 8 \bmod 16$) and $A_S \equiv B_S \equiv N \pmod p$ for every odd $p \mid D$, while the quadratic–reciprocity sign difference carries the exponent $\frac{p-1}{2} \cdot \frac{A_S - B_S}{2}$ with $\frac{A_S - B_S}{2} = 2D$ even; hence $(D \mid A_S) = (D \mid B_S)$ always; one binary certificate per tail family. Running it over all 49,961 admissible bases of Proposition 4.5 proves 31,219 of the one–new–prime families at level 60 empty (62.5%, curiously near 5/8); the remaining 18,742 are Jacobi–inconclusive: their -1–Legendre primes, if any, occur to an even count, visible only through a factorisation of A_S. A factoring attack on those (trial division and Brent–Pollard ρ over a Montgomery core, `hunt/survivor_kill.rs`; the found factors are kept as a pairwise–coprime basis, so a kill certificate (an odd–multiplicity basis piece P with $(D_S \mid P) = -1$) needs no primality testing at all), pushed to ρ–budget 10^6 with a Pollard $p - 1$ stage, closes a further 15,264: in total 46,483 of the 49,961 one–new–prime families at level 60 are proven empty (93.0%). The 3,478 still open resist all of this for a structural reason, not a want of effort: each is Baillie–PSW *composite* yet has no prime factor below $\sim 10^{12}$, so its A_S (or B_S) is a *balanced* semiprime of two $\sim$57–digit primes, splitting one is factoring at cryptographic scale. Two cheap tests confirm no shortcut survives: a product–tree batch–GCD over all 37,484 values A_S, B_S finds only shared primes below 10^9 (already found by ρ, so *no* free kills), and Baillie–PSW leaves exactly 34 families in which A_S and B_S are both *probable* primes; these are immune *conditionally on that primality* (a prime A_S with $(D_S \mid A_S) = +1$ has no inert factor, hence no congruence obstruction for any q). The conditionality is not pedantry and not removable by more of the same computation: Baillie–PSW is rigorous only in its *composite* verdicts, so the 46,483 kills are unconditional while the 34 immunities are not. Upgrading them would require primality *certificates* for thirty-four 114–digit integers (ECPP or a Pocklington factorisation of $A_S - 1$). The elementary route does not reach: immunity needs *both* A_S and B_S prime, and partial factorisation of $n-1$ (trial division to 3×10^6 together with the fully split part) leaves $\log F / \log n$ below 1/3 on at least one side of every one of the 34, so neither Pocklington ($F > \sqrt{n}$) nor Brillhart–Lehmer–Selfridge ($F > n^{1/3}$) applies to any family: 0 of 34 are certifiable this way (`code/immune_certify.py`). This is now settled. Running the Adleman–Pomerance–Rumely –Cohen–Lenstra test on all 68 integers proves every one of them prime in 8 seconds of wall clock, so the 34 immunities upgrade from [C] to [P] and are *unconditional* (`code/immune_prove.gp`, PARI/GP 2.17.4, whose `isprime` is a proof and not a probable–prime test). The run reproduces the whole chain independently: 49,961 admissible bases, 31,219 Jacobi kills, and, among the 81 bases whose A_S and B_S are both prime, exactly the 34 with $(D_S \mid A_S) = +1$; the other 47 carry $(D_S \mid A_S) = -1$ and were already killed. The primality is moreover backed by *independently checkable* evidence rather than a verdict: `code/immune_certify.gp` emits Atkin–Morain ECPP certificates for all 68 integers, all 68 verified by `primecertisvalid`, archived at `certs/immune_ecpp.txt` (512 KB) so a third party can check them without trusting the prover. The structural half of the campaign is now formal: the identity $(D_S \mid A_S) = (D_S \mid B_S)$, which is what makes one binary test decide a family rather than two, is proved in Lean 4 as `Erdos307.jacobiSym_A_eq_B` (`lean/Erdos307/Campaign.lean`), on the three standard axioms and with no `native_decide`; it follows from periodicity, $A_S - B_S = 4D_S$ being exactly the period of $J(a \mid \cdot)$. The primality itself is not formalised and cannot presently be: mathlib's only route to a large prime is `lucas_primality`, which needs a complete factorisation of $n-1$, and that is precisely what the 0 of 34 above rules out. Note the two counts answer different questions and both are correct: 81 is the number of prime pairs, 34 the number of *immune* families among them. A complete kill/immune split of the residual is in any case beyond feasible computation, and the honest state of the single–tail sector is 46,483 *empty* [P], 34 *conditionally immune* [C], 3,478 *factoring–hard*. □

Proposition 4.9 (Sector decomposition of level 60)**.** *Let U be a 60–element prime support with $T(U) > 2$. Then exactly one of the following holds:* (i) $T(U) - 2 \geq 1/\max U$, *and* $U = S \cup \{q\}$ *for an admissible base S (one of the 49,961 of Proposition 4.5); the* single–tail sector*, or* (ii) $T(U) - 2 < 1/\max U$, *and, writing $p > q$ for the two largest elements and W for the rest,* $T(W \cup \{p\}) < 2$ *and* $T(W \cup \{q\}) < 2$*; the* pair sector*, whose members escape every single–tail family.*

Proof. U lies in the tail family of $U \setminus \{r\}$ iff $T(U) - 1/r \geq 2$, which is easiest for $r = \max U$; (i) is that case ($T = 2$ being impossible by rigidity, the base is admissible). In case (ii) no removal reaches 2, and a fortiori the two stated 59–subsets stay below 2. □

Remark 4.10 (The campaign; completeness of the certificate; the arity barrier)**.** The reciprocity certificate captures *every* q–independent congruence obstruction of a tail family: at a prime $\ell \mid A_S B_S$ the system degenerates to the rank–one congruences $x^2 \equiv D_S$ or $y^2 \equiv D_S \pmod{\ell}$, while at $\ell \nmid 2A_S B_S$ the conditions define conics in (x, q) or (y, q) with $\sim \ell$ points, deleting at most half of the residue classes of q and never all (the 2–adic layer pins $q \equiv 5 \bmod 8$ without emptiness); by CRT and Dirichlet, admissible q survive any finite set of such conditions. Hence a tail family admits a congruence kill iff $(D_S \mid \ell) = -1$ for some $\ell \mid A_S B_S$, which the Jacobi symbol detects for 31,219 bases, and partial factorisation (`hunt/survivor_kill.rs`) for a further 15,264: in total 46,483 of the 49,961 single–tail families at level 60 are proven empty (93.0%), 3,478 remaining open (each a balanced $\sim$57–digit semiprime, factoring–hard). The weapon is intrinsically *arity–one*, and provably fails on the pair sector. With $s = p+q$, $t = pq$, a pair–sector member requires the two simultaneous squares $x^2 = A_W t + D_W s$ and $y^2 = B_W t + D_W s$ (the would–be third condition $s^2 - 4t = \square$ is automatic, being $(p-q)^2$), together with the reciprocity–linked constraints $(D_W p \mid q) = (D_W q \mid p) = 1 \pmod{q}$, a solution with $q \mid a$ has $N_U \equiv D_W p$, forcing $D_W p$ to be a square). At a prime $\ell \mid A_W$ the condition $x^2 \equiv D_W s \pmod{\ell}$ constrains only s, which the free second prime absorbs; the pair system is in fact locally soluble at *every* modulus (a theorem, Proposition 4.11 below, corroborated by a direct scan over all $\ell \le 600$) so, unlike the single–tail family, the pair sector admits *no* congruence obstruction at all. It is a genuine two–parameter surface per base, #307 in miniature (Remark 4.13); reciprocity cannot touch it, and only a direct both–squares search or a descent argument can.

Proposition 4.11 (The pair sector admits no congruence kill)**.** *Let W be any pair–sector base and $A = A_W$, $B = B_W$, $D = D_W$, so that a pair–sector solution requires $x^2 = A\,\alpha\beta + D(\alpha+\beta)$ and $y^2 = B\,\alpha\beta + D(\alpha+\beta)$ with α, β the two new primes. For every prime power ℓ^j this system has a solution in units α, β and integers x, y at a nonsingular point. Consequently, by CRT and Dirichlet, no finite system of congruence conditions empties any pair–sector family: the arity barrier of Remark 4.10 is a theorem, and the certificate programme provably ends at arity one.*

Proof. First, no ℓ pins either target to a constant: that needs $\ell \mid A$ (or B) *and* $\ell \mid D$, whence $\ell \mid N$, contradicting $\gcd(N, D) = 1$ (rigidity); so the tail–kill mechanism is unavailable, and solubility is decided in three regimes.

(1) ℓ odd, $\ell \mid D$. This covers every odd $\ell \le 103$: at level 60, omitting a prime p caps $T(U)$ at $T_{61} - 1/p$, and $1/p > T_{61} - 2 = 0.00944\ldots$ exactly for $p \le 103$, so all such primes lie in U, hence in W (the two adjoined primes are the largest). Take $\beta = 1$, $x = 1$, $\alpha = (1-D)/(A+D) \bmod \ell^j$, a unit since $A + D \equiv N \not\equiv 0 \pmod{\ell}$: the first equation holds by construction, and the second target $(B+D)\alpha + D \equiv N \cdot N^{-1} = 1 \pmod{\ell}$ is a unit congruent to a square, so y exists modulo ℓ^j.

(2) ℓ odd, $\ell \nmid 2D$ (in particular every $\ell \ge 107$ outside W). The two conditions are secretly *one*: setting $m = (x^2 - y^2)/4D$ and $u = (x^2 - Am)/D$, the relation $A - B = 4D$ gives the polynomial identity $Bm + Du = y^2$. So any (x, y) with $xy \ne 0$, $x^2 \ne y^2$ and $\delta := u^2 - 4m$ a square delivers α, β as the roots of $z^2 - uz + m$ (nonzero, as $\alpha\beta = m \ne 0$). For each admissible x, δ is a quartic in y and not a square in $\overline{\mathbb{F}}_\ell[y]$ (a factorisation $(Q-R)(Q+R) = 16D^3(x^2 - y^2)$ would force $\ell \mid 4D$), so Weil's bound gives $\frac{1}{2}\ell^2 + O(\ell^{3/2})$ admissible pairs; positive well below $\ell = 107$. The point is nonsingular ($xy \ne 0$; the Jacobian contains $\operatorname{diag}(2x, 2y)$), so it lifts to every ℓ^j.

(3) $\ell = 2$. Both targets are odd and agree modulo 8 (as $\alpha\beta$ is odd, $D(\alpha+\beta) \equiv 0 \bmod 4$, and $(A-B)\alpha\beta = 4D\alpha\beta \equiv 0 \bmod 8$), and a finite table over ($A \bmod 8$, $D \bmod 8$) shows that for every base some achievable pair $(\alpha\beta, \alpha+\beta) \bmod 8$ makes the common target $\equiv 1 \pmod{8}$; a 2–adic square, soluble modulo 2^j for every j.

All three regimes, the collapse identity, the threshold $1/103 > T_{61} - 2 > 1/107$, and the mod–32 layer were verified computationally on three bases and every $\ell \le 600$ (`code/pairlocal.py`; the measured density of admissible (x, y) in regime (2) is 0.497–0.503 against the predicted $\frac{1}{2}$). $\square$

Remark 4.12 (Anatomy of the certificate; the empirical 5/8 law)**.** Writing $D = 2D_1$, the reciprocity signs in the certificate cancel ($\frac{A-1}{2} + \frac{N-1}{2} = N - 1 + 2D_1$ is even), leaving the clean product

$$\left(\tfrac{D}{A}\right) \;=\; \left(\tfrac{2}{A}\right)\left(\tfrac{D_1}{N}\right), \qquad \left(\tfrac{2}{A}\right) = +1 \iff N \equiv 3, 5 \pmod{8},$$

and one layer down $(N \mid D_1) = \prod_{p \mid D_1} \left(\frac{D/p}{p}\right)$, since $N \equiv D/p \pmod{p}$: the cofactor structure of rigidity recurs inside its own certificate. This product evaluates in closed form: with $D/p = 2D_1/p$ and quadratic reciprocity,

$$(N \mid D_1) = (2 \mid D_1) \prod_{\{p,q\} \subset D_1} (-1)^{\frac{p-1}{2}\frac{q-1}{2}} = (-1)^{s + \binom{t}{2}}, \quad s = \#\{p \mid D_1 : p \equiv 3, 5\ (8)\},\ t = \#\{p \mid D_1 : p \equiv 3\ (4)\},$$

a Rédei genus character of D_1, *independent of* N. This is forced, not merely observed: $(N \mid D_1) = \prod_{p|D_1}(N \mid p)$, and cofactor survival gives $N \equiv D/p = 2D_1/p \pmod p$, so $(N \mid p) = (2 \mid p)\,(D_1/p \mid p)$. The first factors multiply to $(-1)^s$ since $(2 \mid p) = -1$ exactly for $p \equiv 3, 5 \pmod 8$; and, each unordered pair $\{p, q\}$ occurring once in each of the two cofactors,

$$\prod_{p|D_1} \left(\tfrac{D_1/p}{p}\right) = \prod_{\{p,q\}\subset D_1} \left(\tfrac{q}{p}\right)\left(\tfrac{p}{q}\right) = \prod_{\{p,q\}\subset D_1} (-1)^{\frac{p-1}{2}\frac{q-1}{2}} = (-1)^{\binom{t}{2}},$$

by quadratic reciprocity, since $\frac{p-1}{2}$ is odd exactly for $p \equiv 3 \pmod 4$. (Checked independently, prime by prime, on 20,000 bases: 0 violations.) The inner symbol is thus not a coin but a genus invariant; its empirical fairness (49.9%, 50.0% over the 49,961 bases) is that character's equidistribution; and $(D_1 \mid N)$ follows by the reciprocity sign $(-1)^{\frac{D_1-1}{2}\frac{N-1}{2}}$, hence depends on N only through $N \bmod 4$. In fact the entire certificate closes into a finite functional. Let $v(S) = (n_1, n_3, n_5, n_7) \bmod 4$ count the odd primes of S in the residue classes $1, 3, 5, 7$ modulo 8. Every factor above is a function of v alone: $s \equiv n_3 + n_5$, $t = n_3 + n_7$ (with $\binom{t}{2}$ needing only $t \bmod 4$), $D_1 \equiv 3^{n_3}5^{n_5}7^{n_7} \pmod 8$, $S(D_1) \equiv (n_1+n_5)-(n_3+n_7) \pmod 4$, and (by the mod–8 law of Proposition 12.18, $D_1' \equiv D_1 S(D_1) \pmod 8$) also $N = D_1 + 2D_1' \equiv D_1 + 2\big(D_1 S(D_1) \bmod 4\big) \pmod 8$. Hence

$$(D \mid A) \;=\; K\big(v(S)\big) :$$

the kill is decided by sixteen residue counts modulo 4. The realized vectors are exactly the coset $n_1 + n_3 + n_5 + n_7 \equiv 58 \equiv 2 \pmod 4$ (all 64 of its classes occur), K is constant on each, and the functional reproduces the direct 113–digit Jacobi computation on every one of the 49,961 bases (`code/kill_classvector.py`). The five–eighths law is then an exact finite statement, and its mechanism is the *reciprocity switch*: for $N \equiv 1 \pmod 4$ the reciprocity sign is inert and the kill reduces to the genus coin; exact conditional rate 12,699/24,975 = 0.5085; while for $N \equiv 3 \pmod 4$ the sign couples the parity of t into the product and the rate jumps to 18,520/24,986 = 0.7412 (by class modulo 8: 0.509, 0.741, 0.508, 0.741 at $N \equiv 1, 3, 5, 7$; the 3/4 classes are $N \equiv 3 \pmod 4$, not the $(2 \mid A)$ classes). The blend $\frac{1}{2}\cdot\frac{24975}{49961} + \frac{3}{4}\cdot\frac{24986}{49961} = 0.62503$ against the exact 31,219/49,961 = 0.62487 is the whole of the "coincidence". The correlation is an ensemble property, a random–subset proxy gives ≈0.45/0.55; and the functional yields no second certificate: it recomputes the same symbol.

Remark 4.13 (The kill–program is one–sided)**.** Every kill is a proof that no solution hides in that family, but if #307 has a solution, the solution's own family is unkillable, so no accumulation of kills can overshoot the truth: the program's asymptote *is* the existence question. Together with the self–similarity of Remark 4.7 (each rung past 60 contains the whole problem), this locates the campaign precisely: it corners where a solution could live, one certified region at a time, and can terminate only if #307 has answer NO, which the heuristics disfavour.

Remark 4.14 (The reciprocity kills are the local factors of the expected count)**.** The divergent naive count of Section 5 and the reciprocity sieve of Proposition 4.8 are the *same* computation. Decompose the expected number of solutions over tail families $S \cup \{q\}$: the summand for a base S is $\kappa_S \left(D_S\sqrt{\text{mass}^2 - 4}\right)^{-1} \sum_q q^{-1}$, where $\kappa_S = \prod_\ell(\text{local solubility density at } \ell)$ is exactly the reciprocity data. A reciprocity–killed base carries an inert $\ell \mid A_S$ with $(D_S \mid \ell) = -1$, so $N_S + 2D_S \equiv D_S \pmod \ell$ is a non–residue for *every* q, forcing $\kappa_S = 0$; the family drops from the count entirely. An immune base (A_S, B_S both prime, both residues) has no inert prime, so $\kappa_S > 0$ and its summand is a divergent $\sum_q q^{-1}$ (verified on an explicit 114–digit immune base: local density 1 at every tested prime, partial sums growing without bound). Hence the campaign of Remark 4.10 is a Selberg–sieve evaluation of the singular series: the kills remove ∼93% of the count's mass, and the residual log–divergence (the entire "weakly favours YES" signal) is carried by the 34 conditionally immune families together with the pair sector. (Conditionally, because their immunity rests on the Baillie–PSW primality of A_S, B_S; the pair sector's contribution, by contrast, is unconditional, being a theorem, Proposition 4.11.) This does not touch the reliability of the count (which ignores the $N(r) \le 1$ rigidity), but it pins down *where* the heuristic's YES lives.

Proposition 4.15 (Local completeness of the reciprocity sieve)**.** *For an admissible base S and an odd prime ℓ, the admissible tail residues $Q_\ell = \{q \bmod \ell : (A_S q + D_S \mid \ell) \ne -1$ and $(B_S q + D_S \mid \ell) \ne -1\}$ fall into exactly three regimes.* (i) *$\ell \mid D_S$, which covers every $\ell \le 167$ uniformly over the* 49,961 *admissible bases, since $1/p \ge (T_{59} - 2) + \frac{1}{281}$ forces every prime $p \le 167$ into every base: then $A_S \equiv B_S \equiv N_S \not\equiv 0 \pmod{\ell}$, the two conditions collapse to the single condition $(N_S q \mid \ell) \ne -1$, and $|Q_\ell| = (\ell+1)/2 \ni 0$.* (ii) *$\ell \mid A_S B_S$: one condition is the constant $(D_S \mid \ell)$, so $Q_\ell = \emptyset$ iff $(D_S \mid \ell) = -1$ (exactly the certificate of Proposition 4.8) and otherwise $|Q_\ell| \ge (\ell-1)/2$.* (iii) *$\ell \nmid A_S B_S D_S$ (hence $\ell \ge 173$): the complete character sums give $|Q_\ell| \ge (\ell-3)/4 - 2 \ge 40$ (the linear sums vanish, the quadratic sum is $-\chi(A_S B_S)$, boundary terms ≤ 2). Consequently $Q_\ell \ne \emptyset$ forces $|Q_\ell| \ge 2$ at every modulus, the surviving local density is positive (e.g. $\prod_{\ell \le 300} |Q_\ell|/\ell \approx 2.2 \times 10^{-19} > 0$ on the canonical base), and by CRT* no finite system of congruences (no covering) can annihilate an unkilled family*: the binary certificate of Proposition 4.8 is the complete local theory of the tail sector, there is no "second kill layer," and the one–sidedness of Remark 4.13 is structural rather than an artefact of method. (All three regimes verified exhaustively on the canonical base: collapse and $|Q_\ell| = (\ell+1)/2$ at $\ell \in \{3, 5, 7, 101, 277\}$; the Weil floor at every prime $281 \le \ell \le 600$; A_S, B_S carry no factor below 10^6, consistent with the balanced–semiprime wall of Remark 4.10.)*

Remark 4.16 (The immune signal is Lehmer–type)**.** The immune families of Remark 4.14 do not merely survive the sieve; they carry a definite arithmetic shape. For a reciprocity–immune base the Pell system of Remark 4.7, $B_S x^2 - A_S y^2 = -4D_S^2$, is locally soluble everywhere ($\kappa_S > 0$), and a Pythagorean pair in the tail family $S \cup \{q\}$ is *exactly* a solution with $q = (x^2 - D_S)/A_S = (y^2 - D_S)/B_S$ a positive prime. Its solutions grow like ε^n in the fundamental unit of the order of discriminant $4A_S B_S$ (224 digits, $A_S B_S$ a nonsquare, on an explicit immune base), so q_n grows exponentially: the surviving question is the primality of a term of an exponential sequence; the Lehmer/Mersenne phenomenon, not a Bunyakovsky one. This is the same residue already isolated over $\mathbb{Z}[\sqrt{2}]$ in Remark 6.8 (primality of $|N(\rho)|$ along Pell–growing sequences), now on rational soil, and it sharpens the "weakly favours YES" of Remark 4.14: the immune contribution to the count is not a generic thin–set $\sum q^{-1}$ but a Lehmer–type expectation; heuristically infinitely many prime q_n, yet exhibiting even one (let alone one meeting the $k = 0$ cycle condition $D(a) = b$ that a genuine two–cycle demands on top of the Pythagorean pair) is open in the way the infinitude of Mersenne primes is open. The return itself, however, is milder than "$k = 0$" suggests: on the Pythagorean locus the deviation $k = (a' - b)/a$ is an *integer* (the identity $v(D(u) - v) = u(u - D(v))$ with $\gcd(u, v) = 1$), forced into $\{0, -1\}$ by the near–balanced immune ratio, so $k = 0$ is a sign choice rather than a measure–zero coincidence, and the naive Wieferich shortcut that would pin q is vacuous, since $x^2 \equiv D_S \pmod{q}$ holds identically from $x^2 = A_S q + D_S$. Whether $k = 0$ is met with positive conditional density is itself unresolved, but the wall is the primality of q_n, not the return. A reciprocity–killed base is the complementary degeneration: there the same Pell system is locally *insoluble* ($\kappa_S = 0$), so no candidate q_n exists at all. The dichotomy of Proposition 4.8 is thus Pell–solubility versus Pell–insolubility, and the residual existence signal is precisely a Lehmer primality question the present methods do not reach. This complements Proposition 8.3, which proves #307 admits *no* polynomial parametrisation: the present Pell orbit exhibits the family that *does* parametrise it, an exponential one. That the parametrisation is forced to be exponential is exactly why the residual question is Lehmer–class (primes in an exponential sequence) and not Schinzel–class (primes in a polynomial), which places it beyond every standard prime–density conjecture. It is worth locating *where* that hardness enters, because neither square condition carries it alone. Imposing only the minus–square on the tail family, $d^2 = B_S q + D_S$, gives $q = (d^2 - D_S)/B_S$; the admissible d are those with $d^2 \equiv D_S \pmod{B_S}$, and along such a class $d = d_0 + tB_S$ one has the identity

$$q(t) \;=\; \frac{d_0^2 - D_S}{B_S} \;+\; 2d_0\, t \;+\; B_S\, t^2,$$

a *quadratic* in t with integer coefficients (verified on toy bases: $3+54t+173t^2$, $7+238t+1693t^2$, $2+446t+17357t^2$). So the minus–square alone asks only for prime values of a quadratic; a Bunyakovsky question, of the same class as the $p = 5$ family of Proposition 12.19(iii), conjecturally routine. The plus–square is likewise a single square condition. It is their *conjunction* (the Pythagorean condition, equivalently the Pell system $B_S x^2 - A_S y^2 = -4D_S^2$) that replaces the quadratic parameter by a fundamental unit and turns the question exponential. The Lehmer–hardness is therefore a property of the *pair* of conditions, not of either one: this is the sandwich that locates the whole difficulty. The class can be named exactly. Since $x_{n+1} = tx_n - x_{n-1}$ has $x_n = (\alpha^n + \beta^n)/2$ with $\alpha\beta = 1$, the squares x_n^2 have characteristic roots $\alpha^2, \beta^2, 1$, and the constant shift is absorbed by the root 1: the candidate sequence

$$q_n = \frac{x_n^2 - D_S}{A_S} \qquad \text{satisfies} \qquad q_{n+3} = (t^2-1)q_{n+2} - (t^2-1)q_{n+1} + q_n,$$

a non–degenerate *ternary linear recurrence* (verified on four Pell parameters). So the existence route asks for a prime term of a ternary linear recurrence, and no unconditional infinitude of prime terms is known for *any* non–degenerate linear recurrence of order ≥ 2. The one unconditional tool that does reach exponentially growing sequences, the primitive divisor theorem of Bilu–Hanrot–Voutier [35] (every term beyond the 30th of a Lucas or Lehmer sequence has a primitive prime divisor), yields *divisors*, not primality, and so does not transfer: q_n must itself be the new prime of the support. The complementary negative route is already closed from the other side; an algebraic factorisation is unavailable (D_S squarefree, so $x^2 - D_S$ is irreducible) and no covering congruence can kill the family by Proposition 4.15. Both directions therefore fail on the same object, for opposite reasons. Quantitatively the wall is *double*–exponential in expectation: immunity makes $A_S B_S$ a product of two distinct primes, hence squarefree, so the conic lives in the near–maximal order of $\mathbb{Q}(\sqrt{A_S B_S})$ of discriminant $\sim A_S B_S \approx 10^{224}$ ($\sqrt{\Delta} \approx 10^{112}$, computed exactly at base scale), and the class–number formula puts the expected regulator at $R = \sqrt{\Delta}\, L(1,\chi)/(2h) \approx 10^{112}$; barring an anomalously large class number, the *least* point of a nonempty fiber sits at height $\exp\bigl(10^{112}\bigr)$; and nonemptiness itself is a class–group condition, since Hasse supplies only rational points. Three nested gates therefore separate an immune family from a solution: fiber nonemptiness (class group), a prime q_n along the orbit (Lehmer), and the $k = 0$ return; a direct sweep to $q \leq 10^{12}$ was structurally incapable of reaching even the first gate. By contrast the twisted fibers of Remark 6.9 have measured least points at two–digit norms and unbounded fiber multiplicity (their linear count is a sum over $\sim X$ Pell fibers, each individually orbit–sparse; the unimodular fibers $\det = \pm 1$ are empty in a 6×10^7 box): the tier gap between the twisted and rational problems is precisely a *fiber–height* gap in one and the same landscape, primality on Pell fibers.

Theorem 4.17 (Parity dichotomy)**.** *Let $a' = b, b' = a$ be a derivative two–cycle (so a, b squarefree,* $\gcd(a,b) = 1$*). Then exactly one of the following holds.*

(i) Mixed parity. *Exactly one of a, b is even; say $a = 2k$ with k odd. Then $b = a'$ is odd and $\omega(b)$ is even.*

(ii) All odd. *Both a, b are odd. Then $P \cup Q$ avoids 2, so reaching reciprocal sum > 2 needs ≥ 1412 odd primes, and* $\min(a, b) > 10^{2519}$.

Proof. They cannot both be even, as $\gcd(a, b) = \gcd(a, a') = 1$. In case (i), write $a = 2k$, k odd squarefree; $b = a' = k + 2k'$ is odd. For odd squarefree $b = \prod_i q_i$ one has $b' = \sum_i b/q_i \equiv \omega(b) \pmod 2$ (each b/q_i is odd). Since $b' = a$ is even, $\omega(b) \equiv 0 \pmod 2$. In case (ii) the union of supports omits 2; the reciprocal sum of the first 1411 odd primes is < 2 while that of the first 1412 exceeds 2, so $|P \cup Q| \geq 1412$ and $\prod_{p \in P \cup Q} p \geq 10^{5040}$ (product of the 1412 smallest odd primes, by direct computation). Applying the same rigidity/ratio argument as in Theorem 4.3(b) (now with $2 \notin U$) gives $\min(a, b) \geq \sqrt{R/T(U)} > 10^{2519}$. The all–odd case is therefore vastly more constrained than the mixed case and, heuristically, empty. □

Proposition 4.18 (Longer cycles are harder; two–cycles are extremal)**.** *Every member of a k–cycle of the arithmetic derivative is squarefree, and for $k \leq 3$ the members are pairwise coprime (each pair is consecutive). Their masses $s(a_i) = a_{i+1}/a_i$ multiply to 1, so by AM–GM $\sum_i s(a_i) \geq k$; for $k \leq 3$ this is the union's reciprocal sum, whence the support contains at least m_k primes, where m_k is least with $\sum_{i \leq m_k} 1/p_i \geq k$. Now $m_2 = 59$ and $m_3 =$ 361,139 (last prime 5,195,977), so a three–cycle has at least 361,139 primes in its support and product exceeding $10^{2.26\times10^6}$ (hence its largest member exceeds $10^{752{,}194}$); by Mertens m_k grows doubly exponentially. Two–cycles are thus the extremal, and only practically relevant; case. (For $k \geq 4$ non–consecutive members may share primes; the bound then holds for the multiset reciprocal sum.)*

5 Local anatomy and a heuristic model

Lemma 5.1 (Local anatomy)**.** *Let a be squarefree and q a prime with $q \nmid a$. Then $q \mid a'$ if and only if $\sum_{p|a} p^{-1} \equiv 0 \pmod q$, where p^{-1} is the inverse of p modulo q. If $q \mid a$, then $q \nmid a'$.*

Proof. $a' = a \sum_{p|a} 1/p$. If $q \nmid a$ then $q \mid a'$ iff $q \mid a \sum_{p|a} p^{-1}$ iff $\sum_{p|a} p^{-1} \equiv 0 \pmod q$. If $q \mid a$, then by Lemma 2.1 applied to the prime set of a, $q \nmid a'$. □

Thus the cycle equation $a' = b$, $b' = a$ factors, prime by prime, into $\omega(a) + \omega(b)$ local congruences of exactly the shape that governs amicable–pair heuristics: for each $q \mid b$ one needs $\sum_{p|a} p^{-1} \equiv 0 \pmod q$, and symmetrically. These local conditions, however, are individually *empty of obstruction*:

Proposition 5.2 (No single–modulus congruence obstruction)**.** *For every modulus m with $2 \leq m \leq 210$, the reduced cross–match system $N_P \equiv D_Q$, $N_Q \equiv D_P \pmod m$ is realisable by disjoint squarefree prime sets P, Q, in both structural branches: $\gcd(m, D_P D_Q) = 1$, and the branch in which a prime divisor of m lies in $P \cup Q$. Hence no individual modulus in this range can rule out a solution of #307.*

Construction. Write $\Sigma_S \equiv \sum_{p \in S} p^{-1} \pmod m$, so $N_S \equiv D_S \Sigma_S$. The reduced system is $\Sigma_P \equiv D_Q D_P^{-1}$, $\Sigma_Q \equiv D_P D_Q^{-1} \pmod m$; its only coupling is $\Sigma_P \Sigma_Q \equiv 1$, automatic, with D_P, D_Q otherwise free. By Dirichlet each unit class mod m contains primes, fixing D_P, D_Q; appending a prime $r \equiv 1 \pmod m$ leaves D_S fixed while sending $N_S \mapsto r N_S + D_S \equiv N_S + D_S$, i.e. incrementing Σ_S by 1, so (when $\gcd(D_S, m) = 1$) every target class is reached with disjoint sets. In the branch $\ell \mid m$, $\ell \in P$, reduction mod ℓ collapses the system to $N_Q \equiv D_P \equiv 0 \pmod \ell$, met by inverse–paired residues in Q; composite m needs no separate gluing, the increment acting directly mod m. We verified both branches for all $2 \leq m \leq 210$ (for prime and prime–power moduli the inside branch requires the directed construction above; seeding a prime divisor of m into P, a blind search missing these as a search–power artefact, not an arithmetic obstruction). □

Remark 5.3 (Precise scope)**.** This excludes only a *single fixed–modulus* congruence disproof. The witnesses are m–dependent (the modulus–6 witness already fails modulo 210), and a single (P, Q) solving the system for *all* m simultaneously is, for bounded integers, equivalent to the integer identities $N_P = D_Q$, $N_Q = D_P$ (that is, to #307 itself) so per–modulus solvability is *no* evidence for existence. Nor does it exclude other finite or local negative arguments (size/Mertens bounds, density, magnitude–congruence hybrids, or a Hasse–type failure across infinitely many moduli); indeed our own $|U| \geq 59$ and 2.09×10^{56} barriers are finite–data results fully consistent with it. Together with Theorem 6.1 (every non–archimedean shadow trivially negative for degree reasons) it brackets the problem: no single–modulus obstruction over $\mathbb{Z}$, fatal non–archimedean analogues over $\mathbb{F}_q[t]$, the difficulty living at the archimedean place.

A first–order count. Model a candidate squarefree a as "P–abundant" ($\sum_{p|a} 1/p > 1$) and ask for $b = a'$ to itself be Q–abundant with $\sum_{q|b} 1/q = 1/s$. Two empirical inputs (Section 9) calibrate this:

- the density of P–abundant n is $\delta = 0.0420$ (and ≈ 0.0176 among squarefree n); a theorem, not merely an observation (Proposition 5.6);
- for a qualifying squarefree a, $b = a'$ is squarefree about 95% of the time, but its mass $\sum_{q|b} 1/q$ has mean ≈ 0.047 against the required ≈ 0.934; the empirical tail probability $\Pr[\sum_{q|b} 1/q \geq 1/s]$ is 0 in 40,920 samples at these scales (Section 9).

The gap is structural: at small scale a P–abundant a hogs the cheap mass–carrying small primes, and $b = a'$, being coprime to a, is both too small and too coprime to assemble reciprocal mass $1/s$. The extremal case is stark: for the primorial $a = \prod_{i \leq k} p_i$, though $\sum_{p|a} 1/p$ rises through 2 at $k = 59$, the cofactor sum a' is essentially prime, with $\sum_{q|a'} 1/q < 0.2$ for every $k < 100$ (and only 0.0014 at $k = 59$): the family that *maximises* the forward mass is the most starved on the return. Equivalently, for squarefree n with n' squarefree, $n'' = n\,\sigma(n)\,\sigma(n')$, so the two–step return ratio $n''/n = \sigma(n)\sigma(n')$ is the #307 product itself; over *all* such $n < 10^6$ it stays strictly below 1, never exceeding 0.54 (maximal at $n = 969969 = 3 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19$), so D is two–step contracting throughout the tested range and must reverse to unit ratio for a solution to exist. The first scale at which b could plausibly carry the required mass is of the order of the barrier ($\sim 10^{56}$); this dynamical mass–gap is *consistent with* the proved barrier rather than an independent derivation of it (the empirical tail probability 0 at small scale is equally consistent with NO and with YES–beyond–the–barrier). A naive "expected number of solutions" built from the local congruences (one factor $\sim 1/q$ per condition) diverges like a constant times $\log X$; this is a divergent naive count of exactly the type known to be unreliable, so we treat it only as very weak, non–rigorous suggestive evidence that solutions could exist at or beyond the barrier.

Remark 5.4 (Calibration)**.** We present the count of the previous paragraph as a heuristic, not a theorem. Heuristics of amicable–pair type are known to be delicate; we report it as weak evidence, consistent with the divergence of the refined expected count, that the answer to #307 is plausibly YES (and hence that the period–one form of the Ufnarovski–Åhlander conjecture is plausibly false).

Remark 5.5 (A cautionary refinement)**.** One can try to sharpen this into a quantitative rate using a Kubilius/Erdős–Wintner model, stratified over which small primes lie in the support of a versus remain available to b. Two lessons emerge, both methodological. First, a plain Monte–Carlo estimate *undercounts* the rate by some 10^{61}: the expectation is dominated by rare, balanced splits of the small primes that sampling almost never sees, so a naive estimate of this kind is misleadingly small and only an explicit stratified sum corrects it. Second, and this is why we record no numerical prediction, the stratified estimate is itself unstable: refining the stratification from the first 8 to the first 14 primes drops the rate by some 17 orders of magnitude and pushes the implied first occurrence from $\sim 10^{10^{38}}$ out past $\sim 10^{10^{55}}$, with no sign of convergence. The only robust conclusion is negative: every version places a hypothetical first solution *doubly exponentially* beyond any conceivable search, consistent with the constructive deficit of Section 8; the model does not reliably decide existence in either direction.

Proposition 5.6 (The abundance density is a theorem)**.** *The additive function $f(n) = \sum_{p|n} 1/p$ satisfies the Erdős–Wintner three–series criterion [17] (here $\sum 1/p^2$ and $\sum 1/p^3$ converge), so f has a limiting distribution and the density $\delta = \lim_{x\to\infty} \frac{1}{x}\#\{n \le x : f(n) > 1\}$ exists, equal to $\Pr\big[\sum_p X_p/p > 1\big]$ for independent $X_p \sim \mathrm{Bernoulli}(1/p)$. Moreover the constant carries a* certified *enclosure:*

$$0.04202 \ \le\ \delta\ \le\ 0.04223.$$

Writing $S_1 = \sum_{p\le P_1} X_p/p$, $P_1 = 3\times 10^4$, and $T \ge 0$ for the tail, monotone upper and lower cdf envelopes of S_1 are propagated through the Bernoulli convolution on a grid of step 10^{-7} with the shift $1/p$ rounded down for the upper and up for the lower envelope (which preserves them, cdfs being nondecreasing) so $\delta \ge 1 - U(1)$ costs nothing (the tail is nonnegative), while $\delta \le 1 - L(1-a) + \Pr[T \ge a]$ with the explicit Chernoff bound $\Pr[T \ge a] \le \exp(1.36 - P_1 a)$, $a = 10^{-3}$ (from $\log \mathbb{E}\, e^{\lambda T} \le \lambda s_2 + 0.72\lambda^2 s_3$ at $\lambda = P_1$, using $s_2 < 1/P_1$, $s_3 < 1/(2P_1^2)$ and $e^x - 1 \le x + 0.72x^2$ on $[0,1]$); float slack 10^{-9} exceeds the accumulated rounding by nine orders, and the machinery reproduces an exact four–prime reference to 2×10^{-16} (`code/delta_cert.py`). The point value 0.0420, at the bottom of the enclosure, agrees to four places with the sieve to 2×10^7; the upper edge is conservative, carried almost entirely by the a–window. As for the density of abundant integers, no closed form is expected.

Remark 5.7 (Two dynamical probes: no second obstruction)**.** Two measurements sharpen the picture beyond the first–order mass count, and both cut towards rather than against a solution. *(i) Image density.* A two–cycle needs *both* members in the image $D(\mathbb{N})$ (each is the other's derivative), so one might fear a second obstruction; that the high–mass numbers a cycle requires are seldom derivatives. The opposite holds: of the integers below 10^7, 68.3% lie in $D(\mathbb{N})$, and the fraction *rises* with mass, from 61.7% at $\sum_{p|n} 1/p < 0.2$ to 78.2% at $\sum_{p|n} 1/p > 1$ and 80% beyond. The high–mass numbers are thus over–represented among derivatives; the second cycle condition is comfortably met. *(ii) The barrier is where the balanced mass–product reaches* 1. Set $r(n) = \big(\sum_{p|n} 1/p\big)\big(\sum_{q|n'} 1/q\big) = D^2(n)/n$, so $r = 1$ is exactly the cycle condition. Exhaustively over $n \le X$ the maximum of r climbs $0.50 \to 0.62$ as X runs $10^3 \to 10^9$ (with no exact cycle, as the barrier demands), with no ceiling; the climb follows the balanced split $r \approx (S(k)/2)^2$, $S(k) = \sum_{i\le k} 1/p_i$, which reaches 1 precisely when $S(k) = 2$, i.e. $k \approx 59$. The dynamical trajectory thus reproduces the $|U| \ge 59$ barrier from the opposite direction. Neither probe decides existence; both *remove* hypothetical obstructions, leaving the exact archimedean closure as the sole gate.

Remark 5.8 (Why the analytic method is starved: $N(r) \le 1$)**.** The rank–one rigidity (Section 11) has a sharp consequence for the circle–method machinery that resolves the *unrestricted* Egyptian–fraction problem. For a rational r let $N(r) = \#\{\, P : \sum_{p\in P} 1/p = r \,\}$ count the finite prime sets with reciprocal sum r. Then

$$N(r) \le 1 :$$

by the Rigidity Lemma (Theorem 2.1) the sum $\sum_{p\in P} 1/p = N_P/D_P$ is already in lowest terms, so the reduced denominator of r equals $D_P = \prod P$ and P is exactly its set of prime divisors; uniquely determined ($N(r) = 1$ precisely when that denominator is squarefree and the numerator of r is its cofactor sum). The circle method needs the opposite: a target hit by a positive proportion of subsets, so a main term dominates. This is exactly what Bloom's shift to $\{p-1\}$ manufactures, $p-1$ being composite and divisor–rich, a rational is the reciprocal sum of *many* subsets of $\{1/(p-1)\}$ (already among the first fourteen shifted primes the multiplicity reaches 3 and grows), whence [6] represents every positive rational there. Over the primes themselves $N(r) \le 1$ leaves nothing to average: the analytic engine that reaches $\{p-h\}$ is provably starved on $\{p\}$, and this is the method–level reason #307 (the rank–one, exact–prime case) lies beyond it.

6 A function–field lens: the obstruction is archimedean

Define an arithmetic derivative on $\mathbb{F}_q[t]$ by $P' = 1$ for monic irreducibles and the Leibniz rule, so $f' = \sum_i e_i\, f/P_i$ for $f = \prod_i P_i^{e_i}$ (the exponents e_i read modulo the characteristic).

Theorem 6.1. *Over $\mathbb{F}_q[t]$ this derivative has no two–cycles and no nonzero fixed points. Indeed $\deg f' \le \deg f - 1$ for every nonconstant f. The analogues of the integer fixed points p^p also vanish: $(P^p)' = p\,P^{p-1}P' = 0$ in characteristic p. The same holds verbatim for every twisted variant ($P' = u_P \in \mathbb{F}_q^\times$, Leibniz).*

Proof. Each term $e_i\, f/P_i$ has degree $\deg f - \deg P_i \le \deg f - 1$, so $\deg f' \le \deg f - 1$. A two–cycle $f' = g$, $g' = f$ would force $\deg g \le \deg f - 1$ and $\deg f \le \deg g - 1$, hence $\deg f \le \deg f - 2$; a fixed point $f' = f$ is excluded likewise. (Verified exhaustively over $\mathbb{F}_2$ to degree 7, $\mathbb{F}_3$ to degree 5, and $\mathbb{F}_5$ to degree 4; twisted variants over $\mathbb{F}_2$ and $\mathbb{F}_5$ on small supports through 2.3×10^6 twist configurations, none closing.) □

Corollary 6.2 (The function–field Erdős–Barbeau equation is insoluble)**.** *For every finite field $\mathbb{F}_q$, all nonempty finite sets P, Q of monic irreducibles, and all unit weights $u_\pi, v_\rho \in \mathbb{F}_q^\times$,*

$$\Big(\sum_{\pi\in P} \frac{u_\pi}{\pi}\Big)\Big(\sum_{\rho\in Q} \frac{v_\rho}{\rho}\Big) \ne 1.$$

Proof. Write $f = \prod P$, $g = \prod Q$, $D_u(f) = \sum_\pi u_\pi f/\pi$; the equation reads $D_u(f)\,D_v(g) = fg$. If some $\pi \in P \cap Q$, then $\pi^2 \mid fg$ while $D_u(f) \equiv u_\pi f/\pi \not\equiv 0$ and $D_v(g) \equiv v_\pi g/\pi \not\equiv 0 \pmod{\pi}$, so $\pi \nmid D_u(f) D_v(g)$; impossible; hence $\gcd(f, g) = 1$. The same congruence gives $\gcd(f, D_u(f)) = \gcd(g, D_v(g)) = 1$, so $g \mid D_u(f)$ and $f \mid D_v(g)$, forcing $D_u(f) = c\,g$, $D_v(g) = c^{-1} f$ with $c \in \mathbb{F}_q^\times$: a twisted two–cycle, excluded by the degree drop. The weighted product equation is thus insoluble over every $\mathbb{F}_q(t)$; it *is* soluble over the unit–rich number rings of Theorem 6.11, since any twisted cycle gives $(D_u(a)/a)(D_v(b)/b) = (b/a)(a/b) = 1$, and over $\mathbb{Q}$, where the admissible weights are ± 1, it is #307 itself with the signed barrier of Remark 6.4, open beyond 2.09×10^{56}. The three regimes (non–archimedean monotonicity (closed, negative), unit–broken monotonicity (closed, positive), ordered–archimedean (open, barred)) now have theorems on two of three doors. □

Over $\mathbb{Z}$, by contrast, $n' = n \sum_i e_i/p_i$ can exceed n precisely because the archimedean absolute value lets a sum of strictly smaller terms outgrow the original; the "mass > 1" phenomenon $\sum 1/p > 1$, which has no degree–theoretic analogue. Theorem 6.1 thus localises the entire existence question to the archimedean place: every non–archimedean shadow of #307 is trivially negative. This is consistent with (and partly explains) both the census finding of Section 9 and Proposition 5.2 (no single modulus obstructs the cross–match system in the tested range); the difficulty is not local. Any eventual proof must therefore distinguish $\mathbb{Z}$ from $\mathbb{F}_q[t]$; over the latter the decisive tools are the degree and the Mason–Stothers/Wronskian inequality [18], neither of which has a $\mathbb{Z}$–analogue. The conjectural analogue, *abc*, does not fill the gap, and quantifiably so. Every cycle inherits, from $N' = a^2 + b^2$ with $N = ab$, the relation $(a-b)^2 + 4ab = (a+b)^2$ on coprime squarefree members, but this triple is *abc*–unexceptional, of quality $q = \log(a+b)^2/\log \operatorname{rad}\big((a-b)\cdot ab \cdot (a+b)\big) \approx \frac{1}{2}$ (median 0.543, never above 0.74 across 6×10^4 coprime squarefree pairs; the only $q > 1$ cases are degenerate smooth–gap pairs such as $(5, 3)$, vanishingly far below the 10^{56} scale a solution must occupy). Granting *abc* in its strong explicit form would therefore yield no bound here: the function–field collapse rests on an inequality genuinely *absent* over $\mathbb{Z}$, not merely unproven, and the size mismatch is by a factor of two in the exponent. #307 lives in the *abc*–comfortable zone.

Remark 6.3 (The free grading, and what evaluation destroys)**.** Theorem 6.1 has a structural reading. In the free Leibniz world (the polynomial ring $\mathbb{Z}[x_p]$ with $x_p' = 1$) the derivative of a squarefree monomial of degree n is the elementary symmetric form e_{n-1}, homogeneous of degree $n-1$: the formal derivative strictly lowers the grading, cycles are impossible, and over $\mathbb{F}_q[t]$ the degree realises this grading faithfully. Over $\mathbb{Z}$ the evaluation $x_p \mapsto p$ destroys homogeneity, and the data show it destroys it completely: for random squarefree a of fixed size, $\omega(a')$ is governed by the size of a' alone (Erdős–Kac), flat in $\omega(a)$ within each parity class (measured: mean $\omega(a') = 2.7$–3.0 as $\omega(a)$ runs over $3, 5, 7$, and 3.6–4.0 over $4, 6, 8$, the offset being the parity anatomy of Theorem 4.17). A two–cycle is therefore a doubly anti–typical event: an additively assembled integer must carry multiplicative rank far in the Erdős–Kac tail *and* on an exactly prescribed support; the two independent prices whose product is the heuristic rarity of Section 5. The deeper additive–multiplicative content collapses, on inspection, into the S–unit form of Proposition 6.7: dividing the cycle equation by D_Q exhibits it as an n–term unit equation; one more coordinate system in which the same exactness reappears unchanged.

Remark 6.4 (The obstruction is *ordered*–archimedean)**.** The barrier is finer than "archimedean": its engine is AM–GM on real, *positive* masses, i.e. the ordering of $\mathbb{R}$. Over $\mathbb{Z}[i]$ (a PID, so the Rigidity Lemma and cross–matching survive verbatim, with cycles defined up to units), the masses $\sum 1/\pi$ are *complex*: phases can align so that $\left|\sum 1/\pi\right| > 1$ with as few as three Gaussian primes (e.g. $\frac{1}{1+i} + \frac{1}{2+i} + \frac{1}{2-i} = 1.3 - 0.5i$), and the 59–prime bound has no analogue. The ordered–archimedean thesis thus predicts that small derivative two–cycles *may* exist over $\mathbb{Z}[i]$ where they are barred over $\mathbb{Z}$, and the prediction is confirmed. Two facts make the contrast precise. *First*, phases do not help over $\mathbb{Z}$: for any *signed* derivative ($p' = \pm 1$, Leibniz) a two–cycle still satisfies $|s||t| = 1$ with $|s| \le \sigma(a)$, $|t| \le \sigma(b)$ and disjoint supports (the rigidity proof is unchanged), so $\sigma(a) + \sigma(b) \ge 2$ and the full 59–prime, 2.09×10^{56} barrier applies to every choice of signs: real phases cannot cancel. *Second*, fourth roots of unity suffice instantly: Leibniz derivatives on $\mathbb{Z}[i]$ with unit prime–values ($\pi' \in \{\pm 1, \pm i\}$) possess genuine two–cycles at norm $\sim 10^4$. The smallest:

$$a = (1+i)(2+7i)(6+11i) = -129 - i, \qquad b = 3\,(2+i)(1+2i)(6+5i) = -75 + 90i,$$

with $a' = b$ under the *standard* values $\pi' = 1$ on the primes of a, and $b' = a$ exactly under $\pi' = i$ on 3, $2+i$, $1+2i$ and $\pi' = 1$ on $6+5i$; seven prime classes ($N(a) = 16{,}642$, $N(b) = 13{,}725$) against the fifty–nine that $\mathbb{Z}$ demands. Exhaustive search finds exactly sixteen such cycles up to conjugation and direction having a member of norm $\le 2 \times 10^7$ with $\omega \le 5$ on the enumerated side (ω–profiles $(3, 4)$, $(4, 4)$, $(5, 4)$, and even $(4, 6)$) and the population grows steadily (4 below 3×10^5, 16 below 2×10^7), consistent with an infinite, slowly growing family; in six of the sixteen one side requires no twist at all. All sixteen were verified exactly, by independent recomputation; no twisted *fixed* point ($a' = ua$) exists in the same range. The strict two–sided normalisation $\pi' = 1$ (first–quadrant representatives) has no cycle up to units (nor any fixed point, zero–derivative, or unit–derivative element) with

$N(a) \le 2 \times 10^9$ (9.4×10^8 candidates, partners by full factorisation); this is the expected order rather than a separate rigidity, a single fixed phase system being heuristically $4^{\omega-1}$–fold rarer than the union of all systems. The contrast with $\mathbb{Z}$ survives this accounting: over $\mathbb{Z}$ *every* one of the $2^{\omega-1}$ sign systems is barred by the theorem above, while over $\mathbb{Z}[i]$ the union of phase systems is demonstrably populated from norm 1.7×10^4 on. The barrier of #307 is thus localised precisely: it is the impossibility of phase cancellation in the ordered field $\mathbb{R}$, where the only available phases ± 1 provably gain nothing; while the moment fourth roots of unity are admitted, the analogous existence problem closes at seven primes. The cycles found come in conjugate pairs (supports not conjugation–stable); a conjugation–*stable* Gaussian cycle would descend to rational data through the norm form $a^2 + b^2$, i.e. precisely the Pythagorean layer of Proposition 12.2; now a concrete target rather than a speculation.

Remark 6.5 (The descended operator; the symmetric sector)**.** Conjugation–stable supports admit an exact descent to $\mathbb{Z}$. Such a squarefree Gaussian integer is an associate of an odd squarefree rational m (split primes entering as whole conjugate pairs; the $(1+i)$–sector is empty by a short parity argument), and a conjugation–equivariant unit assignment makes the twisted derivative rational:

$$D(m) \;=\; \sum_{\substack{p \mid m \\ p \equiv 1\,(4)}} t_p \,\frac{m}{p} \;+\; \sum_{\substack{q \mid m \\ q \equiv 3\,(4)}} \varepsilon_q \,\frac{m}{q}, \qquad t_p \in \{\pm 2x_p, \pm 2y_p\}, \quad \varepsilon_q = \pm 1,$$

where $p = x_p^2 + y_p^2$; a Leibniz derivative on odd squarefree integers whose prime values have size $2\sqrt{p}$ rather than 1, and whose masses $\sim 2/\sqrt{p}$ reach 2 with four primes, so no 59–prime barrier exists for it. A two–cycle of D is exactly a conjugation–stable Gaussian cycle and would plant the phenomenon on rational soil. Parity constrains the hunt: $D(m)$ is odd iff m carries an odd number of primes $\equiv 3 \pmod 4$, so every member must contain such a prime. One level deeper, since $q^{-1} \equiv q \pmod 8$ for odd q, the descended operator obeys $D(m) \equiv m\,W(m) \pmod 8$ with $W(m) = \sum_{p \mid m} t_p\, p$, so a two–cycle forces $W(m) \equiv W(m') \equiv mm' \pmod 8$; a mod–8 constraint on the sign assignment that prunes the $4^{\#\mathrm{split}} 2^{\#\mathrm{inert}}$ phase enumeration severalfold in future searches (verified on 2×10^5 random assignments; `code/peeling_groupoid.py`). An exhaustive search over members $m \le 10^9$ (Gaussian norm 10^{18}, $\omega \le 8$, 1.4×10^9 closure tests) found no cycle, no fixed point $|D(m)| = m$, and no vanishing $D(m) = 0$. We claim no obstruction from this: equivariance prunes the available phases fourfold per split prime, and a phase count makes emptiness at this depth the expected order. The descent target remains open over $\mathbb{Z}[i]$, but not in the real–quadratic world, where the corresponding sector contains a fully rational pair (Remark 6.6). Finally, the descended coefficients are classical objects: at a split prime p the four values $\{\pm 2x_p, \pm 2y_p\}$ are exactly the Frobenius traces of the congruent–number curve $y^2 = x^3 - x$ (CM by $\mathbb{Z}[i]$) and of its twists ($a_p \in \{\pm 2x_p, \pm 2y_p\}$ verified by point counting for all $p < 120$) so the conjugation–stable sector is the arithmetic derivative twisted by the Hecke eigensystem of $\mathbb{Q}(i)$, and its mass statistics are governed unconditionally by Hecke's equidistribution theorem: the first point of contact between #307 and automorphic forms. Concretely: over random squarefree m the three–series conditions converge (the variance term is $\sum 2/p^2$), so the descended mass $D(m)/m$ has an Erdős–Wintner *limiting distribution* whose prime components are arcsine laws scaled by $1/\sqrt{p}$; the required input, equidistribution of the Hecke angles, is visible exactly (Kolmogorov–Smirnov distance 0.0041 from uniform over the 8,977 split primes below 2×10^5, well inside the 95% band 0.0144).

Remark 6.6 (The wall is the orderability of $\mathbb{Q}$)**.** The cycles over $\mathbb{Z}[i]$ permit a dissection of the barrier that no amount of theory over $\mathbb{Z}$ could provide. The entire structural chain of this note (rigidity, cross–matching, the Pythagorean identity, the split discriminant) transfers verbatim to twisted derivatives over $\mathbb{Z}[i]$: Leibniz holds for disjoint supports, so a cycle (a, b) has $D(N) = a^2 + b^2$ for $N = ab$, with $D(N) - 2N = (a - b)^2$ and $2N - D(N) = \big(i(a - b)\big)^2$ (all verified exactly on the cycles found). The last equation exposes the wall: over $\mathbb{Z}[i]$, where $-1 = i^2$ is a square, *both* discriminant factors are squares automatically, and the plus/minus distinction of Proposition 12.17 (which over $\mathbb{Z}$ carries the entire obstruction, via $(a - b)^2 \geq 0 \Rightarrow \sigma(N) \geq 2$) has no content. The unique non–transferring link is the sign–definiteness of squares: the barrier is, in the end, the *formal reality* of $\mathbb{Q}$ in the sense of Artin–Schreier. (The two Artin–Schreier theorems thus bracket the problem: their characteristic–2 map builds the tower of Proposition 12.12, their theory of orderable fields names the wall.) This classifies where the mechanism can operate at all: it needs an ordering making squares nonnegative (formal reality, hence a real embedding) *and* bounded twisting phases (finite unit group); by Dirichlet's unit theorem, unit rank $r_1 + r_2 - 1 = 0$ with $r_1 \geq 1$ forces degree one. $\mathbb{Q}$ *is the only number field over which the mass barrier can operate*: over imaginary quadratic fields squares lose sign–definiteness (over $\mathbb{Z}[i]$ even -1 is a square), and over every other field the unit rank is positive and twisted masses are unbounded. This is a classification of the wall's habitat, not an existence claim elsewhere, but over $\mathbb{Z}[i]$ existence is no longer a claim: it is a list of sixteen. The ladder of imaginary quadratic rings confirms the mechanism quantitatively. Over $\mathbb{Z}[\sqrt{-2}]$, whose only units are ± 1, precisely the twisting freedom that provably gains nothing over $\mathbb{Z}$; sign–twisted cycles nevertheless exist from norm 11,462 on ($a = -90 - 41\sqrt{-2}$, $b = 75 - 45\sqrt{-2}$, seven prime classes): the complex embedding alone carries the phenomenon. Over the Eisenstein ring $\mathbb{Z}[\omega]$, with six units, cycles appear at norm 2,964 with *six* prime classes ($a = -40 + 22\omega$, $b = 56 - 21\omega$); the smallest derivative two–cycle exhibited in any ring, and the population at fixed height grows with the unit group: at norm $\leq 2 \times 10^7$ the censuses read 14, 56, and 278 twisted–cycle instances for unit groups of order 2, 4, 6 (a factor $\sim$4–5 per two extra units, as a phase count predicts) while the strict canonical systems of all three rings remain empty to norm 2×10^9 (3×10^9 candidates in total, partners by full factorisation). Two further objects appeared over $\mathbb{Z}[\sqrt{-2}]$: a composite with *unit* derivative ($a' \sim 1$; impossible over $\mathbb{Z}$, where $n' = 1$ characterises the primes), and an *anti–holomorphic* cycle: $a = -3163 - 1474\sqrt{-2}$ satisfies $a' = -\bar{a}$ exactly, under a single conjugation–equivariant sign assignment closing both directions ($N(a) = N(b) = 14{,}349{,}921$, five prime classes each side). The equation $a' = u\bar{a}$ is vacuous over $\mathbb{Z}$ only in its *one–dimensional* reading, where it says $n' = \pm n$. In the reading that matches the problem it is not vacuous but central: by Proposition 12.6 the Pythagorean layer over $\mathbb{Z}$ *is* the anti–holomorphic equation $\mathcal{D}z = \mu\bar{z}$ for the coordinatewise derivative on $z = a + bi$, and #307 is exactly its unit–multiplier case. The ring ladder and the main problem are therefore instances of one equation rather than analogues; what singles out $\mathbb{Z}$ is the extra constraint $\operatorname{Im}\mu = 1$, which confines the multiplier to a line meeting $\mathbb{Z}[i]^\times$ in the single point $\mu = i$. That the anti–holomorphic sector is thin is uniform across the ladder: over $\mathbb{Z}[\sqrt{-2}]$ the cycle above is still the *only* one after widening the census simultaneously in all three directions; prime–norm $\leq 2 \times 10^4$, $\omega \leq 8$, product–norm $\leq 10^{10}$ (a factor 700 past the cycle itself), 8.09×10^8 supports, `hunt/antiholo_hunt.rs`, and over $\mathbb{Z}$ the sector is empty below 10^{112} by Corollary 12.3. We record the plateau as consistent with genuine finiteness, not as evidence for it: a random model for $a' = -\bar{a}$ predicts a count growing like $\log X$, which the tested range cannot separate from a constant. All cycles were verified exactly by independent recomputation. Finally, the second pillar was tested in isolation: over $\mathbb{Z}[\sqrt{2}]$ (*totally real*, so squares are sign–definite in both embeddings, but with infinite unit group $\pm(1 + \sqrt{2})^{\mathbb{Z}}$) twisted cycles appear already at norm 882: with unit twists of height ≤ 2, $a = 21\sqrt{2}$ (classes $\sqrt{2}$, $-1 + 2\sqrt{2}$, $1 + 2\sqrt{2}$, 3) cycles exactly with $b = 51 + 13\sqrt{2}$, of norm 2263 and only *two* prime classes (against the fifty–nine that $\mathbb{Z}$ demands) and the composite $369 - 15\sqrt{2}$ has derivative exactly $\varepsilon^6 = 99 + 70\sqrt{2}$, a unit. Both pillars of the classification are thus load–bearing, and each is now confirmed by explicit counterexample on the soil where it alone fails: drop formal reality and cycles appear (imaginary quadratic rings, norms 10^3–10^4); keep formal reality but drop unit finiteness and cycles appear (real quadratic, norm 882, the smallest in any ring); over $\mathbb{Q}$, where both hold, the barrier stands at 10^{112} by theorem. The bracketing of $\mathbb{Q}$ is complete from both axes. The deep census sharpens it: at norm $\leq 2 \times 10^7$ the real–quadratic ring carries 30,204 twisted–cycle instances against $14, 56, 278$ for the imaginary rings with $2, 4, 6$ units infinite unit rank dominates by two orders of magnitude, and contains both the *minimal* architecture, $a = \sqrt{2}\,(5 - \sqrt{2})$ cycling with $b = 7$ at norms $(46, 49)$, two prime classes per member (the rational prime 7 being composite in the ring), the smallest derivative cycle found in any ring; and a *fully rational* pair:

$$D(3707) = 1547, \qquad D(1547) = 3707 \qquad (3707 = 11 \cdot 337, \quad 1547 = 7 \cdot 13 \cdot 17),$$

exactly, with unit heights ≤ 2 on both sides, verified independently: two ordinary integers in a derivative two–cycle. The descent phenomenon that is empty over $\mathbb{Z}[i]$ to norm 10^{18} is realised over $\mathbb{Z}[\sqrt{2}]$ at four digits. The canonical system of $\mathbb{Z}[\sqrt{2}]$ remains cycle–free to norm 2×10^9 (7.7×10^8 candidates), now with seven unit–derivative composites.

The cycle phenomenon over rings with units is governed by the theory of unit equations, which supplies its rigidity theorem.

Proposition 6.7 (S–unit rigidity of twisted cycles)**.** *Let $\mathcal{O}$ be the ring of integers of a number field and S a finite set of primes of $\mathcal{O}$. If (a, b) is a twisted two–cycle with* $\operatorname{supp}(ab) \subseteq S$*, then dividing $D(a) = b$ by b exhibits a solution of*

$$x_1 + \cdots + x_{\omega(a)} = 1, \qquad x_i = u_i\, \frac{a/\pi_i}{b},$$

in the S–unit group of the field, and likewise for b. By the theorem of Evertse–Schlickewei–Schmidt [33] (cf. [32]), such an equation has only finitely many nondegenerate solutions; hence every fixed support carries at most finitely many twisted cycles free of vanishing subsums*, and the species can be infinite only through growing supports; precisely the behaviour of every census above. (Vanishing subsums correspond to partial zero–derivative patterns; none occurred in any search. The relation was verified exactly on the $\mathbb{Z}[\sqrt{2}]$ specimen: the four S–unit terms of $a = 21\sqrt{2}$ sum to 1.)*

Remark 6.8 (Galois folding, and the thinness of symmetric sectors)**.** Over a quadratic field with automorphism σ, a σ–equivariant choice of unit values ($u_{\pi^\sigma} = u_\pi^\sigma$) makes D commute with conjugation, so the *single* equation $D(a) = a^\sigma$ already implies that (a, a^σ) is a twisted two–cycle (checked symbolically and on 1,988 random instances). In the two–prime case the partner is forced: $\rho = -\big(\bar{u}v\,\pi + \bar{v}\,(\pi^\sigma)^2\big)/\big(N(u) - N(\pi)\big)$, with integrality a congruence and the entire residue of the problem the primality of $|N(\rho)|$ along Pell–exponentially growing sequences; Lehmer territory rather than Bunyakovsky. Whether a folded sector is starved or rich is decided by its degrees of freedom. When the fold leaves no slack the sector is thin: the folded *two*–prime family over $\mathbb{Z}[\sqrt{2}]$ is empty in the tested box (36,000 parameter triples, none even integral), the conjugation–stable sector over $\mathbb{Z}[i]$ is empty to Gaussian norm 10^{18}, and the anti–holomorphic cycle over $\mathbb{Z}[\sqrt{-2}]$ is unique in its census. But with *three* primes the folded equation $u_1\pi_2\pi_3 + u_2\pi_1\pi_3 + u_3\pi_1\pi_2 = (\pi_1\pi_2\pi_3)^\sigma$ carries three unit unknowns against two real equations, and this underdetermined system is *populated*: an exhaustive box ($p_i \leq 137$, unit heights ≤ 4) contains twenty–four solutions over eleven prime triples, the smallest being

$$a = (3 + \sqrt{2})(5 - 2\sqrt{2})(5 - \sqrt{2}) = 57 - 16\sqrt{2}, \qquad D(a) = a^\sigma = 57 + 16\sqrt{2},$$

with unit values ε^2, ε, $-\varepsilon^{-1}$: *the derivative conjugates the element*, at norm 2737 on both sides, the equivariant return being automatic (verified exactly). Existence of twisted cycles over the four rings tested is a theorem by exhibition; for *every* ring with infinite unit group it is now reduced to the abundance of lattice points on these underdetermined unit systems (measured plentiful already at $k = 3$) a geometry–of–numbers question rather than a primality one.

Remark 6.9 (The forced third prime: a Schinzel–type reduction over $\mathbb{Z}[\sqrt{2}]$, unblocked)**.** Within that abundance the arithmetic is sharper than free lattice–counting. Fixing π_1, π_2 and the units u_1, u_2, u_3, the folded equation rearranges to $\pi_3\,(u_1\pi_2 + u_2\pi_1) - (\pi_1\pi_2)^\sigma \pi_3^\sigma = -u_3\pi_1\pi_2$, a 2×2 integer linear system for $\pi_3 = (x, y)$ whose determinant is $N(u_1\pi_2 + u_2\pi_1) - N(\pi_1\pi_2)$ (an identity, verified on all 168 solutions of the box). So π_3 is *forced* by $(\pi_1, \pi_2, u_1, u_2, u_3)$, and infinitude of twisted cycles is exactly the primality of this forced π_3 for infinitely many such data a Schinzel/Bunyakovsky–type condition over $\mathbb{Z}[\sqrt{2}]$. Crucially it is *not* the regime of Proposition 8.3: the exponential unit freedom $u_i = \pm\varepsilon^{k_i}$ keeps the family non–polynomial yet populated, so the polynomial–rigidity that forbids reducing the *rational* #307 to such a hypothesis does not apply here. A Schinzel–type theorem over $\mathbb{Z}[\sqrt{2}]$ would render twisted infinitude unconditional, whereas none touches #307 itself; the twisted problem sits strictly closer to the reach of current sieve theory than its rational shadow. Concretely the twisted solution set is *abundant* at accessible scale; thousands of cycles at modest norm (Proposition 6.12), and the count of distinct prime–norm signatures grows roughly in proportion to the norm bound in our census (e.g. $6, 12, 20$ at bounds $40, 80, 160$, unit heights saturating); whereas the rational Pythagorean layer is *empty* below 10^{112} (Corollary 12.3). The twisted and rational problems thus differ not in the difficulty of the search but in whether solutions exist to be found at all: for the twist they are plentiful and the only question is their primality, exactly the Schinzel–shaped residue above. That residue is a *pair*, not a triple or a Lehmer sequence: fixing π_1 and bounding the units, the cycle count still grows with $N(\pi_2)$ (verified: $33, 48, 60$ for $\pi_1 = 3+\sqrt{2}$ at $N(\pi_2) \le 60, 120, 240$), so π_1's primality is free and only the *simultaneous* primality of π_2 and the forced π_3 remains. Setting up the sieve pins the difficulty precisely. Writing $C = u_1\pi_2 + u_2\pi_1$, $E' = (\pi_1\pi_2)^\sigma$, $F = -u_3\pi_1\pi_2$, the forced third prime is $\pi_3 = (\bar{C}F + E'\bar{F})/\det$ with $\det = N(C) - N(E')$ a binary quadratic form in $(c, d) = \pi_2$. The first condition is the clean form $N(\pi_2) = c^2 - 2d^2$; the second, however, is *not* a second form but the rational condition $N(\pi_3) = N(G)/\det^2$ ($G = \bar{C}F + E'\bar{F}$ quadratic, $N(G)$ quartic, $\gcd(N(G), \det) = 1$), evaluated on the integral locus $\det \mid G$, and $N(\pi_3)$ is unbounded, spiking as $\det \to 0$ (values to $\sim 10^{11}$ already at $N(\pi_2) \le 2 \times 10^4$). On this *frozen* (π_1, units) slice the question is primality along the folded conic bundle, and the slice is thin: its unimodular fibres $\det = \pm 1$ are *empty* by a genus obstruction (there $\det = \pm 1$ would need $x^2 - 2y^2 \in \{-55, -43\}$, both non–residues). But the thinness is an artefact of freezing exactly the parameters that carry the abundance. The *full* twisted variety $D(a) = a^\sigma$ (π_1, π_2, π_3 and their units all free) is a cubic threefold whose solution count grows with positive exponent; a positive–density attack must run there, not on a fixed fibre, so the fixed–fibre conic sieve is a closed dead end. The honest placement is then: the reachable object is the full threefold at the Heath–Brown $x^3 + 2y^3$ frontier, compounded by a *coupled* second primality ($N(\pi_2)$ and $N(\pi_3)$ simultaneously prime); the two–sparse–conditions/parity barrier where the circle method loses minor–arc cancellation. This is strictly below the exponential/Lehmer tier of the immune families (Remark 4.16, where no sieve reaches at all), but a generation beyond present technology; it is where genuinely new analytic input for the twist would first have to bite.

The natural attempt to *prove* that infinitude (forcing one prime to be the conjugate of another so that a single free prime remains, whence Dirichlet would suffice) is blocked, and the block is a clean structural law.

Proposition 6.10 (No conjugate prime pair in a twisted cycle)**.** *Let D be a Leibniz derivative with unit prime–values over a quadratic ring with automorphism σ, and a a squarefree twisted cycle $D(a) = a^\sigma$. Then no two prime factors of a are σ–conjugate. Equivalently a twisted cycle's support is disjoint from its conjugate; it is never conjugation–stable, which is exactly why the exhibited $\mathbb{Z}[i]$ cycles are asymmetric, coming in conjugate pairs of* distinct *supports rather than on one symmetric support. This* separates *two problems easily conflated: the twisted equality $D(a) = a^\sigma$ treated here, and the symmetric target of Remark 6.5, which is a genuine two–cycle $D(m) = m'$ ($m \neq m'$, both conjugation–fixed) of the* descended *rational operator; not a twisted equality. The present proposition bars the first from the symmetric locus; it does not bear on the second, which remains genuinely open (the exhaustive emptiness there, Remark 6.5, is unexplained by it).*

Proof. Suppose a split prime π and its distinct conjugate π^σ both divide a. Reducing $D(a) = \sum_i u_i\,(a/\pi_i)$ modulo π kills every term but the one at $\pi_i = \pi$, so $D(a) \equiv u_\pi\,(a/\pi) \not\equiv 0 \pmod{\pi}$ (u_π a unit and $\pi \nmid a/\pi$ by squarefreeness). Yet $a^\sigma = \prod_i \pi_i^\sigma$ carries the factor $(\pi^\sigma)^\sigma = \pi$, so $a^\sigma \equiv 0 \pmod{\pi}$; hence $D(a) \neq a^\sigma$. An exhaustive $\mathbb{Z}[\sqrt{2}]$ search confirms no cycle carries a conjugate pair. $\square$

We codify what is theorem and what remains open in this circle.

Theorem 6.11 (Existence of twisted derivative two–cycles)**.** *Leibniz derivatives with unit prime–values admit genuine two–cycles over $\mathbb{Z}[i]$, $\mathbb{Z}[\sqrt{-2}]$, $\mathbb{Z}[\omega]$, and $\mathbb{Z}[\sqrt{2}]$. The proof is by exhibited certificate, each verified exactly by independent recomputation: the smallest instances are, in order of norm, $\big(\sqrt{2}(5-\sqrt{2}), 7\big)$ over $\mathbb{Z}[\sqrt{2}]$ at norms* (46, 49)*; the Eisenstein cycle at norm* 2,964*; the sign–twisted cycle over $\mathbb{Z}[\sqrt{-2}]$ at norm* 11,462*, and the cycle over $\mathbb{Z}[i]$ at norm* 16,642*; together with the rational pair $D(3707) = 1547$, $D(1547) = 3707$ over $\mathbb{Z}[\sqrt{2}]$.*

Proposition 6.12 (Geometry of the solution set)**.** *Over $\mathbb{Z}[\sqrt{2}]$, consider the Galois–folded system $u_1\pi_2\pi_3 + u_2\pi_1\pi_3 + u_3\pi_1\pi_2 = (\pi_1\pi_2\pi_3)^\sigma$ of Remark* 6.8. (i) *For* every *triple of split primes the system is solvable with the u_i on the real unit hyperbola $\{|st| = 1\}$: solving the second embedding for u_2 and letting the third coordinate tend to zero, the first–embedding residual sweeps $\pm\infty$ while the remaining terms stay bounded, so a root exists by continuity; the real solution set is a nonempty curve* (*verified numerically on* 200 *random triples, all solvable*)*. The volume of the solution variety never vanishes.* (ii) *By Proposition* 6.7 *each fixed support carries finitely many integral solutions, so infinitude can only come from growing supports.* (iii) *The measured integral hit rates are* 11 *of* 455 *prime triples in the smallest box, and* 30,204 *cycles of norm $\leq 2 \times 10^7$ at twist height ≤ 2.* (iv) *What remains open is precisely whether the discrete unit lattice meets the real curve in integer points for infinitely many supports, inhomogeneous Diophantine approximation on exponentially parametrised curves. Two shortcuts are closed: bounded–unit polynomial families are obstructed (the rational component of the return forces unit coefficients to track growing prime coordinates, which bounded units cannot), and even the cleanest forward family ($D\big(\sqrt{2}(x-\sqrt{2})\big) = x+2$* identically in *$x$, supplying a candidate cycle whenever x^2-2 and $x+2$ are prime) has sporadic returns: the closing two–term unit equation, effectively finite for each x, is solvable at $x = 5$ and provably insolvable at $x = 15$ and $x = 21$. Those failures are, however, exactly the rigid case: a* prime *return $x \pm 2$ leaves the closing system no slack. The identity has four unit branches $D\big(\sqrt{2}(x \pm \sqrt{2})\big) = x \pm 2$ (independent signs, twists* 1 *and $\pm\varepsilon^{\pm 1}$), and when $x \pm 2$ is composite squarefree the closing system gains one unit unknown per prime class against the same two equations, and does close: $x = 107, 121, 387, 1693$ yield cycles with rational partners $b = 105, 119, 385, 1695$, each verified exactly. Infinitude along this family is thereby reduced to Bunyakovsky for x^2-2 together with recurring composite–return closure: the softest form the gap takes anywhere in this paper. Even the primality half of that reduction can be removed. By the* *half–dimensional sieve [14, 15] (applied to* $(2y+1)^2-2=4y^2+4y-1$ *to keep* x *odd),* x^2-2 *is a* P_2 *infinitely often; every prime factor of* x^2-2 *has* 2 *as a quadratic residue and therefore splits in* $\mathbb{Z}[\sqrt{2}]$*, so* $\sqrt{2}(x-\sqrt{2})$ *is infinitely often squarefree and all–split with* $\omega\in\{2,3\}$*. The candidate supports thus exist* unconditionally*; all that remains open on this route is whether the unit gates open along that supply infinitely often. Exactness migrates; it does not disappear.*

Proposition 6.13 (No vanishing derivatives, no transport, and the price of exactness)**.** (i) *No squarefree* z *with* $\omega(z)\geq 1$ *has* $D(z)=0$*, for any twisted derivative over any of the rings considered (indeed over any UFD): modulo a prime* $\pi\mid z$ *every cofactor term but one vanishes, leaving* $D(z)\equiv u_\pi\, z/\pi\not\equiv 0 \pmod{\pi}$*. The zero–derivative counters of every census above are therefore theorems, not observations.* (ii) *Consequently cycles cannot be transported multiplicatively: for squarefree* c *coprime to* ab*, Leibniz gives* $D(ca)=D(c)\,a+c\,D(a)$*, so* (ca,cb) *is a cycle exactly when* $D(c)=0$*; impossible. Cycles do not breed; each one requires fresh exactness.* (iii) *The two cycle equations can be bought by construction, at a conserved price. Buying* neither *leaves the generic stratum: two exact unit conditions (*30,204 *specimens). Buying* one *is free in two dual ways; forward,* $\pi:=q-\varepsilon^m\sqrt{2}$ *makes* $D(\sqrt{2}\,\pi)=q$ *an identity (the four–branch family above is the case* $m=\pm1$*); return,* $\pi:=(\rho+\varepsilon^{-1}\rho^\sigma)/\sqrt{2}=(c+2d)-d\sqrt{2}$ *for* $\rho=c+d\sqrt{2}$ *closes* $D(b)=\sqrt{2}\,\pi$ *identically for every associate* b *of* $\rho\rho^\sigma$*, and the residue is one primality,* $|N(\pi)|$*, plus one exact unit gate, measured at* 2 *of* 50 *pairs with both primalities in the smallest box. The two hits re–derive, with no search, precisely the two census minimal–shape cycles with prime rational partner: norms* $(46,49)$ *and* $(206,49)$*. Buying* both *collapses the parameters to unit–indexed quadratics in* q *with a square–discriminant condition; the resulting* $q=7$ *line carries the known cycles at* $m=1$ *and* $m=-2$ *(discriminants* 25 *and* 81*) together with one Galois image, and every other prime–norm rung out to* $|m|\leq 14$ *fails its return, verified exactly. Each purchased identity is repaid as a sparser arithmetic condition elsewhere: the hardness of exactness is conserved, never destroyed.*

Remark 6.14 (The two faces of the gap: unit rank dichotomises the obstruction)**.** The invariant that governs abundance in Remark 6.6 (the unit rank $r=r_1+r_2-1$ of Dirichlet) also governs the *type* of the residual open problem, and the two types are mutually exclusive. When $r=0$ (the rings $\mathbb{Z}$, $\mathbb{Z}[i]$, $\mathbb{Z}[\sqrt{-2}]$, $\mathbb{Z}[\omega]$) the unit group is finite, so a given pair of supports admits only $|\mathcal{O}^\times|$ twist assignments and hence finitely many candidate cycles: the solution locus is a set of *isolated points*, zero–dimensional, and infinitude can arise only from infinitely many *distinct* supports independently passing a closing test; a Bateman–Horn/Schinzel–type recurrence over prime tuples, with no continuous family to deform along. The constituent primalities are individually classical (each support prime is a value of a binary norm–form, so by Fermat–Dirichlet (and, for $\mathbb{Z}[i]$, by the Green–Sawhney theorem with *both* coordinates prime) there is no shortage of supports); it is their *simultaneous* closure that is not classical. This was tested directly: the minimal shape $a=(1+i)\pi$ over $\mathbb{Z}[i]$ yields *no* cycle for any Gaussian prime π of norm ≤ 4000 under any of the sixteen twists, and the sixteen genuine cycles are irregular mixtures of inert and split classes ($\omega=3,4,5$) with no one–parameter family among them; the signature of isolated points, not a curve. When $r\geq 1$ (every real quadratic ring, e.g. $\mathbb{Z}[\sqrt{2}]$), Proposition 6.12 makes the real solution locus a positive–dimensional curve in each support, but the unit lattice is an infinite rank–r discrete set, and infinitude requires it to meet that curve in *integer* points infinitely often; inhomogeneous Diophantine approximation on an exponentially parametrised curve. No unit rank delivers both a classical gate and a continuous family: rank zero gives the former and denies the latter, positive rank the reverse. This is why no single ring lets the infinitude be *finished* by present methods; a structural account of the obstruction's universality, not a barrier theorem. It also locates $\mathbb{Z}$ exactly: the unique ring carrying *both* the rank–zero sporadicity *and*, by formal reality, the mass barrier; the worst cell of the trichotomy of Corollary 6.2, which is precisely why #307 is the hard representative of its family.

7 A Thābit–type rule and its honest limits

The bridge recasts the search for a solution as the search for a derivative two–cycle, and the derivative's Leibniz structure makes the cycle equations *linear* in the "closing" primes. This yields a constructive calculus analogous to Thābit ibn Qurra's rule for amicable pairs and to Euler's divisor method.

Theorem 7.1 (Frame Rule)**.** *Let M, N be coprime squarefree integers with derivatives M', N', and set $\Delta := MN - M'N'$. Suppose*

$$p = \frac{MN' + N^2}{\Delta}, \qquad q = \frac{M'N + M^2}{\Delta}$$

are integers, both prime, each coprime to MN, with $p \neq q$. Then $a := Mp$, $b := Nq$ form a derivative two–cycle $a' = b$, $b' = a$; equivalently (P, Q) with P the prime set of Mp and Q that of Nq solves Erdős #307.

Proof. With p prime and $p \nmid M$, $a = Mp$ is squarefree and $a' = M'p + M$; similarly $b' = N'q + N$. The cycle conditions $a' = b$, $b' = a$ read

$$M'p + M = Nq, \qquad N'q + N = Mp,$$

a linear system in (p, q) with determinant $-(MN - M'N') = -\Delta$. Solving gives exactly $p = (MN' + N^2)/\Delta$ and $q = (M'N + M^2)/\Delta$; substituting back verifies both equations identically. (Both identities were checked symbolically.) □

A two–primes–per–side variant ("Rule 2") is obtained identically: with $a = Mp_1p_2$, $b = Nq_1q_2$ and elementary symmetric functions e_1, e_2, f_1, f_2, the cycle equations are linear in the f_i,

$$f_2 = \frac{Me_1 + M'e_2}{N}, \qquad f_1 = \frac{MNe_2 - MN'e_1 - M'N'e_2}{N^2},$$

and a solution is realised whenever $f_1^2 - 4f_2$ is a perfect square with prime roots.

An Euler-style divisor trick. Fix N and let $M = M_0r$ with r a new prime, so $M' = M_0'r + M_0$. Put $\Delta_0 := M_0N - M_0'N'$.

Proposition 7.2 (Divisor trick)**.** *With the above notation and $p_{\text{num}} := MN' + N^2$:*

(a) $\Delta(r) := MN - M'N' = r\,\Delta_0 - M_0N'$ (linear in r)*;*

(b) $\Delta_0\, p_{\text{num}} - M_0N'\,\Delta(r) = K$ *where* $K := \Delta_0 N^2 + (M_0N')^2$ *is* constant in r*;*

(c) *consequently* $\Delta(r) \mid p_{\text{num}} \Rightarrow \Delta(r) \mid K$*, and the converse holds provided* $\gcd(\Delta(r), \Delta_0) = 1$ (*equivalently* $\gcd(M_0N', \Delta_0) = 1$*, since* $\Delta(r) \equiv -M_0N' \bmod \Delta_0$)*;*

(d) *each positive divisor $d \mid K$ with $d \equiv -M_0N' \pmod{\Delta_0}$ yields a candidate $r = (d + M_0N')/\Delta_0$ and recovers $\Delta(r) = d$; this enumeration is complete (it captures every genuine solution). The resulting $p = p_{\text{num}}/d$ is integral when* $\gcd(M_0N', \Delta_0) = 1$*, and otherwise must be checked.*

Proof. (a) and (b) are direct expansions (verified symbolically; $\partial K/\partial r = 0$). (c): from (b), $\Delta(r) \mid p_{\mathrm{num}} \Rightarrow \Delta(r) \mid \Delta_0 p_{\mathrm{num}} = M_0 N' \Delta(r) + K \Rightarrow \Delta(r) \mid K$. For the converse, $\Delta(r) \mid K$ gives $\Delta(r) \mid \Delta_0 p_{\mathrm{num}}$, whence $\Delta(r) \mid p_{\mathrm{num}}$ iff $\gcd(\Delta(r), \Delta_0) = 1$. (d) inverts (a). Excluded degeneracies: $\Delta_0 = 0$ (r undefined) and $K \leq 0$. □

Thus, exactly as in Euler's amicable–pair machinery, one factors the single number K once and reads off candidates from its divisors lying in a fixed residue class modulo Δ_0; only the primality of r, p, q then remains. Two caveats deserve emphasis: the converse in (c) and the automatic integrality of p require $\gcd(M_0 N', \Delta_0) = 1$ (it fails, e.g., for $M_0 = 2$, $N = 19 \cdot 47$, $r = 5$, where $\Delta(r) \mid K$ but $\Delta(r) \nmid p_{\mathrm{num}}$), and the count of admissible classes is governed by Δ_0, not merely by $\tau(K)$.

When the closing side carries a *single* new prime, the rule collapses to a normal form that separates the two layers of the problem.

Proposition 7.3 (One–prime normal form; the exactness stratum)**.** (a) *Erdős #307 has a solution if and only if there exist squarefree M, N with*

$$MN - M'N' = M^2,$$

such that $p := N'/M$ is a prime not dividing M. Indeed the equation forces $M \mid M'N'$, hence $M \mid N'$ by $\gcd(M, M') = 1$, and $N = M + M'p$; then $a = Mp$, $b = N$ satisfy $a' = M'p + M = N$ and $b' = N' = Mp = a$ (coprimality and $a \neq b$ are automatic). Conversely every solution (a, b) and every prime $p \mid a$ yield such a pair with $M = a/p$, so each solution carries $\omega(a)$ witnesses. The problem thus splits into an exactness *layer (the Diophantine equation) and a single* primality *layer (one prime value of the determined quantity N'/M).* (b) The exactness layer alone carries the full barrier. *Call (M, N) a* relaxed witness *if $MN - M'N' = M^2$ with $p = N'/M$ composite. The identity $\left(\sigma(M) + \frac{1}{p}\right)\sigma(N) = \frac{N}{Mp} \cdot \frac{Mp}{N} = 1$ holds for* every *witness, so by strict AM–GM ($\sigma(N) = 1$ being impossible) $\sigma(M) + \sigma(N) > 2 - \frac{1}{p}$. Rigidity forces three structural facts. The supports of M and N are* automatically *disjoint: a common prime q would divide $N' = Mp$ through $q \mid M$, against* $\gcd(N, N') = 1$*. Likewise* $\gcd(p, MN) = 1$*: a prime shared by p and M propagates to $N = M + M'p$ and lands in the previous case, while one shared by p and N divides N'. And if p is even, both members are odd: $2 \mid N' = Mp$ forces N odd by the same lemma, while M even would make $N = M + M'p$ even. The union of supports therefore avoids every prime divisor of p, and for even p avoids 2; exact greedy computation then gives, for* every *composite p,*

$$\omega(MN) \ \geq \ 59, \qquad MN \ > \ 8.77 \times 10^{112}$$

the cycle barrier itself; with strictly stronger floors at every finite p: $\omega \geq 194$, $MN > 10^{497}$ at $p = 9$; $\omega \geq 230$, $MN > 10^{609}$ at $p = 4$; the even–p floors rising towards the all–odd bound 10^{5040}, and the infimum 8.77×10^{112} approached only as $p \to \infty$ through composites free of prime factors ≤ 277. Deleting primality thus lowers the barrier by nothing*: the 59–prime, 10^{112} wall is carried entirely by the exactness equation, and the primality of N'/M is a separate demand on top of it. (No witness, prime or composite, exists with $M \leq 120$, $p \leq 2000$, exhaustively.)*

The decisive obstruction: a circular objective. The heuristic yield of the divisor trick from one frame is

$$\mathbb{E}[\text{cycles}] \ \approx \ A \cdot \frac{1}{\ln r \, \ln p \, \ln q}, \qquad A = \#\{d \mid K : \ d \equiv -M_0 N' \ (\mathrm{mod}\ \Delta_0)\} \ \approx \ \frac{\tau(K)}{\Delta_0},$$

the last step being an equidistribution heuristic. To make $\mathbb{E} \geq 1$ at the barrier scale ($1/\ln^3 \approx 5 \times 10^{-7}$) one needs $A \gtrsim 2 \times 10^6$ admissible divisors per frame, hence $\Delta_0 = O(1)$ with $\tau(K) \sim 10^6$. The natural objective is therefore "engineer frames with Δ_0 small." But here is the obstruction, an exact identity:

$$\boxed{\Delta_0 \;=\; M_0 N\big(1 - s_{M_0}\, s_N\big), \qquad s_{M_0} = \sum_{p \mid M_0} \frac{1}{p}, \quad s_N = \sum_{p \mid N} \frac{1}{p}.}$$

So Δ_0 is small *relative to* M_0N exactly when $s_{M_0}s_N \approx 1$ (which is the Erdős #307 condition for the frames themselves) and $\Delta_0 = O(1)$ in absolute terms at scale requires $s_{M_0}s_N = 1 - O(1/M_0N)$, i.e. a reciprocal–product within $O(1/M_0N)$ of 1, *strictly harder* than #307. The frame–engineering objective is thus circular: the Frame Rule is a faithful *parametrisation* of solutions, not a reduction in difficulty.

Remark 7.4 (Honest status of the constructive program)**.** The Frame Rule (Theorem 7.1) and divisor trick (Proposition 7.2) are exact theorems; they make the existence of a solution *falsifiable in principle by finite computation per frame*, and they place #307 in the same constructive lineage as Thābit's and Euler's rules. They do not, as they stand, lower the difficulty below #307 itself: by the boxed identity the bottleneck (small Δ_0) is the original problem in disguise, and a complete small–frame sweep (2.2×10^5 frames) produces zero cycles with total expected count < 1, as the barrier demands. We regard the calculus as an exact parametrisation of solutions whose central obstruction we have now identified precisely; not a pathway that lowers the difficulty, and not evidence that a solution is within reach.

8 A breeder form and the feasibility of certificate search

Allowing *two* free primes on one side turns the divisor trick into a cleaner bilinear form, Euler's amicable–pair method, in the shape used by te Riele's breeding algorithms [12], and lets us state precisely the strongest honest conclusion: a *conditional* reduction, which nonetheless does not become a feasible finite search.

Proposition 8.1 (Bilinear breeder form)**.** *Fix coprime squarefree M_0, N and seek a two–cycle $a = M_0rp$, $b = Nq$ with r, p, q distinct primes coprime to M_0N. Put $G := NM_0 - N'M_0'$ $(= \Delta_0)$, $c := M_0N'$, and $K := GN^2 + c^2$. Then the cycle equations $a' = b$, $b' = a$ are equivalent to*

$$(Gr - c)(Gp - c) = K, \qquad q = \frac{M_0rp - N}{N'}.$$

Proof. Expanding, $(Gr - c)(Gp - c) - K = -N'G\,(a' - b)$ (verified symbolically), so the bilinear equation is equivalent to $a' = b$; the second cycle equation $b' = a$ then determines q as stated. □

Each factorisation $K = d \cdot (K/d)$ with $d \equiv -c \pmod G$ proposes a candidate $(r, p) = \big((d + c)/G, (K/d + c)/G\big)$; a solution is any such candidate whose r, p, q are all prime. A dual, partition–side form is occasionally useful:

Proposition 8.2 (Subset form)**.** *For a support U with $D = \prod_{p \in U} p$ and $\Sigma = \sum_{p \in U} D/p$, a partition $U = P \sqcup Q$ solves #307 iff $S = P$ satisfies $A(\Sigma - A) = D^2$, where $A := \sum_{p \in S} D/p$. Moreover $A(\Sigma - A) - D^2 = -(1 - s_Ps_Q)\, D^2$.*

(Both identities verified symbolically and on 2×10^4 random supports.) The last identity says the "analytic" residual $1 - s_Ps_Q$ and the "arithmetic" residual $A(\Sigma - A) - D^2$ are one equation up to the factor D^2: the two obvious steering handles carry no independent information.

A conditional reduction. Several ingredients of a construction are unconditional. Greedy Egyptian–fraction choice gives disjoint P, Q with $0 < 1 - s_P s_Q = O(1/T)$ (solvability to any precision); a prime r nearest $M_0 N'/\Delta_0$ gives $G \le \Delta_0^{0.475}(M_0 N')^{0.525}$ unconditionally (Baker–Harman–Pintz [13]), and each small prime ℓ can be forced to divide K by one congruence on a free prime slot of N (Dirichlet then supplies the prime). Granting these, #307 reduces to a single prime–density statement,

(H) among the admissible candidates so produced, some triple (r, p, q) is prime,

and (H) $\Rightarrow$ #307 is YES. This (H) is a Hardy–Littlewood/Bateman–Horn–type assertion for an explicit, computable, *non–polynomial* family; we know of no named conjecture implying it. That is not an accident of our construction:

Proposition 8.3 (No polynomial families)**.** *Let $p_1(t), \ldots, p_k(t)$, $q_1(t), \ldots, q_l(t)$ be integer polynomials, each taking prime values for infinitely many integers t (a fixed–shape Schinzel–type family), such that $\left(\sum_i 1/p_i(t)\right)\left(\sum_j 1/q_j(t)\right) = 1$ for infinitely many t. Then every p_i and q_j is constant: the family is a single fixed solution. Consequently no polynomial parametrisation can reduce #307 to Bunyakovsky, Schinzel's Hypothesis H, or Bateman–Horn.*

Proof. Holding at infinitely many t, the relation holds identically as rational functions. A nonconstant p_i taking infinitely many prime (hence positive) values has positive leading coefficient on an infinite branch, so $p_i(t) \to +\infty$ there; split each side into constants and nonconstant members, $\sum_i 1/p_i = A_P + \varepsilon_P(t)$ with $\varepsilon_P(t) \to 0^+$. Letting $t \to \infty$ in the identity gives $A_P A_Q = 1$ (in particular both sides contain constants), and subtracting, $A_P \varepsilon_Q(t) + A_Q \varepsilon_P(t) + \varepsilon_P(t)\varepsilon_Q(t) = 0$ for all large t. Every term on the left is nonnegative, and strictly positive as soon as one nonconstant member exists; hence there are none. $\square$

This upgrades the earlier computational observation (that the breeder family is not polynomially parametrised) to a theorem for *all* fixed–shape families: the prime–density hypothesis behind #307 is irreducibly non–polynomial, which is why no standard conjecture applies.

Why (H) is not a finite search. It is tempting to read (H) as "run a certificate hunt that is expected to succeed." It is not, for a quantitative reason. The expected number of certificates from one frame is $\mathbb{E} \approx \left(\tau(K)/G\right) \cdot (\ln r \, \ln p \, \ln q)^{-1}$, and *both* factors are pinned by the geometry.

- *Demand.* $G = \Delta_0 = M_0 N(1 - s_{M_0} s_N)$ exactly, so G is small only in the circular regime $s_{M_0} s_N \approx 1$; over 2×10^4 random coprime squarefree frames the ratio $G/(M_0 N)$ has median 1 and best value $\approx 10^{-0.23}$, and the honest Baker–Harman–Pintz reduction still leaves $\log_{10} G \approx 13$ at the barrier.

- *Supply.* $\tau(K)$ is bounded by the *size* of $K = GN^2 + c^2$ ($\ge c^2$), which is an *output* of the frame, not a free dial. The maximal order of the divisor function [16, Thm. 317] caps $\log_2 \tau(K)$ at ≈ 36 for $K \le 10^{54}$ (the barrier geometry) and at ≈ 112 even for $K \le 10^{240}$; reaching $\tau(K) = 2^{120}$ needs $K \gtrsim 10^{263}$, i.e. frame scale far beyond the barrier.

Enlarging a frame grows $K \propto N^2$ but gains divisors only at rate $\log 2/\log\log K < 1$ per log–unit of K, while it grows G at rate 1 per log–unit of N; so $\tau(K)/G$ cannot be driven up. Optimising $\mathbb{E}$ over all (M_0, N) splits with geometry–consistent K and the maximal–order τ, and granting the most favourable demand and singular–series constants, yields $\log_{10} \mathbb{E} \le -4$ at the barrier and *decreasing* with scale; empirically 385 factorable frames realise a median of 0 and a maximum of 1 admissible divisor, against the $\sim 10^5$ a single productive frame would need. A self–consistent accounting; capping the number w of forced prime factors of K by the frame's own residue freedom (counted generously), sharpens this to $\log_{10} \mathbb{E} \leq -8$ at $B = 56$, falling to ≈ -260 at $B = 3000$: the deficit is *scale–invariant*, so no choice of scale, slot count, or budget split reaches expectation 1. (The construction is even self–similar: engineering K's algebraic; e.g. Gaussian–integer; factorisation to supply divisors reduces to prescribing N and N' simultaneously, an arithmetic–antiderivative inverse problem of the same difficulty class as #307.) The breeder thus makes #307 conditional on (H) but does *not* render it a feasible finite computation: the supply–demand conflict through $K = GN^2 + c^2$ is exactly the circularity of Section 7 in quantitative form. There is also a structural reason the analogy with amicable pairs was bound to fall short: te Riele's breeding amplifies a *known seed* pair into new ones, whereas the barrier (Theorem 4.3) forbids any seed below 2.09×10^{56} the method imports a precondition the problem cannot supply.

9 Computations

All verdicts use exact integer/rational arithmetic; floating point is used only for pre–screening.

- *Exhaustive non–existence below* 10^8*.* Enumerating all prime sets P with $\sum 1/p > 1$ and $\prod P \leq 10^8$ (forcing Q as the prime support of N_P and testing $N_Q = D_P$): 1,075,419 candidate sets, no solution. Subsumed by Theorem 4.3 but an independent sanity check.

- *Derivative sieve.* No solution of $n'' = n$ with $n' \neq n$ and both members $\leq 2 \times 10^7$ (smallest–prime–factor sieve of the derivative).

- *Function field.* The arithmetic derivative on $\mathbb{F}_q[t]$ has no two–cycles and no nonzero fixed points, verified exhaustively over $\mathbb{F}_2$ (degree ≤ 7), $\mathbb{F}_3$ (≤ 5), $\mathbb{F}_5$ (≤ 4), confirming Theorem 6.1.

- *Density.* The sieve gives density 0.042029 to 2×10^7 (0.017623 among squarefree); the independent Erdős–Wintner convolution over primes $\leq 10^5$ gives $\delta = 0.04202$, agreeing to four places (Proposition 5.6).

- *Conditional mass.* Of 43,087 qualifying squarefree a sampled, $b = a'$ is squarefree 95.0% of the time; on the resulting 40,920 squarefree b the mean reciprocal mass is 0.047 vs. the required $1/s \approx 0.934$, with empirical tail probability $\Pr[\sum_{q|b} 1/q \geq 1/s] = 0/40{,}920$.

- *Frame/ledger sweep.* Over 2.2×10^5 factorable small frames: the divisor–trick identities hold on every instance; the smallest Δ_0 observed is 22; the equidistribution $A \approx \tau(K)/\Delta_0$ holds only in aggregate ($\sum A = 127$ vs. $\sum \tau(K)/\Delta_0 = 39.8$) and fails per frame ($A = 0$ for 99.95% of frames); total expected cycles < 1; zero cycles, as the barrier requires.

- *μ–Sondow line census.* Enumerating the lines $n' = n \pm d$ (squarefree members coprime to d) finds no adjacent opposite–sign pair at additive distance d: exhaustively to 2×10^7 for $|d| \leq 200$, and along the small–$|\delta|$ genealogy tree out to members of more than a thousand digits. Every $\delta = \pm 1$ member encountered is a known Giuga or primary pseudoperfect number; the closest opposite–sign approach recorded is $|y - (x + d)| = 11$ (at $d = 1$, 30 vs. 42, and $d = 37$, 210 vs. 258), attained only among small members; the two trees do not come close at height.

Code and logs accompany this note.

10 Formalisation in Lean 4

The rigidity and barrier results are formalised against mathlib (Lean 4, toolchain `v4.30.0`). The following are machine–checked and `sorry`–free, depending only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`:

- Lemma 2.1 (`rigidity_coprime`) and Theorem 2.3 (`solution_structure`);
- Lemma 4.1 (`reciprocal_sum_gt_two`); the exact thresholds $\sum_{\text{first } 58} 1/p < 2 < \sum_{\text{first } 59} 1/p$ grounding $|U| \geq 59$ (`recipSum58_lt_two`, `recipSum59_gt_two`); the numeric core $\Pi_{59}^2 \geq (4 \times 10^{112})N_{59}$ (`barrier_numeric`), and the algebraic core of the barrier (`barrier`), giving $D_P^2 \geq 4 \times 10^{112}$.

The single non–elementary ingredient, that the k smallest primes minimise $\prod$ and maximise $\sum 1/p$ among k–element prime sets, is isolated as an explicit hypothesis of `barrier` (the ratio bound $R/T(U) \geq \Pi_{59}/T_{59}$), so the dependency is transparent rather than hidden; its standalone proof via `Nat.nth Nat.Prime` is routine and left as future work.

11 Status and questions

Erdős #307 remains **open**. We have shown it is equivalent to the existence of a non–trivial two–cycle of the arithmetic derivative; established rigidity and a barrier (Rosen's $|P \cup Q| \geq 59$, together with our $\min(\prod P, \prod Q) \geq 2.09 \times 10^{56}$) that voids all direct search and improves the $n'' = n$ record by ~ 47 orders of magnitude; given a parity dichotomy, a proof that longer cycles are strictly harder, an Erdős–Wintner density theorem, and a function–field theorem localising the obstruction to the archimedean place; developed an exact Thābit/Euler–type constructive calculus and a bilinear breeder form whose central obstruction we identify precisely, and extracted the maximal honest consequence of the construction; a conditional reduction (H) $\Rightarrow$ YES to an explicit prime–density hypothesis, together with a quantitative ledger showing that (H) is *not* a feasible finite search (the supply–demand conflict through $K = GN^2 + c^2$ keeps the expected certificate count per reachable frame below 1). The rigidity result and the barrier are fully Lean–verified (extremality proved, not assumed), and the finite verification of Proposition 4.5 sharpens the barrier to $|P \cup Q| \geq 60$, $\min(\prod P, \prod Q) > 3.50 \times 10^{57}$. The heuristics weakly favour YES (existence), hence weakly against the period–one Ufnarovski–Åhlander conjecture, but we claim no proof in either direction.

Open problems: (i) close the residual 3,478 single–tail families at level 60, and close the pair sector of Proposition 4.9; for which Proposition 4.11 rules out every congruence kill, so only global, per–family input (factorisation or direct search) can serve (the canonical tail family is closed, Proposition 4.8; the finite verification of Proposition 4.5 is Lean–checked, `erdos307_sixty`; the conditional 3/4 law is explained; it is the reciprocity switch of Remark 4.12, made exact by the class–vector functional there); (ii) decide the parity dichotomy's all–odd case (we expect it empty); (iii) sharpen the certified enclosure $0.04202 \leq \delta \leq 0.04223$ of Proposition 5.6, whose width is carried almost entirely by the tail window a; three digits are certified, and a longer envelope convolution with a smaller window certifies more; (iv) settle the density hypothesis (H) of Section 8, or exhibit a non–circular frame family escaping the $\tau(K)/G$ ceiling; (v) find a $\mathbb{Z}$–side structure detecting the archimedean obstruction of Section 6; one that must be non–local, since by Proposition 5.2 no single modulus obstructs; (vi) settle #307.

Remark 11.1 (The place of the problem: a local–global dictionary)**.** The pieces above assemble into a dictionary. (i) For a fixed support U, the subset form of Proposition 8.2, $A(\Sigma - A) = D^2$, satisfies the *Hasse principle as an equation*: an integer solution A exists iff one exists over $\mathbb{R}$ and over every $\mathbb{Z}_\ell$ (the discriminant $\Sigma^2 - 4D^2$ must be a nonnegative square, and squares obey local–global). Its real–place condition is $\Sigma \geq 2D$, i.e. $\sigma(U) \geq 2$: *the barrier is precisely the real–place solvability condition of the subset form.* (ii) The barrier never needed primality: pairwise coprime integers have distinct smallest prime factors, so $\sum 1/a_i \leq \sum 1/p_i$, and *any* pairwise–coprime support of mass ≥ 2 has at least 59 elements and product at least Π_{59}. Coprimality and the ordering of $\mathbb{R}$ carry the wall; primality enters only through rigidity. (iii) What remains unsolved is the *realization* of the root $A^* = \left(\Sigma + \sqrt{\Sigma^2 - 4D^2}\right)/2$ as a sub–multiset sum of the cofactors D/p. Its local conditions are exactly the local anatomy of Lemma 5.1: modulo ℓ^2 for $\ell \in U$ the subset sums occupy precisely the two cosets $\ell\mathbb{Z} \cup (D/\ell + \ell\mathbb{Z})$ (2ℓ residues, observed exactly (6 of 9, 10 of 25, 14 of 49, 22 of 121)) and these are the same constraints the equation itself forces: no completion separates the equation from its realization. In the language of integral points, then: #307 is everywhere locally soluble, with the barrier as its archimedean height condition, and its resolution is a *strong approximation* question for the thin, prime–structured set of subset sums a failure mode invisible to Brauer–Manin–type obstructions, consistent with every no–go above and with the existence of cycles over the neighbouring rings. We record this as a dictionary, not a method.

Attribution. We are careful to separate prior, concurrent, and present contributions (details in Section 12). *Prior/concurrent:* the two–cycle reformulation is due to Seva [19] (2019) and, independently, Bado [20]; the bound $|P \cup Q| \geq 59$ to J. Rosen and the disjointness valuation $v_p(\sum 1/p_i) = -1$ to R. Israel [9]; the source identification to "Woett" [9]. Bado [20] independently obtained the rigidity cross–matching, the local anatomy, the parity restriction, a one–sided exact test, approximate solvability, and the union criterion of Proposition 12.1. *Present note:* the identification of the map with the arithmetic derivative and the two–way bridge to the Ufnarovski–Åhlander–Kovič literature; the product barrier 2.09×10^{56}; the function–field theorem and archimedean localisation; the constructive deficit analysis and seed obstruction; the Erdős–Wintner density theorem, and the k–cycle barriers. Both external sources [19, 20] have been examined directly and the overlaps above reflect their actual contents.

Acknowledgments. This work was carried out by the author with substantial assistance from Anthropic's Claude. The AI assistant contributed to the derivations, the exact and heuristic computations, the Lean 4 formalisation, and the drafting of this note; it was also used to stress-test and attempt to refute each claim. All statements, proofs, attributions, and the final text were reviewed and are the responsibility of the author. The author thanks the contributors to the MathOverflow thread [9]; in particular R. Israel and J. Rosen, whose observations are credited above.

12 Note added: prior and concurrent work

After this note was drafted, two items came to light; having now examined both directly, we record the overlaps precisely.

Seva (MathOverflow 323194, 2019) [19]. Defines $s(Q) = Q \sum_{p|Q} 1/p$, derives the valuation property $v_p(s(Q)) = v_p(Q) - 1$, asks for $s(P) = Q$, $s(Q) = P$, and states this is Problem #307 "in disguise." Priority for the two–cycle reformulation belongs here. He further asks whether every s–trajectory stabilises at 0; the s–analogue of the Ufnarovski–Åhlander conjecture, posed independently of it. (Note $s \neq n'$ globally, e.g. $s(p^\alpha) = p^{\alpha-1}$, but $s = n'$ on squarefree integers, where both two–cycle questions live.)

Bado (preprint, Aix–Marseille, 2026) [20]. Working with $A(S) = \sum_{p\in S} M(S)/p$ and $M(S) = \prod_{p\in S} p$ (our N_S, D_S), Bado independently obtains: the lowest–terms lemma (his Lemma 3.1, our Lemma 2.1); the forcing identities $A(P) = M(Q)$, $A(Q) = M(P)$ with $P \cap Q = \varnothing$ (his Thm. A, our Theorem 2.3); the two–cycle reformulation $T(T(P)) = P$ (his Thm. B); the valuation signature (his Thm. 6.1, = Israel's); the local zero–sum congruences (his Prop. 8.1, our Lemma 5.1); the parity restrictions (his Prop. 8.2, the parity part of Theorem 4.17); an exact one–sided test (his Thm. 9.1); the approximation theorem (his Thm. 10.2, sharper than our greedy version in that all primes exceed a given bound), and the union discriminant / Pythagorean structure (his Thms. 7.1, 7.4). He cites only Barbeau, Bloom, and Erdős–Graham: crucially, he does *not* identify the map with the arithmetic derivative, and his note contains none of the present note's $n'' = n$ bridge, explicit barrier constant, function–field theorem, density theorem, k–cycle bounds, or constructive deficit analysis. We regard the overlap on the shared structural layer as strong independent validation, and import his union criterion with credit:

Proposition 12.1 (Single–integer test; derivative form of Bado's Thms. 7.1, 7.4)**.** *If (a, b) is a two–cycle of the arithmetic derivative, then $N = ab$ is squarefree and satisfies $N' = a^2 + b^2$, whence $N'^2 - 4N^2 = (a^2 - b^2)^2$ is a perfect square. Conversely, a squarefree N with $N'^2 - 4N^2 = \square$ determines a unique unordered factorisation $N = ab$ with $N' = a^2 + b^2$, on which the cycle condition reduces to a single equation (Bado's deviation parameter $t = 0$, his Thm. 7.6). This gives a one–integer necessary test; computationally, no squarefree $N < 2 \times 10^5$ with $\omega(N) \geq 2$ passes it (*0 *of* 103,596*, consistent with Theorem 4.3, which forces $N > 4 \times 10^{112}$).*

Proof. $N = ab$ with a, b coprime squarefree gives N squarefree and, by Leibniz with $a' = b$, $b' = a$, $N' = a'b + ab' = b^2 + a^2$; then $N'^2 - 4N^2 = (a^2 + b^2)^2 - 4(ab)^2 = (a^2 - b^2)^2$. The converse is Bado's [20]; the emptiness of the test below 10^{112} is upgraded from a sieve to a theorem in Corollary 12.3. $\square$

The Pythagorean equation $N' = a^2 + b^2$ deserves to be isolated from the (harder) cycle condition.

Proposition 12.2 (Pair form; the one–equation relaxation)**.** *For coprime squarefree a, b, the single equation $a^2 + b^2 = (ab)'$ holds if and only if there is an integer k with $a' = b + ak$ and $b' = a - bk$; a two–cycle is exactly the case $k = 0$. We call a solution of $a^2 + b^2 = (ab)'$ (any k) a* Pythagorean pair*.*

Proof. The Leibniz identity $a^2 + b^2 - (ab)' = a(a - b') - b(a' - b)$ (a polynomial identity, verified symbolically and on 2000 random pairs) makes the equation equivalent to $a(a - b') = b(a' - b)$; since $\gcd(a, b) = 1$ this forces $a \mid a' - b$ and $b \mid a - b'$ with equal quotients k, i.e. $a' = b + ak$, $b' = a - bk$. Setting $k = 0$ recovers $a' = b$, $b' = a$. (Equivalently, in $\mathbb{Z}[i]$: #307 asks for $z = a + bi$ with coprime coordinates and ab squarefree satisfying $|z|^2 = (\operatorname{Im} z^2/2)'$ and $k = 0$.) $\square$

Corollary 12.3 (The single–integer test is provably empty below 10^{112})**.** *Every Pythagorean pair satisfies $\sum_{p|a} 1/p + \sum_{q|b} 1/q = a/b + b/a \geq 2$ (divide $a^2 + b^2 = (ab)'$ by ab and apply AM–GM), so its support carries at least* 59 *primes and $N = ab \geq \prod$(first* 59 *primes) $> 7.9 \times 10^{112}$. Hence no squarefree $N < 10^{112}$ with $\omega(N) \geq 2$ satisfies $N'^2 - 4N^2 = \square$: the empirical "*0 *of* 103,596*" of Proposition 12.1 is a theorem, and the barrier of Theorem 4.3 extends from two–cycles to the strictly weaker one–equation (Pythagorean) problem.*

Remark 12.4. (a) As $N' = a^2 + b^2$ is a sum of two coprime squares, every odd prime dividing N' is $\equiv 1 \pmod 4$; a cheap union–side pre–filter complementing Bado's congruence filters. (When exactly one of a, b is even this is the hypotenuse of the primitive triple $(|a^2 - b^2|, 2ab, a^2 + b^2)$.) (b) Since Seva's s agrees with n' on squarefree integers, and his valuation property forces two–cycle members squarefree, his question is *exactly* #307, not merely an analogue. (c) Primitive Pythagorean triples form the Barning–Hall ternary tree (the orbit structure of a reflection group in $O(2, 1; \mathbb{Z})$), and a #307 solution's triple $(|a^2 - b^2|, 2ab, a^2 + b^2)$ occupies one node with generator $(a, b) = (\prod P, \prod Q)$. The tree nonetheless yields no descent on the problem: its moves $(a, b) \mapsto (2a \mp b,\, a)$, $(a + 2b,\, b)$ are linear, preserving coprimality but not squarefreeness (over random squarefree coprime generators only 37% of children keep both members squarefree) so the locus of #307–eligible nodes (squarefree generators of disjoint prime support meeting $a^2 + b^2 = (ab)'$) is not tree–invariant. The additive tree is transverse to the multiplicative constraint; the arithmetically natural recursion is instead the prime–append δ–tree of Proposition 14.41, $\delta(Mp) = p\,\delta(M) + M$, which preserves squarefreeness and tracks the line parameter, which is why the classical members organise on *that* tree, not Barning–Hall.

The deviation parameter of Proposition 12.2 organises the one–equation problem into a short graded family.

Proposition 12.5 (The deviation ladder)**.** *Let (a, b) be a Pythagorean pair with $a < b$ and parameter k. Writing $\sigma(m) = \sum_{p|m} 1/p$, one has $k = \sigma(a) - b/a = a/b - \sigma(b) \in \mathbb{Z}$, and:*

(1) *$k \le 0$, with $k = 0$ exactly the (open) two–cycle case. Indeed $b' = a - bk > 0$ and $b > a \ge 1$ force $k \le 0$ (if $k \ge 1$ then $b' \le a - b < 0$). Moreover $|k| < \min(b/a, \sigma(b)) \le \sigma(b) \le \log\log b + O(1)$, so the ladder has finitely many rungs; and, since $\sigma(b)$ barely exceeds 2 at the relevant scales, in practice only one or two.*

(2) (Parity.) *For odd squarefree m, $m' \equiv \omega(m) \pmod 2$; for even squarefree m, m' is odd. Hence if exactly one member is even, $k \equiv \omega(\text{odd member}) \pmod 2$; if both are odd, $\omega(a) \equiv \omega(b) \pmod 2$ and $k \equiv \omega(a) + 1 \pmod 2$. The case $k = 0$ recovers the parity dichotomy of Theorem 4.17.*

(3) *Each rung is the system $a' = b + ak$, $b' = a - bk$, equivalently $(\sigma(a) - k)(\sigma(b) + k) = 1$ (a shifted copy of the cross–match $st = 1$) so every rung is exactly as rigid as $k = 0$. Since $\sigma(a) + \sigma(b) = a/b + b/a$ is independent of k, Corollary 12.3 holds on* every *rung: the whole ladder is empty below 10^{112}. (We make no quantitative claim that $k = 0$ is easier than the rest.)*

(4) *On the rung $k = -1$ the equations read $a' = b - a$, $b' = a + b$ and $a'^2 + b'^2 = 2(a^2 + b^2)$: the derivative acts on $z = a + bi$ as the orientation–reversing $\sqrt{2}$–similarity $z \mapsto (-1 + i)\,\bar{z}$.*

(5) *By uniqueness of the splitting (Bado, Thm. 7.5), each squarefree N with $N'^2 - 4N^2 = \square$ carries, under the convention $a < b$, a well–defined invariant $k(N) \in \mathbb{Z}_{\le 0}$; #307 holds iff $k(N) = 0$ for some N, and by (3) the invariant lives on a domain that is empty below 10^{112}.*

The whole ladder is one equation in $\mathbb{Z}[i]$, and the rung index is a Gaussian multiplier.

Proposition 12.6 (Gaussian multiplier form of the ladder)**.** *For coprime squarefree a, b put $z = a + bi \in \mathbb{Z}[i]$ and let $\mathcal{D}z := a' + b'i$ be the coordinatewise arithmetic derivative. Then, unconditionally,*

$$z \cdot \mathcal{D}z \;=\; (aa' - bb') \;+\; i\,(ab)'.$$

Consequently (a,b) is a Pythagorean pair if and only if $\mathcal{D}z = \mu\bar{z}$ for a Gaussian integer μ with $\operatorname{Im}\mu = 1$; that multiplier is $\mu = k + i$ with k the deviation of Proposition 12.5, and

$$k = \frac{aa' - bb'}{a^2 + b^2}, \qquad a'^2 + b'^2 = (k^2+1)(a^2+b^2) = N(\mu)\,(a^2+b^2), \qquad \gcd(a', b') \mid k^2 + 1.$$

Hence #307 asks for a Pythagorean pair whose multiplier is a unit. *Since $\{k+i : k \in \mathbb{Z}\} \cap \mathbb{Z}[i]^\times = \{i\}$, the two–cycle rung $k = 0$ is the unique rung on which $\mathcal{D}$ acts as an isometry ($a'^2 + b'^2 = a^2 + b^2$); every other rung is an expanding anti–similarity of ratio $\sqrt{k^2+1} > 1$, and so cannot be periodic. The condition $\operatorname{Im}\mu = 1$ is the Pythagorean equation and $\operatorname{Re}\mu = 0$ is #307: one coordinate of a single Gaussian integer each.*

Proof. Leibniz gives $\operatorname{Im}(z\,\mathcal{D}z) = a'b + ab' = (ab)'$ and $\operatorname{Re}(z\,\mathcal{D}z) = aa' - bb'$, which is the identity (verified symbolically and on 7,501 coprime squarefree pairs). If (a,b) is Pythagorean then $(ab)' = a^2+b^2 = N(z)$, so $z\,\mathcal{D}z = (aa'-bb') + i\,N(z)$; dividing by $N(z) = z\bar{z}$ gives $\mathcal{D}z/\bar{z} = (aa'-bb')/N(z) + i$, which lies in $\mathbb{Z}[i]$ with real part the integer k of Proposition 12.2. Conversely $\mathcal{D}z = \mu\bar{z}$ with $\operatorname{Im}\mu = 1$ forces $(ab)' = \operatorname{Im}(z\,\mathcal{D}z) = \operatorname{Im}(\mu)N(z) = a^2+b^2$. Substituting $a' = b + ak$, $b' = a - bk$ gives $\mathcal{D}z = (k+i)\bar{z}$ identically, whence the norm identity $|\mathcal{D}z|^2 = |\mu|^2|z|^2$. Finally $g = \gcd(a', b')$ divides both $k(b+ak) + (a-bk) = a(k^2+1)$ and $(b+ak) - k(a-bk) = b(k^2+1)$, and $\gcd(a,b) = 1$. $\square$

The ladder of Proposition 12.5 has a twin, visible only after extending the derivative to $\mathbb{Q}$ by the quotient rule $(a/b)' = (a'b - ab')/b^2$ [7].

Proposition 12.7 (The co–ladder and the arithmetic Riccati equation)**.** *For coprime squarefree $a \neq b$ the following are equivalent:* (R) *$u = a/b$ satisfies the arithmetic Riccati equation*

$$u' \;=\; 1 - u^2;$$

(W) *$a'b - ab' = b^2 - a^2$;* (L) *there is $k \in \mathbb{Z}$ with $a' = b + ak$* and *$b' = a + bk$; the ladder of Proposition 12.2 with the sign of one equation flipped. The deviation cancels in the quotient: on* (L)*, $a'b - ab' = (b+ak)b - a(a+bk) = b^2 - a^2$ identically, so the Riccati equation is the k–free form of this* second *ladder. On it:*

- (i) (Positivity barrier.) *$k \geq 0$ always: with $a < b$, $k \leq -1$ would give $b' = a + bk \leq a - b < 0$. Hence $\sigma(a) + \sigma(b) = a/b + b/a + 2k > 2$, so every co–ladder pair has $\omega(ab) \geq 59$ and $ab > 7.9 \times 10^{112}$: the co–ladder is empty at every reachable scale* by positivity alone*; a second transversal relaxation of the two–cycle carrying the full barrier through a mechanism independent of the minus square.*
- (ii) (The cycle is the intersection.) *A two–cycle is exactly a common point of the two ladders at $k_{\text{sym}} = k_{\text{anti}} = 0$: #307 asks for $u \in \mathbb{Q}_{>0} \setminus \{1\}$ satisfying the Riccati equation together with the Pythagorean equation of Proposition 12.2. In multiplier form the two ladders read $\mathcal{D}z = kz + i\bar{z}$ (symmetric; norm $(k^2+1)N(z) + 4kab$) and $\mathcal{D}z = (k+i)\bar{z}$ (antisymmetric; norm $(k^2+1)N(z)$), meeting at $\mathcal{D}z = i\bar{z}$.*
- (iii) (Inversion symmetry.) $(1/u)' = 1 - (1/u)^2$ *is automatic from* (R)*; the Riccati layer is symmetric in $a \leftrightarrow b$, as* (L) *is.*

Proof. (W)$\Leftrightarrow$(L): the equation rearranges to $b(a' - b) = a(b' - a)$, and $\gcd(a,b) = 1$ forces $a \mid a' - b$, $b \mid b' - a$ with equal quotients; the converse is the displayed cancellation. (R)$\Leftrightarrow$(W) is the quotient rule. In (i), summing $\sigma(a) = b/a + k$ and $\sigma(b) = a/b + k$ gives the mass bound (strict, as $a \neq b$), and mass > 2 forces ≥ 59 primes as in Theorem 4.3. In (iii), $(1/u)' = -u'/u^2 = (u^2-1)/u^2$. All identities were verified symbolically; exhaustively, no co–ladder pair with $ab \le 10^7$ exists (3,918,188 candidate pairs through the $a \mid a' - b$ prefilter; the barrier's prediction), and a first small–box probe over $\mathbb{Z}[i]$ (canonical $\pi' = 1$) also finds none, though there the positivity mechanism is unavailable and the layer's status is genuinely open (`code/riccati_coladder.py`). $\square$

The two ladders embed in a plane, and the barrier becomes a half–plane law.

Proposition 12.8 (The deviation half–plane)**.** *For coprime squarefree $a \neq b$ call the pair* integral *if both deviations $s = (a' - b)/a$ and $t = (b' - a)/b$ are integers, and place it at $(s, t) \in \mathbb{Z}^2$; the antisymmetric ladder is the line $t = -s$, the co–ladder is $t = s$, and a two–cycle is the origin. Then*

$$\sigma(ab) \;=\; \frac{a}{b} + \frac{b}{a} + s + t \;>\; 2 + (s + t),$$

so every *stratum in the closed half–plane $s + t \ge 0$ carries the full barrier: $\omega(ab) \ge 59$ and $ab > 7.9 \times 10^{112}$. Both ladder barriers are the two central lines of this one inequality, and #307 sits at the origin; on the boundary of the protected region, at the crossing of its two distinguished lines. The open half–plane $s + t \le -1$ is unprotected, and is populated from $ab = 30$ on: over $ab \le 10^7$ exactly* 21 *strata occur, all obeying the law, the nearest being $(0, -1)$ and $(-1, 0)$ at $(a, b) = (6, 5)$ and $(5, 6)$; off–axis strata occur ($(1, -13)$ at $(690, 53)$), so integrality does not force one exact side. The $(0, -1)$ stratum with $b = a'$ is the chain equation*

$$a'' + a' = a, \qquad \textit{equivalently} \qquad \sigma(a)\big(1 + \sigma(a')\big) = 1,$$

a companion of the cross–match $\sigma(a)\sigma(b) = 1$ with the same shape and one shifted factor. Its solutions up to 10^7 are $\{6,\ 42,\ 15590,\ 47058\}$ and they split in two. Every primary pseudoperfect number whose predecessor is prime *solves it, trivially: then $a' = a - 1$ and $a'' = 1$ ($6, 42, 47058$; 1806 fails, its predecessor $1805 = 5 \cdot 19^2$ being composite, and 2 fails as $a'' = 0$). The remaining solution $15590 = 2 \cdot 5 \cdot 1559$ is not on the pseudoperfect line at all: $15590' = 10923 = 3 \cdot 11 \cdot 331$ and $10923' = 4667$, a genuine two–step closure. It is the first solution of the chain equation whose derivative is composite; the sequence $6, 42, 15590, 47058$ does not appear in the OEIS. All census claims verified exhaustively (`code/deviation_halfplane.py`).*

Proof. $\sigma(a) = a'/a = b/a + s$ and $\sigma(b) = a/b + t$ sum to the display, with strict inequality by AM–GM ($a \neq b$); mass > 2 forces ≥ 59 primes as in Theorem 4.3. The ladder identifications are Propositions 12.2 and 12.7; the census is a finite computation. $\square$

Corollary 12.9 (Below mass 5/2 the rung is not a choice but a parity)**.** *Let (a, b), $a < b$, be a Pythagorean pair with $\sigma(ab) < 5/2$. Then $k \in \{0, -1\}$, and k is* determined *by the parity of ω:*

$$\begin{array}{ll} \textit{exactly one member even:} & k = 0 \iff \omega(\textit{odd member})\ \textit{even}; \\ \textit{both members odd:} & k = 0 \iff \omega(a)\ \textit{odd}. \end{array}$$

Hence, at these masses, the single–integer test of Proposition 12.1 together with one parity check is equivalent *to #307, not merely necessary for it: a squarefree N with $N'^2 - 4N^2 = \square$ whose splitting has the stated ω–parity* is *a two–cycle. In particular the finite verifications of Proposition 4.5 and Remark 4.7 are decisive in both directions; a surviving candidate would have been a solution, not a candidate, and the whole level–59/60 regime lies in this range, its masses being confined to $(2,\ 2.00235]$, where $r = b/a \le 1.0497$.*

Proof. $\sigma(ab) = r + 1/r$ with $r = b/a > 1$, and $r + 1/r < 5/2 \iff r < 2$. By Proposition 12.5(1), $k \le 0$ and $|k| < b/a = r < 2$, so $k \in \{0, -1\}$; a set of two consecutive integers is separated by parity, and Proposition 12.5(2) supplies that parity from ω. The stated equivalences are the two cases of that law at $k \equiv 0$. The mass range for $|U| \in \{59, 60\}$ is Theorem 4.3 and Proposition 4.5. $\square$

Corollary 12.10 (Diagonal stratification: the cost of the last lattice step)**.** *Let $m(x)$ be least with $\sum_{i \le m(x)} 1/p_i > x$. The diagonal $s+t = -j$ carries the barrier $\sigma(ab) > 2-j$, hence $\omega(ab) \ge m(2-j)$ and $ab \ge \Pi_{m(2-j)}$:*

$$\begin{array}{llll} j \ge 2: & \textit{mass} > 0, & \textit{no barrier}, & ab \ge 2; \\ j = 1: & \textit{mass} > 1, & \omega \ge 3, & ab \ge 2 \cdot 3 \cdot 5 = 30; \\ j = 0: & \textit{mass} > 2, & \omega \ge 59, & ab \ge \Pi_{59} > 8.77 \times 10^{112}. \end{array}$$

The deviation plane is thus stratified by mass, one diagonal at a time, and #307 sits on the first diagonal carrying a nontrivial barrier. The $j = 1$ floor is attained*: the census minimum on that diagonal is exactly* 30*, at $(a, b) = (5, 6)$ and $(6, 5)$; so the barrier is sharp one step off the origin. The final lattice step therefore costs*

$$\Pi_{59}/30 \;\approx\; 2.9 \times 10^{111},$$

one hundred and eleven orders of magnitude for a single unit of $s + t$. This is the barrier of Theorem 4.3 in its most compressed form: not a bound on solutions, but the price of the last step of an integer lattice walk, paid because the diagonal index is a mass threshold and the mass thresholds 1 *and* 2 *sit at* 3 *and* 59 *primes.*

Remark 12.11 (The populated strata calibrate the heuristic model)**.** The unprotected strata are the first family of #307–type exact coincidences where the average–case model can be tested against exhaustive truth. Pricing the second exactness condition of a candidate pair at $1/b$ and re–running the census enumeration with identical filters gives expected counts 11.4, 18.3, 26.1, 34.2 at $ab \le 10^4, \dots, 10^7$ against observed $6, 12, 18, 23$: the ratio is stable near $2/3$ over four decades; the model has the right order of growth and a constant local correction, i.e. it is *calibrated* at the one–condition level. Composing two levels, however, already breaks the naive pricing: for the chain equation $a'' + a' = a$ the crude spread price predicts $E = 48.8$ against the observed 4; a twelvefold bias at reachable height, because the composition requires mass–window conditioning between the levels (the stratified–versus–naive lesson of Section 5, here measured rather than inferred). The reliability of averaging thus degrades measurably with each composed exact condition, and the two–cycle at the origin composes many. This is the quantitative face of the average–to–instance gap: the model earns trust stratum by stratum, and loses it condition by condition (`code/strata_calibration.py`).

The Pythagorean layer behaves very differently over function fields, and refines into two arithmetic sub–layers over $\mathbb{Z}$.

Proposition 12.12 (The Pythagorean layer over function fields)**.** *Over $\mathbb{F}_q[t]$ with q odd, $a^2 + b^2 = (ab)'$ has no coprime squarefree monic solutions: $\deg(a^2+b^2) = 2\max(\deg a, \deg b) > \deg a + \deg b - 1 \ge \deg((ab)')$, the leading coefficient* 1 *or* $2 \ne 0$ *preventing cancellation. Over $\mathbb{F}_2[t]$ the layer is nonempty: up to degree* 8 *(exhaustively,* 23,548 *coprime squarefree pairs) its only solutions are the three pairs of the Artin–Schreier tower $x_{j+1} = x_j(x_j + 1)$, namely $(t, t+1)$, (t^2+t, t^2+t+1), (t^4+t, t^4+t+1); each has $a' = (a+1)' = 1$ and $(a(a+1))' = (a+1) + a = 1$, and the climb halts at level* 4 *(there $(x_3+1)' = t^2 + t \ne 1$). These pairs lie off the cycle stratum $k = 0$, on the side the integer normalisation of Proposition 12.5(1) forbids. Cycles ($k = 0$) remain impossible over every $\mathbb{F}_q[t]$ (Theorem 6.1): the function–field world separates the Pythagorean layer from the cycle layer, sharpening the integer obstruction to "the archimedean order forces $k \le 0$."*

The function–field immunity is more fragile than the characteristic: it is destroyed by inverting a single place.

Proposition 12.13 (Inverting one place destroys the function–field obstruction)**.** *Over the S–integers $\mathbb{F}_q[t, t^{-1}]$, whose unit group $\mathbb{F}_q^\times \cdot t^{\mathbb{Z}}$ is infinite, unit–valued derivative two–cycles exist. Over $\mathbb{F}_3$ the smallest has degree 3 and only* two *prime classes:*

$$a = t^3 + t^2 + 1 = (t-1)(t^2 - t - 1), \qquad b = t^3 - t^2 - 1 = (t+1)(t^2 + t - 1),$$

with $D(a) = b$ and $D(b) = a$ under the assignment $(u_1, u_2) = (t, 1)$ on each side, a, b coprime squarefree and both coprime to t.

Proof. $D(a) = t\,(t^2-t-1)+1\cdot(t-1) = t^3-t^2-1 = b$ and $D(b) = t\,(t^2+t-1)+1\cdot(t+1) = t^3+t^2+1 = a$, both exact in $\mathbb{F}_3[t]$; the factorisations are as displayed and share no factor. The values t and 1 are units of $\mathbb{F}_q[t, t^{-1}]$, so this is a unit–valued Leibniz derivative in the sense used throughout Section 6. Verified independently by exhaustive search and by direct recomputation (`hunt/ff_twist.rs`). □

Remark 12.14 (What the function–field theorem is really about)**.** Proposition 12.12 is a *degree* argument: with $\pi' = 1$ one has $\deg D(a) \leq \deg a - 1$, so $D(a) = b$, $D(b) = a$ forces $\deg b \leq \deg b - 2$. Sign twists cannot disturb it, because $\mathbb{F}_q^\times$ consists of constants and degree is blind to them, exactly as, over $\mathbb{Z}$, twisting by ± 1 retains the full barrier (Remark 6.4). But allowing the values t^k, units once t is inverted, makes $\deg D(a) \leq \deg a - 1 + K$, and the two–cycle inequalities collapse to $0 \leq 2K - 2$: the obstruction is vacuous from $K = 1$, and cycles duly appear at once. A census over $\mathbb{F}_3$ confirms both halves: at $K = 0$ the count is 0 at every degree tested, and at $K = 1, 2$ it is $4, 16$ / $4, 48$ / $12, 122$ for degrees $\leq 4, 5, 6$. The moral sharpens Section 6's thesis rather than softening it. What immunises $\mathbb{F}_q[t]$ is not the characteristic, nor the finiteness of its unit group (cycles exist over $\mathbb{Z}[\sqrt{-2}]$, whose unit group has order 2) but the presence of a *single* distinguished place carrying a degree function that the derivative strictly lowers. Adjoining a second place destroys it. This is the exact structural analogue of the claim made for $\mathbb{Z}$: one archimedean place, one mass function the derivative must respect, and a barrier that survives every twist available in the ring.

Proposition 12.15 (No place of $\mathbb{Q}$ carries the function–field descent)**.** *The mechanism isolated in Remark 12.14 (a place whose valuation the derivative strictly lowers) is unavailable over $\mathbb{Q}$, and this is a classification statement rather than a failure of ingenuity.*

(i) Archimedean place. *$|D(n)| = n\,\sigma(n)$, which exceeds n for every n of mass > 1; the derivative* expands*, first at $n = 30$, and on 1.79% of squarefree $n \leq 10^6$. The absolute value is in any case not ultrametric, so the second half of the degree argument ($\deg(x+y) \leq \max$) is unavailable too.*

(ii) p–adic places. *For squarefree n with $p \mid n$, cofactor survival gives $v_p(D(n)) = 0 < 1 = v_p(n)$: the derivative lowers v_p there. But it* raises *it elsewhere, there are squarefree n with $p \nmid n$ and $p \mid D(n)$ (e.g. $p = 2$, $n = 15$; $p = 3$, $n = 14$; $p = 5$, $n = 6$), in comparable abundance to the lowering cases at every p tested. So no p–adic place contracts uniformly.*

(iii) *By Ostrowski's theorem* (i) *and* (ii) *exhaust the places of $\mathbb{Q}$.*

Hence no valuation–theoretic descent of the type that proves Proposition 12.12 can exist over $\mathbb{Q}$: the function–field argument fails to transfer for a structural reason internal to the classification of absolute values, not for want of a cleverer place. (Counts and witnesses: `code/no_place.rs`.)

Proposition 12.16 (Deviation calculus for k–cycles)**.** *For pairwise coprime squarefree $a_1, \dots, a_k$ with product N and deviations $\kappa_i = (a_i' - a_{i+1})/a_i$ (indices mod k): the ratios a_{i+1}/a_i have product 1 (telescoping), and $\sigma(N) = \sum_i a_{i+1}/a_i + \sum_i \kappa_i$. On the relaxed layer $\sum_i \kappa_i = 0$ (which contains the cycles $\kappa_1 = \cdots = \kappa_k = 0$) AM–GM gives $\sigma(N) \geq k$, so the distinct–prime barrier m_k of Proposition 4.18 extends to the relaxed problem for every k, and to genuine k–cycles for $k \leq 3$; for $k \geq 4$ it inherits the pairwise–coprimality caveat of Proposition 4.18 (the bound then holds for the multiset reciprocal sum). At $k = 2$, $\kappa_1 = -\kappa_2$ is the deviation of Proposition 12.5 and $\sum \kappa_i = 0$ is $a^2 + b^2 = (ab)'$.*

Proposition 12.17 (Split discriminant)**.** *For coprime squarefree a, b with $N = ab$,*

$$N' + 2N - (a+b)^2 = N' - 2N - (a-b)^2 = (ab)' - a^2 - b^2.$$

Hence N is the product of a Pythagorean pair iff $N' - 2N$ and $N' + 2N$ are both *perfect squares d^2, s^2 (their product is Bado's discriminant $N'^2 - 4N^2$), with $a, b = (s \pm d)/2$. The minus factor carries the obstruction: $N' - 2N = (a-b)^2 \geq 0$ forces $\sigma(N) \geq 2$, hence the 59–prime barrier, indeed* 0 *of* 103,596 *squarefree $N < 2 \times 10^5$ satisfy $N' - 2N = \square$ (Corollary 12.3 in one line). The plus factor is satisfiable at small scale (*114 *such $N < 2 \times 10^5$). In characteristic* 2 *the obstructing factor vanishes ($a - b = a + b$) and the criterion degenerates to $(a+b)^2 = N'$: the permissiveness of Proposition 12.12 and the barrier of Corollary 12.3 are the absence and presence of one and the same algebraic factor.*

Both squares of the split criterion are pinned modulo 8 by an elementary law that seems worth recording on its own.

Proposition 12.18 (The mod–8 law of the derivative)**.** *For odd squarefree m write $S(m) = \sum_{p \mid m} p$. Then*

$$m\,m' \;\equiv\; S(m) \pmod 8.$$

Consequently: (a) *modulo 2 this is the parity law $m' \equiv \omega(m)$ used in Theorem 4.17;* (b) *an all–odd two–cycle $a' = b$, $b' = a$ forces*

$$S(a) \;\equiv\; S(b) \;\equiv\; ab \pmod 8$$

the two prime sums are congruent to each other and to the product; refining Theorem 4.17(ii); (c) *in the mixed case $a = 2k$ (k odd), the odd member obeys $b \equiv k\big(1 + 2S(k)\big) \pmod{16}$ and $S(b) \equiv 2kb \pmod 8$, and the even member the companion law $(2k)\,(2k)' \equiv 2 + 4S(k) \pmod{16}$;* (d) *for odd squarefree N the plus quantity satisfies $N' + 2N \equiv N\big(S(N) + 2\big) \pmod 8$, so an odd plus–hit must have $N(S(N) + 2) \equiv 1 \pmod 8$ when $\omega(N)$ is odd and $\equiv 0, 4 \pmod 8$ when $\omega(N)$ is even. None of these congruences is an obstruction, consistently with Proposition 5.2, but they thin every hunt over the odd sector by a fixed factor.*

Proof. $q^2 \equiv 1 \pmod 8$ for odd q, so $m/q \equiv mq \pmod 8$; summing over $q \mid m$ gives $m' \equiv m\,S(m) \pmod 8$, and multiplying by m (with $m^2 \equiv 1$) gives the law. (a) is the law modulo 2, since $S(m) \equiv \omega(m)$ for odd primes. (b) is the law at a and at b with $a' = b$, $b' = a$. In (c), Leibniz gives $b = (2k)' = k + 2k'$, and the law at k gives $k' \equiv kS(k) \pmod 8$, whence $b \equiv k + 2kS(k) \pmod{16}$; the second congruence is the law at b with $b' = 2k$, and the companion follows from $2k(k + 2k') \equiv 2k^2 + 4kk' \pmod{16}$ with $k^2 \equiv 1 \pmod 8$ and $kk' \equiv S(k) \pmod 8$. In (d), $s^2 \bmod 8 \in \{0, 1, 4\}$ and $s \equiv N' \equiv \omega(N) \pmod 2$. All parts were verified exhaustively over the 405,285 odd squarefree $m \leq 10^6$ and all odd plus–hits below 10^6: zero exceptions (`code/mod8_plusthin.py`). $\square$

Proposition 12.19 (Structure of the plus–layer). *Call squarefree N with $\omega(N) \geq 2$ a* plus–hit *if $N' + 2N$ is a perfect square; the first are* $130, 145, 305, 689, 1745, 1870, 1970, 2353, \ldots$ (114 *below* 2×10^5*; OEIS [31]*). (i) *For a semiprime $N = pq$, $2(N' + 2N) + 1 = (2p + 1)(2q + 1)$, so a plus–hit forces $(2s)^2 \equiv -2 \pmod{\ell}$ for every prime $\ell \mid (2p + 1)(2q + 1)$, whence $\ell \equiv 1, 3 \pmod 8$ and* $p \equiv q \equiv 1 \pmod 4$ *(verified on all* 59 *semiprime instances; this explains the dominance of* 5 *and* 13*).* (ii) [*empirical*] $\#\{$*plus–hits* $\leq X\} \approx 0.25\sqrt{X}$ *(ratios* $0.250, 0.253, 0.255$ *at* $X = 5\times 10^4, 10^5, 2\times 10^5$*): the plus factor behaves like a random integer of its size, so (with Proposition 12.17) the whole integer obstruction sits in the minus factor.* (iii) *Fixing $p = 5$, $q = (s^2 - 5)/11$ with s even and $s \equiv \pm 4 \pmod{11}$ gives $q \equiv 1 \pmod 4$, and each prime value yields the plus–hit $N = 5q$ (*68 *with $s < 2000$, up to $N = 1{,}774{,}805$); Bunyakovsky's conjecture for this quadratic thus implies infinitely many plus–hits. The two discriminant factors separate cleanly: the plus–layer is Bunyakovsky–class, the minus–layer is the barrier.* (iv) *[Remark] The full cycle condition states that (d^2, N', s^2) is a three–term arithmetic progression with common difference $2N$ whose outer terms are squares; a congruent–number–shaped configuration; the elliptic–curve machinery of that theory does not transfer because N' is not an algebraic function of N. A* 251*–term b–file to $N < 10^6$ (counting ratio* 0.251*) and an OEIS submission draft accompany this note; the plus–hit sequence matched no OEIS entry in our search.* (v) *[P, C]* General local law (QR tournament). *For every odd prime $\ell \mid N$ of a plus–hit, $N' + 2N \equiv N/\ell \pmod{\ell}$ (all cofactor terms N/p with $p \neq \ell$ carry a factor of ℓ; only N/ℓ survives), so N/ℓ is a nonzero quadratic residue mod ℓ. Verified with zero violations on all* 114 *plus–hits below* 2×10^5 *and all* 251 *terms to* 10^6*; the semiprime $p \equiv q \equiv 1 \pmod 4$ theorem of* (i) *is its $\omega = 2$ shadow via quadratic reciprocity.*

The empirical $\frac{1}{4}\sqrt{X}$ law of part (ii) can be proved, up to logarithms, on the stratum that dominates the small–scale census.

Proposition 12.20 (The semiprime plus–layer is provably thin). *The number of semiprime plus–hits $N = pq \leq X$ is $O\big(\sqrt{X}\,(\log X)^2\big)$, by a fully elementary argument; Erdős's bound $\sum_{s\leq Y} \tau(2s^2+1) \ll Y \log Y$ [34] sharpens this to $O(\sqrt{X}\log X)$. Under Bunyakovsky's conjecture the $p = 5$ family of Proposition 12.19(iii) alone contributes $\gg \sqrt{X}/\log X$, so conditionally*

$$\frac{\sqrt{X}}{\log X} \;\ll\; \#\{\textit{semiprime plus–hits} \leq X\} \;\ll\; \sqrt{X}\,\log X :$$

the $\sqrt{X}$ order of the empirical law is genuine on the semiprime stratum, not an artefact of the tested range.

Proof. For a semiprime plus–hit $N = pq \leq X$ one has $s^2 = N' + 2N = p + q + 2pq \leq 3X + 1$ (as $p + q \leq pq + 1$), and Proposition 12.19(i) gives $2s^2 + 1 = (2p + 1)(2q + 1)$: the map $N \mapsto \big(s, \{2p + 1, 2q + 1\}\big)$ into pairs (root, unordered nontrivial factorisation) is injective, so the count is at most $\sum_{s\leq Y} \tau(2s^2 + 1)$ with $Y = \sqrt{3X + 1}$. The polynomial $2s^2 + 1$ has discriminant -8, prime to every odd modulus, so its roots modulo an odd prime are simple and lift uniquely by Hensel: $2s^2 + 1 \equiv 0 \pmod u$ has at most $2^{\omega(u)} \leq \tau(u)$ solutions modulo u (none for even u). Pairing each divisor of $2s^2 + 1$ with its cofactor and keeping the smaller member $u \leq \sqrt{2s^2 + 1} \leq \sqrt{2}\,Y$,

$$\sum_{s\leq Y} \tau(2s^2 + 1) \;\leq\; 2\sum_{u\leq\sqrt{2}Y} \#\{s \leq Y : u \mid 2s^2 + 1\} + Y \;\leq\; 2\sum_{u\leq\sqrt{2}Y} \tau(u)\Big(\frac{Y}{u} + 1\Big) + Y \;\ll\; Y(\log Y)^2,$$

by Dirichlet's $\sum_{u\leq Z} \tau(u) \sim Z\log Z$ and $\sum_{u\leq Z} \tau(u)/u \sim \frac{1}{2}(\log Z)^2$. For the conditional lower bound, the family of Proposition 12.19(iii) has $\asymp \sqrt{X}$ eligible $s \leq \sqrt{2.2X + 5}$ (two residue classes modulo

22), of which Bunyakovsky for $(s^2-5)/11$ makes $\gg \sqrt{X}/\log X$ prime. The injection, the Hensel root count, and the censuses were verified numerically (59 semiprime hits below 2×10^5, all reproduced; $\sum_{s\le Y}\tau(2s^2+1)/(Y\log Y)$ stable near 1.05–1.09 for $Y=500$–2000; `code/mod8_plusthin.py`). The $\omega\ge 3$ stratum remains empirical. $\square$

The semiprime argument generalises: the slot calculus of Proposition 14.26 counts.

Proposition 12.21 (Slot counting: every fixed–ω stratum of the plus–layer is sparse)**.** *Fix $r\ge 2$. The number of plus–hits $N\le X$ with $\omega(N)=r$ is $O\big(X^{1-1/r+o(1)}\big)$.*

Proof. Write $p=P^+(N)$ for the largest prime of N and $M=N/p$, so $\omega(M)=r-1$ and, since $N<p^r$, $M\le N^{1-1/r}\le X^{1-1/r}$. The slot identity of Proposition 14.26 (base M, slot p) gives $N'+2N=p\,E_0(M)+M$ with $E_0(M)=M'+2M$, so a plus–hit $s^2=N'+2N$ forces

$$s^2\equiv M \pmod{E_0(M)}, \qquad p=\frac{s^2-M}{E_0(M)},$$

and the map $N\mapsto(M,s)$ is injective. For squarefree M one has $\gcd(M,M')=1$ (modulo $q\mid M$ only the cofactor M/q survives in M'), hence $\gcd(M,E_0(M))=1$: at every odd prime $\ell\mid E_0(M)$ a root of $s^2\equiv M$ has $\ell\nmid s$, is nonsingular, and lifts uniquely (Hensel), so the congruence has at most $4\cdot 2^{\omega(E_0)}=X^{o(1)}$ roots modulo $E_0(M)$. Since $s^2=N\big(\sigma(N)+2\big)$ and $\sigma(N)\ll\log\log X$, each base carries at most $X^{o(1)}\big(X^{1/2}/E_0(M)+1\big)$ admissible s, and with $E_0(M)\ge 2M$,

$$\#\{N\}\;\le\sum_{\substack{M\le X^{1-1/r}\\ \omega(M)=r-1}} X^{o(1)}\Big(\frac{X^{1/2}}{M}+1\Big)\;\ll\; X^{1/2+o(1)}+X^{1-1/r+o(1)}\;=\;X^{1-1/r+o(1)}.$$

(For $r=2$ this recovers the order of Proposition 12.20, which is sharper by logarithms.) The injection, the identity, $\gcd(M,M')=1$ and $P^+(N)\ge N^{1/r}$ were verified on all 251 plus–hits below 10^6: zero exceptions (`code/strata_count.py`). $\square$

Remark 12.22 (What separates the strata from the full layer)**.** The bound is honest but far from the truth: at $X=10^6$ the strata $\omega=2,\dots,5$ hold 124, 69, 50, 8 hits (every one below $\sqrt{X}$) because the congruence $s^2\equiv M \pmod{E_0(M)}$ is usually *insoluble*: over random odd semiprime bases the average number of roots is 0.34, the no–residue branch of the trichotomy of Proposition 14.26 dominating. Summing the proposition over r does *not* give the full layer density zero, because the $X^{o(1)}$ root–count factors are not uniform in r; an average bound for $\tau(M'+2M)$ (a Titchmarsh–divisor–type sub–problem) remains the route to the true ($\sqrt{X}$–order) count. Density zero itself, however, needs less: a smooth/rough dichotomy bypasses the average entirely, because roughness buys a factor $X^{-1/\log\log X}$ against which the divisor bound costs only $X^{\log 2/\log\log X}$, and $\log 2<1$.

Proposition 12.23 (The plus–layer has density zero)**.** *Unconditionally, with $\log_3=\log\log\log$,*

$$\#\{\text{plus–hits } N\le X\}\;\ll\;\frac{X}{(\log X)^{(1+o(1))\log_3 X}}.$$

In particular the plus–layer has density zero. The proof applies verbatim to $\{N$ squarefree : $N'+cN=\square\}$ for any fixed $c\ge 1$.

Proof. Split at $y=X^{1/u}$ with $u=\log\log X$. *Smooth part.* The number of $N\le X$ with $P^+(N)\le y$ is $\Psi(X,y)=X\,u^{-u(1+o(1))}=X(\log X)^{-(1+o(1))\log_3 X}$ (Rankin's bound; [17], III.5). *Rough part.* If

$P^+(N) = p > y$, the injection of Proposition 12.21 applies with $M = N/p \le X/y = X^{1-1/u}$: the hit determines (M, s) with $s^2 \equiv M \pmod{E_0(M)}$, a congruence with at most $\rho(E_0) \le 4 \cdot 2^{\omega(E_0)} \le \exp\big((\log 2 + o(1)) \log X / \log\log X\big)$ roots by the maximal order of the divisor function [16]; each base therefore carries at most $\rho(E_0)\big(X^{1/2+o(1)}/E_0(M) + 1\big)$ values of s. Summing, with $E_0(M) \ge 2M$:

$$\#\{\text{rough hits}\} \ \ll\ X^{1/2+o(1)} + X^{1-1/u} \exp\Big((\log 2 + o(1)) \frac{\log X}{\log\log X}\Big) \ =\ X^{1/2+o(1)} + X \exp\Big(-(1 - \log 2 - o(1)) \frac{\log X}{\log\log}$$

and $1 - \log 2 = 0.3068\ldots > 0$: the rough part is smaller than every $X/(\log X)^A$. The smooth part dominates and gives the stated rate. (At $X = 10^6$ the split reads $251 = 225$ rough $+\,26$ smooth, the injection collision–free; `code/pluszero.py`.) ☐

On a derivative *line* the slot is not merely constrained; it is pinned, and the counting sharpens to every line at once.

Proposition 12.24 (The determined slot: all derivative lines are thin, uniformly)**.** *For integers $e \ge 1$ and c,*

$$\#\{N \le X \text{ squarefree}:\ N' = eN + c\} \ \ll\ \frac{X}{(\log X)^{(1+o(1)) \log_3 X}},$$

uniformly in e and c. In particular the primary pseudoperfect line $n' = n - 1$, the Giuga quotient–one line $n' = n + 1$, every line $n' = n \pm d$ of [27], and every slope–two line $n' = 2n + c$ obey this bound. (Continuity of the limit law of Proposition 5.6 already forces each line to have density zero; the content is the rate, far beyond what concentration inequalities give, and the uniformity. Classical lines with extra divisibility structure may well admit sharper bounds; we claim none.)

Proof. Split at $y = X^{1/u}$, $u = \log\log X$; smooth members number $\Psi(X, y) = X(\log X)^{-(1+o(1)) \log_3 X}$ (Rankin). For a rough member $N = Mp$, $p = P^+(N) > y$, Leibniz gives $N' - eN = p\,(M' - eM) + M$, and $M' - eM \ne 0$: $\sigma(M) = e$ would make the auto–reduced fraction of Theorem 2.1 an integer, forcing $D_M = 1$, $M = 1$. Hence

$$p \ =\ \frac{c - M}{M' - eM}$$

is *determined* by M (the line has no free root at all) so $N \mapsto N/P^+(N)$ is injective on rough members, whose number is at most $X^{1-1/u}$. The smooth part dominates. Two classical objects fall out of the formula: on the stratum $M' - M = -1$ of the two slope–one lines it reads $p = M + 1$ ($c = -1$) and $p = M - 1$ ($c = +1$) (the oblong chains $6 \mapsto 42 \mapsto 1806$ and $30 = 6 \cdot 5$, $1722 = 42 \cdot 41$) while off–stratum members stay slot–determined ($858 = 66 \cdot 13$ with $M' - M = -5$; $47058 = 1518 \cdot 31$ with $M' - M = -49$). Verified: the formula reproduces $P^+(N)$ on all listed members and identically on the 51,201 squarefree $N \le 10^5$ with $\omega \ge 2$ (`code/lines_slot.py`). ☐

The members of a line carry far more structure than thinness: every prime factor is pinned modulo its cofactor, and the top of the factorisation is capped by the bottom.

Proposition 12.25 (Line anatomy: pinning, coupling, caps)**.** *Let M be squarefree on the line $M' = eM + c$, let $p > q$ be its two largest primes and $R = M/(pq)$.*

(a) (Pinning.) *$M/t \equiv c \pmod t$ for* every *prime $t \mid M$; the Giuga congruence of the line, independent of e. Equivalently, each prime factor lies in one determined residue class modulo its cofactor: $t \equiv c\,(M/(tr))^{-1} \pmod r$ for all primes $r \mid M/t$.*

(b) (Top cap.) *If $M/p \ne c$ then $p \le \sqrt{M} + |c|$: line members are $(\sqrt{M} + |c|)$–smooth.*

(c) (Coupling.) $pq \mid R(p+q)-c$. *Here* $\gcd(pq,c)=1$ *is automatic:* $p \mid c$ *together with* $p \mid qR-c$ *would give* $p \mid qR$*, impossible as* p *exceeds every prime of* qR*.*

(d) (Second cap.) *If* $R(p+q) \neq c$ *then* $q < R+\sqrt{R^2+|c|}$*; the degenerate stratum* $R(p+q)=c$ *forces* $c \geq 2\sqrt{RM}$*, so it is empty on every line with* $c < 2\sqrt{M}$*; in particular on all the near–critical minus lines.*

Proof. (a) Cofactor survival gives $M' \equiv M/t \pmod t$ while $eM \equiv 0$; the second form is CRT. (b) $M/p-c$ is a multiple of p; if nonzero, $M/p \geq p-|c|$. (c) $p \mid qR-c$ and $q \mid pR-c$ give $pq \mid (qR-c)(pR-c) = pqR^2 - cR(p+q)+c^2$, so $pq \mid c\,(R(p+q)-c)$, and the displayed coprimality removes c. (d) If $R(p+q) \neq c$ then $pq \leq |R(p+q)-c| \leq R(p+q)+|c| < 2Rp+|c|$, whence $q^2 < 2Rq+|c|$; if $R(p+q)=c$ then $c \geq 2R\sqrt{pq} = 2\sqrt{RM}$ by AM–GM. All parts were verified on the 7,184,234 squarefree $n \leq 2\times 10^7$ with $\omega \geq 3$, for both $e=1,2$ with $c := n'-en$ (zero exceptions, the second cap attained with slack 1 at $n=70$) and on the classical members, where the coupling quotient $\big(R(p+q)-c\big)/pq$ equals 1 on the oblong–chain members and 5 on both off–stratum members 47058 and 66198 (`code/line_anatomy.py`). ☐

The lines themselves close into a family under peeling, and the closure explains why no reorganization of the counting has broken through.

Proposition 12.26 (The peeling groupoid)**.** *For integers* $a \geq 1$*,* b*,* c *let* $L(a,b,c) = \{n \text{ squarefree} : an' = bn + c\}$*.*

(i) (Closure; the slope is a mass.) *If* $n = qm \in L(a,b,c)$ *with* $q = P^+(n)$*, then* $m \in L(aq,\ bq-a,\ c)$*, and the slope moves by* $b/a \mapsto b/a - 1/q$*: the full peel chain of* n *descends the partial sums of* $\sigma(n)$*; the classical lines, their* μ*–Sondow shifts, and the slope–two lines of the minus square are one orbit family.*

(ii) (Rigidity at every level.) *Along the chain of* n *on an integer line* $(a,b)=(1,e)$*, the level–*j *denominator vanishes iff* $\sigma(m_j) = b_j/a_j$*, which telescopes to the single* j*–independent condition* $\sigma(n)=e$*; impossible, since an integer mass forces* $n=1$ *by Theorem 2.1. So every denominator on every genuine chain is nonzero.*

(iii) (Determined slot at every level.) *Consequently* $q = (c-am)/(am'-bm)$ *at each step: a line member is a tower of single determined candidates over its own core.*

Every partial–peel reorganization of the line counts (peeling one prime, several, in any order, from either layer of the split criterion) stays inside this groupoid, where part (iii) *reproduces the one–candidate–per–cofactor obstruction verbatim. The obstruction is therefore* invariant *under the natural changes of viewpoint: a proof must inject input from outside the groupoid; primality along determined thin sequences, or a decomposition of integers not indexed by their primes.*

Proof. (i) is Leibniz: $a(qm'+m) = bqm+c$ gives $aq\,m' = (bq-a)m+c$, and $(bq-a)/(aq) = b/a-1/q$. In (ii), $\sigma(m_j) = b_j/a_j = e - \sum_{i<j} 1/q_i$ says $\sigma\big(m_j \prod_{i<j} q_i\big) = \sigma(n) = e$; by the auto–reduced fraction of Theorem 2.1 an integer mass forces $D=1$. (iii) is the display above (i). Verified: closure and slot on 10^5 random (n,a,b) (the only slot exceptions being exactly the $\sigma(m)=b/a$ locus, excluded on genuine chains, 73 of 10^5 random triples, 0 of 310,788 genuine chain steps at $e=1,2,3$), and the slope telescope exact on every chain (`code/peeling_groupoid.py`). ☐

The determined slot bounds more than itself: unwound one level further it constrains the *second* prime by a mass.

Proposition 12.27 (Mass–defect bound on the second prime)**.** *Let M be squarefree with $\omega(M) \geq 3$ on the line $M' = eM + c$, and write $p = P^+(M)$, $m = M/p$, $P = P^+(m)$, $n = m/P$. If $\sigma(m) < e$ then*

$$P\big(e - \sigma(n)\big) \;<\; 2 - \frac{c}{m}, \qquad \textit{equivalently} \qquad P^2\,(en - n') \;<\; 2Pn - c.$$

For $e = 2$ the hypothesis is automatic below Π_{59}; that is, throughout the range in which the minus layer is counted. On the near–critical slope–two lines ($|c|$ small) the bound reads $P < 2/(2 - \sigma(n)) + O(|c|/n)$: the second–largest prime of M is bounded by twice the reciprocal of the mass defect of the remaining cofactor.

Proof. M squarefree gives $p > P$. The determined slot of Proposition 12.26 is $p = (c-m)/(m' - em)$, and $\sigma(m) < e$ makes the denominator negative, so $p = (m-c)/\big(m(e - \sigma(m))\big)$ and $p > P$ reads $P\,m(e - \sigma(m)) < m - c$, i.e. $P(e - \sigma(m)) < 1 - c/m$. Since $m = Pn$ and $\sigma(m) = \sigma(n) + 1/P$, the left side is $P(e - \sigma(n)) - 1$, which gives the first display; multiplying by Pn clears denominators. Verified with 0 violations on every squarefree $M \leq 2 \times 10^6$ with $\omega \geq 3$, for $e = 1, 2, 3$; at $e = 1$ the 21,146 exceptions are exactly the $\sigma(m) \geq 1$ cases excluded by the hypothesis, and the bound is within a factor 2 of equality for 11% of members and within 100 for 59% (`code/slot_massbound.rs`). □

Remark 12.28 (The exact form, and a window that is real but inert)**.** Proposition 12.27 has an exact ancestor. With $N = pm$, $p = P^+(N)$, $k = 2m - m'$, Leibniz gives $N' = pm' + m$, so on the minus layer

$$d^2 \;=\; N' - 2N \;=\; m - pk, \qquad \text{whence} \qquad \sigma(N) \;=\; 2 + \frac{1}{P^+(N)} - \frac{k}{m}, \qquad k \geq 1.$$

Since $k \geq 1$ this is a *two–sided* window; the barrier supplies only the lower half:

$$2 \;<\; \sigma(U) \;<\; 2 + \frac{1}{\max U}, \qquad \text{equivalently} \qquad \sigma\big(U \setminus \{\max U\}\big) < 2,$$

so a minus–layer support is *mass–minimal*: deleting any one of its primes drops the mass below 2 (the largest is the binding case). The upper half is nonetheless *inert* where it can be tested: all 49,961 admissible level–59 supports already satisfy it, the tightest margin being 1.26×10^{-3}, so it prunes nothing there. We record it because it is exact, because it is the identity behind Proposition 12.27, and because it may bind at higher levels, where the largest admissible prime grows. (The enumeration also reproduces the count 49,961 of Proposition 4.5 independently, from a different traversal; `code/masswindow.rs`.)

Remark 12.29 (The minus layer is the anatomy of a quadratic polynomial)**.** The identity $d^2 = m - pk$ of Remark 12.28 can be read as a parametrisation rather than a constraint. On the minus layer $d^2 = N' - 2N \geq 0$ is a square, so

$$m \;=\; pk + d^2,$$

and for *fixed* (p, k) the cofactor m runs over the values of the quadratic polynomial $f_{p,k}(d) = d^2 + pk$. The minus layer is therefore not an arbitrary exact coincidence but a question about the *anatomy of quadratic–polynomial values*: for which d does $f_{p,k}(d)$ carry mass $\sigma = 2 - k/f_{p,k}(d)$, i.e. $\omega \approx 58$ prime factors concentrated on the small primes, with $P^+(f_{p,k}(d)) < p$?

This is the first place in this circle where a *named* toolkit applies. The anatomy of polynomial values is a developed subject; Erdős–Kac for $\omega(f(n))$, Halberstam–Richert sieve bounds for almost–primes in polynomial sequences, and the divisor sums of [34] already used in Proposition 12.20;

whereas "primality of a determined value" (Remark 14.24) had none. What the toolkit must supply is unusual: not the typical behaviour of $\omega(f(d))$, which is log log by Erdős–Kac, but a *large deviation* to $\omega \approx 58$ with the prime factors forced onto the bottom of the spectrum, since only the small primes carry mass. Existing polynomial–anatomy results control typical and almost–prime behaviour, not deviations of this shape; so the reformulation names the tool without yet applying it. We record it as the most promising translation of the residual question, and note that the $p = 5$ family of Proposition 12.19(iii) is the plus–layer shadow of the same quadratic mechanism. (Identity verified on all 1,606,956 squarefree $M \leq 3 \times 10^6$ with $\omega \geq 2$: 0 violations.)

Remark 12.30 (The residue amplifier is real, and exactly paid for)**.** Sieving $f_c(d) = d^2 + c$ by its roots exposes a mechanism the union–side reading of Corollary 14.25 misses. A prime ℓ divides some value iff $-c$ is a quadratic residue modulo ℓ, which removes half the primes; the $2^{-\omega}$ penalty the corollary records. But an *admissible* odd ℓ has *two* roots, so it divides values with density $2/\ell$ rather than $1/\ell$: on the primes that survive, the weight is doubled. Only the penalty has ever been counted, and one might expect the doubling to tilt the heuristic. It does not: the two effects cancel exactly.

Per fixed c the effect is genuine. Writing σ_B for the mass on primes $\leq B$, measured over $d \leq 2 \times 10^6$ with $B = 2 \times 10^4$: a generic $c = 1009$ gives mean $\sigma_B = 0.368$, a friendlier $c = 4966$ (admitting 35 of the 46 primes below 200) gives 0.400, against 0.452 for random integers; a *perfectly* friendly c, admitting every odd prime, would give $\frac{1}{4} + 2\sum_{\ell>2} \ell^{-2} = 0.654$, a factor 1.45 above random. So friendliness helps, and in the limit beats a random integer outright.

But friendliness is priced. Each odd ℓ is admissible for only half of the c, so a c admitting the first 58 odd primes has density 2^{-58}, exactly cancelling the 2^{58} gained from the doubled densities:

$$2^{-58} \cdot \tfrac{1}{2} \prod_{i=2}^{59} \frac{2}{\ell_i} \;=\; \prod_{i=1}^{59} \frac{1}{\ell_i},$$

the random–integer probability. Averaging over $c = 1, \ldots, 400$ confirms it numerically: the mean mass ratio is 0.9990, and the tail ratios $\Pr[\sigma_B > T]$ are 0.998, 0.996, 0.978 at $T = 0.8, 1.0, 1.2$; unity to within sampling, while individual c range from 0.257 to 0.635 in mean. *The quadratic reformulation is heuristically neutral*: it neither strengthens nor weakens the YES lean of Remark 4.14, and the 2^{-59} filter of Corollary 14.25 is correctly priced after all. What Remark 12.29 buys is a toolkit, not a bias (`code/quad_anatomy.rs`, `code/amplifier_cost.rs`).

The bound has enough force to regenerate the barrier on its own.

Corollary 12.31 (A second, independent derivation of the barrier)**.** *Let N be a minus–layer hit with $\omega(N) \geq 3$; more generally a member of any line $N' = 2N + c$ with $c \geq 0$. Write $p = P^+(N)$, $m = N/p$, $P = P^+(m)$, $n = m/P$. Then $N > 10^{112.94}$, without using the arithmetic–geometric mean inequality.*

Proof. If $\sigma(m) \geq 2$ then m carries at least 59 primes and $N > m \geq \Pi_{59}$ already. Otherwise Proposition 12.27 applies, and $c \geq 0$ makes its right side at most 2: $P\big(2 - \sigma(n)\big) < 2$. Since $P > P^+(n)$ this gives

$$2 - \sigma(n) \;<\; \frac{2}{P^+(n)}, \qquad \text{i.e.} \qquad \sigma(n) > 2 - \frac{2}{P^+(n)}.$$

But $\sigma(n) \leq \sum_{q \leq P^+(n)} 1/q$, so $P^+(n)$ must satisfy $\sum_{q \leq p} 1/q > 2 - 2/p$. The least such prime is $p = 269$, the 57th: at $p = 269$ the sum is $1.99505 > 1.99257$, while at $p = 263$ it is $1.99133 < 1.99240$. Hence $\omega(n) \geq 57$, $P^+(n) \geq 269$, and $n \geq \Pi_{57} > 10^{108.07}$; with $p > P > P^+(n)$ the two further primes give $N > \Pi_{57} \cdot 271 \cdot 277 > 10^{112.94}$. $\square$

Remark 12.32 (Two mechanisms, one constant)**.** Corollary 12.31 reaches $10^{112.94}$, and $\Pi_{59} > 8.77 \times 10^{112}$ is $10^{112.94}$: the two derivations agree to two decimal places. They share no step. Theorem 4.3 divides the defining equation by ab and applies AM–GM to $a/b + b/a$; the present route never forms that quotient, using instead the determined slot of Proposition 12.26, the trivial inequality $p > P^+(N/p)$, and the additivity $\sigma(m) = \sigma(n) + 1/P$. What they share is the arithmetic input underneath (how many primes Mertens needs to reach mass 2) which is why the constants coincide rather than merely being comparable. The barrier is thus not an artefact of the mass inequality one happens to use: it is the same Mertens threshold seen through two independent structures, and the critical set $S_2 = \{n : 2 - \sigma(n) < 2/P^+(n)\}$ of the second structure is empty below 10^{108} (verified empty for $n \le 5 \times 10^6$, `code/critical_set.rs`).

Remark 12.33 (The minus side: coercive versus degenerate moduli)**.** Proposition 12.23 rests on the plus modulus being *coercive*: $E_0^+(M) = M' + 2M \ge 2M$. The minus–square obeys the same slot identity with $E_0^-(M) = M' - 2M = M(\sigma(M) - 2)$, a modulus that *vanishes exactly on the mass–2 wall*: the degeneration locus of the counting method is the barrier stratum itself; the counting incarnation of the form degeneration $N_0(2 - \sigma) \to 0$ of Proposition 14.27. The minus–layer is the union of the slope–two lines $N' = 2N + d^2$ over square heights, and three facts are unconditional. (i) A rough minus–hit ($P^+(N) = p > y$) forces its base upward: if $\sigma(M) < 2 - 1/y$ then $d^2 = p\, E_0^-(M) + M < M(1 - p/y) \le 0$, impossible; so $\sigma(M) \ge 2 - 1/y$, and once $y > (2 - T_{58})^{-1} = 793.7\ldots$ this yields $\omega(M) \ge 59$ and $M \ge \Pi_{59}$: *the* 59*–prime wall propagates from the hit to its base*, and $N \ge \Pi_{59}\, y$. (ii) Bases with $\sigma(M) \ge 2 + 1/y$ have moduli $E_0^- \ge M/y$, and contribute as in Proposition 12.23. (iii) The remaining bases are near–critical, $\sigma(M) \in [2 - 1/y, 2 + 1/y)$; exactly the low slope–two lines $M' - 2M = \pm k$, $k \le M/y$. Each is thin by Proposition 12.24, but at a degenerate modulus the d–freedom is $\sqrt{X}$–sized, and summing the uniform bound does not close the layer. What would: *square–root bounds on slope–two lines*, $\#\{M \le Z : M' - 2M = \pm k\} \ll Z^{1/2+o(1)}$ uniformly in k. There is a second, sharper route to that bound, obtained by abandoning the prime–indexed decomposition altogether. Split $M = uv$ by *size* rather than by largest prime, u the coprime divisor nearest $\sqrt{M}$; Leibniz gives the exact bilinearisation

$$M' - 2M \;=\; v\,\delta(u) + u\,\delta(v), \qquad \delta(n) = n' - n,$$

so a slope–two line becomes a bilinear equation in (u, v) with both variables of comparable size and, for fixed u, the exact line $u\,v' - (2u - u')\,v = c$, i.e. $v \in L\big(u,\, 2u - u',\, c\big)$ in the notation of Proposition 12.26. Writing $N(u)$ for the number of v on that line in the balanced window $v \asymp M/u$,

$$\#\{M \le Z \text{ on the line}\} \;\le\; \sum_{u \le Z^{1/2}} N(u),$$

so the missing square–root bound follows from $N(u) \ll Z^{o(1)}$ on average; a *short–window line–multiplicity* hypothesis. This is a strictly sharper target than the one–class–per–cofactor count: the summation length drops from Z cofactors to $Z^{1/2}$ balanced divisors (empirically $Z^{0.43}$, the median of $\log u/\log M$ over nearest–to–$\sqrt{M}$ coprime splits being 0.432), and the surviving object is bilinear in a short window; the shape on which dispersion and Fouvry–Iwaniec methods operate, rather than a worst–case single–class count. The measured multiplicity is exactly 1 (mean 1.00, no sampled line carrying two solutions), so the hypothesis is not merely plausible but numerically flat; proving it, however, runs back into the determined slot, since $L(u, 2u - u', c)$ is itself a groupoid object. All identities verified exhaustively on the 591,423 squarefree $M \le 10^6$, in exact rationals for the line form (`code/bilinear_balanced.py`). By Proposition 12.25(a) that missing bound is a *one–class–per–cofactor* prime count: a line member is $M = pm$ with p prime in a single determined class modulo its own cofactor m, so the count is $\sum_m \#\{p \le Z/m : p \equiv \alpha_c(m) \pmod m\}$, and the trivial estimate Z/m^2+1 per cofactor fails exactly on its $+1$: one candidate integer per m, thinned only by the primality of that single determined value. Bounding worst–case single–class counts summed over all moduli lies beyond Bombieri–Vinogradov and beyond Linnik uniformly; the same species of count guards the true $\sqrt{X}$ order of the plus layer. The two layers' remaining questions are one wall. The ledger: plus layer closed by coercivity; minus layer reduced to the near–critical spectrum. (A small unconditional companion, from Proposition 12.18: an odd member of $N' = 2N + c$ has $S(N) \equiv cN + 2 \pmod 8$.)

Remark 12.34 (The primality can be dropped: the missing bound is a divisibility count)**.** The formulation above asks for a bound on how often a determined value is *prime*, which is what places it beyond Bombieri–Vinogradov. That requirement can be removed. The slot $(c-m)/(m'-em)$ is an integer at all only when

$$(m' - em) \bigm| (c - m),$$

and line members inject into the set of m satisfying that divisibility, so

$$\#\{m \le Z :\ (m' - em) \mid (c - m)\} \ \ll\ Z^{1/2+o(1)} \quad \text{uniformly in } (e, c)$$

already implies the square–root bound, with no prime–distribution input whatever.

The substitution costs almost nothing, because integrality is where the thinness lives. Counting both conditions separately on the two lines whose members are known, over $m \le 2 \times 10^7$ (so $\log Z = 16.8$): the line $n' = n - 1$ has 21 integral slots of which 16 are prime, and $n' = n + 1$ has 29 of which 19 are prime. Integrality alone is already $O(\log Z)$; primality removes a further factor of about 1.3 and no more. The prime–counting requirement was carrying a constant, while the divisibility was carrying the whole saving.

What remains is therefore a purely multiplicative question about the derivative, of the same shape as Proposition 12.26 but with the primality stripped out, and with roughly four orders of slack against the target at the ranges where the count can be observed. We record the substitution because it changes which literature the problem faces: divisibility counts for $m' - em$ rather than primes in determined classes.

13 Two closures on the existence route

Atom A10 asks for a prime term of the ternary recurrence of Remark 4.16. Before attacking it, two families of approach can be removed: the relaxations that would give the construction more freedom, and the algebraic methods that would avoid the archimedean place. Both are closed outright.

Proposition 13.1 (No free slot)**.** *Let P, Q be finite sets of primes with $\sigma(P)\sigma(Q) = 1$, and put $a = \prod P$, $b = \prod Q$. Then $b = a'$ and $a = b'$ exactly. In particular Q is* determined *by P: it is the set of prime factors of a', and the existence route carries exactly one parameter, the set P.*

Proof. $\sigma(a) = a'/a$ and $\sigma(b) = b'/b$, both in lowest terms since $\gcd(m, m') = 1$ for squarefree m. Then $\sigma(a)\sigma(b) = 1$ reads $a'b' = ab$. From $\gcd(a', a) = 1$ and $a' \mid ab$ we get $a' \mid b$, say $b = a'k$; symmetrically $b' \mid a$, say $a = b'j$. Substituting, $a'b' = ab = b'j \cdot a'k$, so $jk = 1$ and $j = k = 1$. □

Corollary 13.2 (The multi-slot relaxations are vacuous)**.** *There is no version of the construction in which two or more primes of Q may be chosen freely. In particular one cannot fix a base $B \subset Q$ and solve $1/q_1 + 1/q_2 = r/s$ for the remaining two primes by ranging over the $\tau(s^2)$ factorisations $(rq_1 - s)(rq_2 - s) = s^2$: by Proposition 13.1 the pair (q_1, q_2) is already forced to be the corresponding* *part of the factorisation of a'. The exponential parametrisation of Remark 4.16 is therefore intrinsic, not an artefact of the one-free-prime ansatz, and no amount of extra slots converts the Lehmer question into a Bunyakovsky one.*

The second closure is not new here, but it belongs in this list because it is what forces the whole existence route to be analytic. Theorem 6.1 of Section 6 shows that over $\mathbb{F}_q[t]$ the arithmetic derivative satisfies $\deg f' \leq \deg f - 1$, hence has no two–cycles and no nonzero fixed points, and Corollary 6.2 concludes that the function–field Erdős–Barbeau equation is insoluble outright. (An independent brute-force recheck, 8192 squarefree monics over $\mathbb{F}_2$ to degree 13 and 6561 over $\mathbb{F}_3$ to degree 8, 0 violations and 0 two–cycles: `code/ff_nocycle.rs`.) What is added here is not that statement but its consequence for *method*, recorded next.

Remark 13.3 (Why the existence route cannot be algebraic)**.** The mechanism of Theorem 6.1 is the ultrametric inequality. The cycle equation is $\sigma(a)\sigma(b) = 1$, so one of the two masses must reach 1; over $K[t]$ the mass satisfies $|\sigma(f)| = \max_i |1/\pi_i| < 1$ by ultrametricity and can never do so, whereas over $\mathbb{Z}$ the archimedean absolute value permits $\sum_{p|n} 1/p > 1$. So #307 is possible at all only because $|\cdot|_\infty$ is not ultrametric.

This has a consequence for method. Any argument for *existence* that is purely algebraic, in the sense that it applies verbatim with $\mathbb{Z}$ replaced by $K[t]$, would prove a statement that Theorem 6.1 refutes; hence no such argument exists. Together with Proposition 8.3 (no polynomial parametrisation) and Proposition 12.15 (no place of $\mathbb{Q}$ carries the descent) this closes the algebraic and geometric toolkit on *both* routes: the negative route has no place to descend at, and the positive route has no place to construct at. What remains for A10 is archimedean and analytic, which is precisely the regime in which lower bounds for primes in sparse sequences are unavailable. That is a sharper statement of the obstruction than Remark 4.16 gives, and it is the reason the present work does not resolve the existence direction.

There is a third closure, and it removes the one concrete attack that survives the other two. The 34 immune families of Remark 4.10 are precisely the tail families the congruence campaign cannot kill, so each is a live candidate and the obvious move is to test them one at a time. That move is not available, and the reason is quantitative rather than a want of effort.

Proposition 13.4 (The immune families cannot be tested)**.** *Let S be one of the 34 immune bases. Deciding whether the tail family $S \cup \{q\}$ contains a Pythagorean pair is, by Remark 4.16, deciding whether $B_S x^2 - A_S y^2 = -4D_S^2$ has an integral solution with $q = (x^2 - D_S)/A_S$ prime. Neither of the two routes to that decision is within reach.*

(i) Global route. *Producing a solution requires the fundamental unit, equivalently the class group, of the real quadratic order of discriminant $4A_S B_S$. On the first immune base that discriminant has 225 digits and is a nonsquare, so the subexponential index-calculus cost $L_\Delta(1/2) = \exp\big(\sqrt{\log\Delta \, \log\log\Delta}\big)$ evaluates to $10^{24.7}$ operations.*

(ii) Parametric route. *Since A_S is prime and $(D_S \mid A_S) = +1$, a square root $x_0^2 \equiv D_S \pmod{A_S}$ exists and the family is genuinely parametrised by $x = x_0 + tA_S$, with $q(t) = \big((x_0 + tA_S)^2 - D_S\big)/A_S$. But a solution additionally needs $B_S q(t) + D_S$ to be a square, and that quantity has 223 digits already at $t = 0$, so the heuristic hit rate is 10^{-112} per trial. Checked at $t = 0, 1, 2, 3$: not a square* (`code/a10_pell_feasibility.gp`)*.*

One hope survives all of this on its face: a lower bound for *some term of some sequence in a family* is formally weaker than one for a single sequence, and Proposition 4.5 supplies 49,961 of them. It fails, and it fails for a reason no enlargement of the family can repair.

Proposition 13.5 (Multiplicity cannot help)**.** *Every known lower-bound method for primes in a set requires that set to have counting function at least X^θ for some fixed $\theta > 0$. The union over all* 49,961 *bases of the terms of their sequences below X has counting function $C \log X$ with $C = 49{,}961/\log\varepsilon$, since each sequence grows like ε^n and so contributes only $O(\log X)$ terms. Hence*

$$\frac{\log(\textit{union below } X)}{\log X} = \frac{\log\bigl(C\log X\bigr)}{\log X} \longrightarrow 0,$$

and the family size C is powerless because it sits inside a logarithm.

Measurement. On an immune base ε has 224 digits, so at $X = 10^{500}$ each sequence contributes 2.2 terms and the whole family contributes $10^{5.05}$, a density exponent of 0.0101. Reaching $\theta = \frac{1}{2}$ at that X would need $10^{249.7}$ bases against the $10^{4.7}$ available, a shortfall of 10^{245}. Even the absolute ceiling on the family at *every* level combined, all $2^{139} = 10^{41.8}$ subsets of the primes below 800, falls short by $10^{207.8}$ (`code/a10_multiplicity.gp`). □

Each of the four blocks was then attacked on its own terms. None fell; each is now located more precisely, and two further routes are closed.

Proposition 13.6 ($\mathbb{Q}$ is optimal for the barrier)**.** *Theorem 6.1 kills the algebraic route by ultrametricity, which suggests replacing $K[t]$ by a number ring $\mathcal{O}_K$, where the absolute value is archimedean and the obstruction disappears. It does disappear, but nothing is gained: for every number field K the least $\prod N(\pi)$ over prime ideals with $\sum 1/N(\pi) \geq 2$ is at least that of $\mathbb{Q}$.*

Proof and measurement. Mass per unit log-norm is $1/(p\log p)$ in every field, but that is not what the barrier measures: doubling the mass carried by each rational prime would let one *stop* at a far smaller prime, and since $\sum 1/p$ diverges only like $\log\log$, halving the required sum from 2 to 1 would cut the primorial from $e^{p_{59}}$ to e^{p_3}. The gain would therefore be enormous if it were attainable. It is not: by Chebotarev, primes split completely with density only $1/d$ in a degree-d Galois field, so the split primes carry mass $d \cdot (1/d)\log\log Y = \log\log Y$, exactly as over $\mathbb{Q}$, while each costs $d\log p$ rather than $\log p$; inert primes contribute $1/p^d$, that is, nothing. Measured: $\mathbb{Q}$ needs 59 primes and $10^{112.9}$, whereas $\mathbb{Q}(i)$ needs 226 prime ideals and $10^{735.1}$ (`code/a10_fieldbarrier.gp`). The hypothetical field in which every rational prime splits completely would need only 6 primes and $10^{3.0}$, which is precisely why Chebotarev's density $1/d$ is the operative fact. □

Remark 13.7 (The three other blocks, attacked)**.** (i) Against Proposition 13.1. The rigidity $\prod Q = a'$ would dissolve if disjointness were an assumption one could relax. It is not: $P \cap Q = \varnothing$ is *derived* in Theorem 2.3 by the Israel mechanism, $r \in P \cap Q$ forcing $v_r(st) = -2 \neq 0$. There is nothing to relax.

(ii) Against Proposition 13.4. Homogenised, $B_S x^2 - A_S y^2 = -4D_S^2$ is the ternary form $B_S x^2 - A_S y^2 + 4D_S^2 z^2$, a *conic*, and conics obey Hasse–Minkowski. Since Remark 4.7 already gives local solubility everywhere, *rational* solubility follows unconditionally, with no class group anywhere in the argument. This sharpens the block rather than removing it: what is missing is not solubility but *integrality*, and extracting integral points from the rational parametrisation returns the fundamental unit. Worse, the constructive step is itself blocked, since Simon's algorithm must factor the determinant $-4A_S B_S D_S^2$, whose cofactor after removing the known primes of D_S is the 224-digit balanced semiprime $A_S B_S$: the same cryptographic wall as Remark 4.10, reached from a new direction.

(iii) Against Proposition 13.5. The template that exploits a family without density is Heath-Brown's: Artin's conjecture holds for all but at most two primes, with no means of saying which. It does not transfer. That argument needs each failure to impose a *detectable* structure, which several failures then cannot share. Here a failing family would be one all of whose q_n are composite, and the only known mechanism for that is a covering system, which Proposition 4.15 proves does not exist. A failure would therefore be structureless, and structureless failures cannot be played against one another.

All of the above bounds the negative direction. The positive direction admits a quantitative shadow that had not been measured: a solution is exactly $r(a) := \sigma(a)\sigma(a') = 1$, so one may ask how close r actually comes to 1. The answer is instructive, and it disposes of the usual way such heuristics are calibrated.

Proposition 13.8 (The near-miss spectrum, and why the heuristic cannot be calibrated)**.** *Over all* 4,385,727 *integers* $a \le 10^7$ *with* a *and* a' *both squarefree, the maximum of* $r(a)$ *is* 0.5535*, attained at* $a =$ 3,972,254*; the frontier advances by decade as* 0.333, 0.345, 0.502, 0.502, 0.536, 0.554*. In particular* $r \ge 1$ *never occurs, the defect* $|a'' - a|$ *is of size* $a^{0.9}$ *or larger in all but three cases, and the smallest relative defect is* 0.4465*. Consequently the constant in the heuristic count of two-cycles,* $f(1)\log X$ *with* f *the density of* r *at* 1*, is* not measurable*: by Theorem* 4.3 *the entire region* $r \ge 1$ *lies above* $10^{112.9}$*, so no computation can sample it* (`code/a10_nearmiss.rs`, `code/a10_density1.rs`)*.*

The situation is not the usual one. Heuristics of Bateman–Horn or Hardy–Littlewood type are normally checked against data, and the agreement is what justifies confidence in them. Here the prediction that solutions exist rests on a constant that no computation can estimate, even in principle, because the barrier and the sampleable range are disjoint. The expected-count heuristic for #307 is therefore not merely unproved but uncalibrated.

The measurement does expose a sharp structural fact, which turns out to be exactly the barrier seen statistically.

Proposition 13.9 (The barrier is statistically complete)**.** $\mathbb{E}[\sigma(a')]$ *falls by an order of magnitude as* $\sigma(a)$ *grows, from* 0.2035 *unconditionally to* 0.0207 *on* $\sigma(a) \ge 1.3$*. The whole of this suppression is coprimality. Writing the prediction for an integer chosen at random subject to* $\gcd(n, a) = 1$*, namely* $\sum_{p \nmid a} p^{-2}$*, the ratio of measured to predicted is*

$$0.65, \quad 0.82, \quad 0.91, \quad 0.92, \quad 0.95, \quad 1.01$$

at $\sigma(a) \ge 0$, 0.5, 0.9, 1.0, 1.2, 1.3*: it converges to* 1 *precisely in the regime a solution would inhabit* (`code/a10_anticorr.rs`)*.*

So a' is mass-poor for one reason only, that it is coprime to a and a has absorbed the small primes, and there is no further effect to exploit. This is an independent confirmation of Proposition 14.34: the barrier is not merely near-optimal as a bound, it already contains the entire statistical structure of the obstruction, and no refinement by correlation is available.

Remark 13.10 (Four independent blocks on A10)**.** The existence route is therefore obstructed at four separate levels, and no two of them are the same obstruction. Proposition 13.1 removes the freedom: the construction carries one parameter and cannot be relaxed into a denser family. Theorem 6.1 removes the method: no argument that survives the passage to $K[t]$ can prove existence, because there the statement is false. Proposition 13.4 removes the computation: even the 34 most reduced objects in the whole problem, the ones every sieve has failed to kill, cannot be examined. Proposition 13.5 removes the last refuge, the hope that a large family of sequences is easier than one: the union stays logarithmically thin however many sequences it has. What is left is a lower bound for primes in a sparse exponential sequence, which exists for no non-degenerate linear recurrence at all. We state plainly that the present work does not decide the existence direction and does not contain a route to deciding it.

14 The square sieve at the mass-two wall

Remark 14.21 left the residue inside $\{\sigma(m) \geq 2 - \varepsilon\}$. We now attack that set directly, by a change of question. The minus layer is $m' - 2m = d^2$, so $\sigma(m) = 2 + d^2/m > 2$: it lives *above* the wall, and the layer is exactly the set on which the arithmetic function $m \mapsto m' - 2m$ takes square values. Atom A9 is therefore an instance of *how often is an arithmetic function a perfect square*, and for that question there is a tool: Heath-Brown's square sieve. It has not previously been applied to the arithmetic derivative, because m' is not a polynomial and no Weil bound is available. The obstruction dissolves because the relevant symbol factorises.

Lemma 14.1 (Factorisation of the symbol)**.** *Let r be odd and let m be squarefree with* $\gcd(m, r) = 1$ *and no prime factor of m dividing r. Put $\sigma_r(m) = \sum_{p|m} p^{-1} \in \mathbb{Z}/r$. Then $m' \equiv m\,\sigma_r(m) \pmod r$ and*

$$\Big(\frac{m' - 2m}{r}\Big) = \Big(\frac{m}{r}\Big)\Big(\frac{\sigma_r(m) - 2}{r}\Big).$$

Proof. $m' = \sum_{p|m} m/p$ and $m/p \equiv m\,p^{-1} \pmod r$ for each $p \mid m$, since p is invertible modulo r. Summing gives $m' \equiv m\sigma_r(m)$, hence $m' - 2m \equiv m(\sigma_r(m) - 2)$, and the Jacobi symbol is completely multiplicative in its numerator. ☐

Verified exhaustively for $r = 101, 1009, 10007$ over all admissible squarefree $m \leq 3 \times 10^7$: 54,514,790 checks, 0 violations (`code/sqsieve_lemmas.rs`).

The point of Lemma 14.1 is that σ_r is *additive* over the distinct prime divisors, so after a Gauss-sum expansion the second factor becomes a product over $p \mid m$ and the whole sum turns multiplicative. That is what puts it inside Halász's theorem.

Lemma 14.2 (Cancellation in the character sums)**.** *Fix an odd squarefree $r > 1$. Then*

$$T_r(N) \;:=\; \sum_{\substack{m \leq N,\ \mu^2(m)=1 \\ \gcd(m,r)=1}} \Big(\frac{m' - 2m}{r}\Big) \;=\; o_r(N) \qquad (N \to \infty).$$

Proof. Take $r = q$ prime; the general squarefree r follows by expanding $(\cdot/r) = \prod_{q|r}(\cdot/q)$ and running the argument in each factor. Let $g = \sum_t (t/q) e_q(t)$, so $|g| = \sqrt{q}$ and $(x/q) = g^{-1} \sum_{t \bmod q} (t/q) e_q(tx)$ for every x. By Lemma 14.1,

$$T_q(N) = g^{-1} \sum_{t \bmod q} \Big(\frac{t}{q}\Big) e_q(-2t) \sum_{m \leq N} h_t(m), \qquad h_t(m) = \Big(\frac{m}{q}\Big) e_q\big(t\,\sigma_q(m)\big),$$

where h_t is extended to all integers by $h_t(p^k) = 0$ for $k \geq 2$ and $h_t(p) = 0$ for $p \mid q$. Since σ_q is additive over distinct primes, h_t is multiplicative with $|h_t| \leq 1$ and $h_t(p) = (p/q)\, e_q(t\,p^{-1})$, a function of $p \bmod q$ alone.

For $t = 0$, $h_0 = (\cdot/q)$ is a non-principal real character and the sum is $o(N)$.

For $t \not\equiv 0$ we apply Halász's theorem in the Montgomery form: with $M = \min_{|\tau| \leq \log N} \sum_{p \leq N} \big(1 - \operatorname{Re} h_t(p) p^{-i\tau}\big)/p$ one has $\sum_{m \leq N} h_t(m) \ll N\big((1 + M)e^{-M} + (\log N)^{-1/2}\big)$. If $|\tau| \geq 1/\log N$ then $\big|\sum_{p \leq N} p^{-1-i\tau}\big| \ll \log\log(3 + |\tau|)$, so $M \geq \log\log N - O(\log\log\log N)$. If $|\tau| < 1/\log N$ then $p^{-i\tau} = 1 + O(1)$ for $p \leq N$ and, by Mertens in progressions,

$$M \;=\; \log\log N\Big(1 - \operatorname{Re} \frac{1}{\varphi(q)} \sum_{u \in (\mathbb{Z}/q)^\times} \Big(\frac{u}{q}\Big) e_q(t\,u^{-1})\Big) + O_q(1).$$

The inner sum is a Salié sum, of modulus at most $2\sqrt{q}$ for $t \not\equiv 0$, so the bracket is at least $1-3/\sqrt{q} \ge \frac{1}{2}$ for $q \ge 36$. Either way $M \gg \log\log N$, whence $\sum_{m\le N} h_t(m) \ll N \log\log N(\log N)^{-1/2} = o(N)$. Summing the q values of t and dividing by $|g| = \sqrt{q}$ gives $T_q(N) \ll \sqrt{q}\cdot o(N) = o_q(N)$. $\square$

Measured at $r = 1009$, $N = 3\times 10^7$: $|\sum_m h_t(m)|/N$ equals 3.6×10^{-6}, 1.3×10^{-3}, 2.1×10^{-3}, 5.7×10^{-4}, 1.5×10^{-3} at $t = 0, 1, 2, 17, 500$, all of size $\sqrt{N}$ rather than merely $o(N)$. The sums $T_r(N)$ themselves were measured at $N = 5\times 10^7$ for $r = 101, 1009, 10007, 100003$ and $r = pq = 101\cdot 1009$, $1009\cdot 10007$, giving $|T_r|/\sqrt{N} = 8.1$, 0.1, 1.2, 1.0, 1.0, 0.2 (`code/sqsieve_test.rs`).

Lemma 14.3 (Local densities at p^2)**.** *For each fixed odd prime p, $\#\{m \le N : \mu^2(m) = 1,\ p^2 \mid (m' - 2m)\} \ll N/p^2$.*

Proof. For $p \nmid m$ the condition is $\sigma_{p^2}(m) \equiv 2 \pmod{p^2}$ by Lemma 14.1 applied with modulus p^2. Detecting it by additive characters, the non-trivial frequencies give multiplicative sums $\prod_{\ell\mid m} e_{p^2}(t/\ell)$ whose values at primes depend only on $\ell \bmod p^2$ and which are not Dirichlet characters, because $u \mapsto e_{p^2}(t/u)$ is not multiplicative; the Halász argument of Lemma 14.2 applies verbatim with the Salié bound replaced by the Weil bound for Kloosterman sums to prime-power modulus. Hence the density is $p^{-2} + o(1)$. The case $p \mid m$ does not arise at all: writing $m = pn$ with $p \nmid n$ gives $m' = pn' + n$, so $m' - 2m = n + p(n' - 2n) \equiv n \not\equiv 0 \pmod{p}$, and p does not even divide $m' - 2m$, let alone p^2. $\square$

Measured ratios of the count to N/p^2 at $N = 3\times 10^7$: 0.842, 0.887, 0.930, 0.925, 0.993 for $p = 5, 7, 11, 13, 101$, approaching 1 as predicted.

Theorem 14.4 (The minus layer has density zero)**.** *$\#\{m \le N :\ m' - 2m$ is a positive perfect square$\} = o(N)$. Equivalently, atom* A9 *holds: the minus layer has density zero. The same argument with $m' + 2m$ reproves Proposition* 12.20*, so the full Pythagorean layer has density zero unconditionally.*

Proof. Write $a_m = m' - 2m$ and fix P, letting $\mathcal{P}$ be the primes in $(P, 2P]$ and $\Pi = |\mathcal{P}| \sim P/\log P$. If a_m is a perfect square coprime to every $p \in \mathcal{P}$ then $(a_m/p) = 1$ for all such p, so $\sum_{p\in\mathcal{P}} (a_m/p) = \Pi$. Hence

$$\Pi^2\,\#\{m \le N : a_m = \square,\ \gcd(a_m, \textstyle\prod\mathcal{P}) = 1\} \ \le\ \sum_{m\le N}\Big(\sum_{p\in\mathcal{P}}\Big(\frac{a_m}{p}\Big)\Big)^2 \ =\ \sum_{p,q\in\mathcal{P}}\ \sum_{m\le N}\Big(\frac{a_m}{pq}\Big).$$

The diagonal $p = q$ contributes $(a_m/p^2) = (a_m/p)^2 \in \{0, 1\}$, hence at most ΠN; the off-diagonal is $\sum_{p\ne q} T_{pq}(N)$, which is $o(N)$ for each of the finitely many pairs by Lemma 14.2. A square divisible by p is divisible by p^2, so the excluded m number at most $\sum_{p\in\mathcal{P}} O(N/p^2) \ll N/(P\log P)$ by Lemma 14.3, using that $p \mid m$ is impossible there. Finally $a_m = 0$ means $m' = 2m$, a single line, of density zero by Proposition 12.24. Altogether

$$\limsup_{N\to\infty} \frac{1}{N}\#\{m \le N : a_m = \square > 0\} \ \ll\ \frac{1}{\Pi} + \frac{1}{P\log P}.$$

The left side does not depend on P; letting $P \to \infty$ gives 0. $\square$

The proof of Theorem 14.4 sends $P \to \infty$ only after N, which costs all quantitative information. Keeping P a function of N turns it into an explicit rate, and what makes that legitimate is the uniformity already established in Lemma 14.2: the pretentious distance there satisfies $M \ge (1 - 3/\sqrt{r}\,)\log\log N \ge \frac{1}{2}\log\log N$ for $r \ge 36$, with no dependence on r beyond that.

Corollary 14.5 (An explicit rate for A9)**.**

$$\#\{m \le N:\ m' - 2m \textit{ is a positive perfect square}\} \ \ll\ \frac{N \log\log N}{(\log N)^{1/4}}.$$

Proof. Take $\mathcal{P}$ to be the primes in $(P, 2P]$ with $P = P(N)$ to be chosen, so $\Pi \sim P/\log P$. The three terms of Theorem 14.4 are then

$$\frac{N}{\Pi} \ \ll\ \frac{N \log P}{P}, \qquad \frac{1}{\Pi^2} \sum_{p \ne q} |T_{pq}(N)| \ \le\ \max_{p \ne q} |T_{pq}(N)|, \qquad \sum_{p \in \mathcal{P}} O\Big(\frac{N}{p^2}\Big) \ \ll\ \frac{N}{P \log P}.$$

By Lemma 14.2 with its uniform M, Halász gives $T_r(N) \ll \sqrt{r}\, N(\log N)^{-1/2}$, and $r = pq \le 4P^2$ gives $\sqrt{r} \le 2P$, so the middle term is $\ll PN(\log N)^{-1/2}$. Balancing it against the first, $N \log P/P = PN(\log N)^{-1/2}$, gives $P \asymp (\log N)^{1/4}$, at which both equal $N \log\log N(\log N)^{-1/4}$ up to constants; the third term is then $O\big(N(\log N)^{-1/4}/\log\log N\big)$ and is absorbed. $\square$

Remark 14.6 (Where the exponent comes from)**.** The exponent $\frac{1}{4}$ is not intrinsic to the arithmetic. It is what remains after balancing against the floor $N(\log N)^{-1/2}$ in the Montgomery form of Halász's theorem, which holds however large the pretentious distance M becomes. Any improvement of that floor transfers here without change: a floor of $N(\log N)^{-\alpha}$ yields the exponent $\alpha/2$. The measured character sums are of size $\sqrt{N}$ rather than $N(\log N)^{-1/2}$ (Section 14), so the truth is far stronger than what the general theorem can deliver, and the gap is entirely in the analytic input rather than in the sieve.

The exponent $\frac{1}{4}$ is an artefact of the general theorem, not of the arithmetic, and the gap can be closed off exactly. Everything in the sieve except the character-sum input is elementary, so the reachable exponent is a function of one measurable quantity: how $T_r(N)$ grows with r.

Hypothesis 14.7 (Square-root cancellation, uniform in the modulus)**.** There is an absolute $A \ge 0$ with $|T_r(N)| \ll r^A N^{1/2+o(1)}$ for every odd squarefree r.

Theorem 14.8 (Conditional power saving, and the crux exactly)**.** *Assume Hypothesis* 14.7*. Then*

$$\#\{m \le N:\ m' - 2m \textit{ is a positive perfect square}\} \ \ll\ N^{1-\delta}, \qquad \delta = \frac{1}{2(2A+1)}.$$

In particular $A = 0$ gives $N^{1/2+o(1)}$, which is exactly the bound CRUX-A9 *asks for.*

Proof. Run the sieve of Theorem 14.4 with $\mathcal{P}$ the primes in $(P, 2P]$, $\Pi \sim P/\log P$, and dispose of the m with $p \mid a_m$ by the divisor bound rather than Lemma 14.3: a perfect square a_m has at most $\log(2N)/\log P$ prime factors in $\mathcal{P}$, so $\sum_{p \in \mathcal{P}} (a_m/p) \ge \Pi/2$ once $\Pi \ge 2\log(2N)/\log P$. Then

$$\#\{\,\cdot\,\} \ \ll\ \frac{N}{\Pi} + \frac{1}{\Pi^2} \sum_{p \ne q} |T_{pq}(N)| \ \ll\ \frac{N \log P}{P} + P^{2A} N^{1/2+o(1)},$$

since $pq \le 4P^2$. Balancing the two terms gives $P = N^{1/(2(2A+1))}$ and the stated exponent; the side condition $\Pi \ge 2\log(2N)/\log P$ holds comfortably at that P. For $A = 0$ take $P = N^{1/2}$, whence both terms are $N^{1/2+o(1)}$. $\square$

Remark 14.9 (The hypothesis is measured, and A reads as zero)**.** Hypothesis 14.7 is not a wish. At $N = 4 \times 10^7$, so $\sqrt{N} = 6.325 \times 10^3$, the ratio $|T_r(N)|/\sqrt{N}$ across twelve moduli spanning ten orders of magnitude is

r prime :	101	1009	10007	100003	$10^6{+}3$	$10^7{+}19$
$\lvert T_r\rvert/\sqrt{N}$:	7.38	0.17	0.84	1.07	0.05	0.15
$r = pq$:	101909	10097063	1000730021	100003300009		
$\lvert T_r\rvert/\sqrt{N}$:	0.95	0.21	1.30	0.64		

together with 2.34 and 1.00 at the squarefree 1155 and 1113121. The implied exponent $A = \log(|T_r|/\sqrt{N})/\log r$ is $+0.006$, -0.004, $+0.013$, -0.018 and -0.000 at the five largest moduli; the outlying $+0.433$ occurs at $r = 101$, where $\log r$ is too small for the ratio to be meaningful. Note that the last two moduli *exceed* N, and the sums are still of size $\sqrt{N}$ (`code/sqsieve_uniform.rs`).

So the sharp form of A9 is not blocked by the sieve, which is elementary, nor by the arithmetic, which cooperates: it is blocked by the absence of a Weil-type bound for a character sum whose argument is not a polynomial. That is a single, isolated, and precisely stated analytic deficit, and Lemma 14.1 has already reduced it to a sum over a multiplicative function of σ_r.

The obvious route to Hypothesis 14.7 is the one already used in Lemma 14.2: expand by Gauss sums and bound each multiplicative sum $\sum_{m\le N} h_t(m)$ separately, where square-root cancellation would follow from a Weil-type input or, in the analytic form, from GRH applied to $\prod_\chi L(s,\chi)^{\widehat{\alpha}(\chi)}$. That route is closed. The reason is not the one a first look suggests.

Proposition 14.10 (The decomposition destroys the cancellation)**.** *Write $\alpha(u) = (u/r)e_r(tu^{-1})$, so that $h_t(p) = \alpha(p \bmod r)$. The principal Fourier coefficient $\widehat{\alpha}(\chi_0) = \varphi(r)^{-1}\sum_u \alpha(u)$ is a Salié sum; measured, it has modulus exactly $\sqrt{r}$ before normalisation, at $r = 1009$ and $r = 10007$ giving* 31.8 *and* 100.0 *against $\sqrt{r} = 31.76$ and* 100.03*. It is therefore nonzero, and $\prod_\chi L(s,\chi)^{\widehat{\alpha}(\chi)}$ retains a ζ-type branch point at $s = 1$. This does* not *by itself decide the matter, since the measured sums fall* 50 *to* 350 *times below what that branch point predicts. What decides it is the growth in N: at $r = 10007$, $t = 1$,*

N :	10^6	3×10^6	10^7	3×10^7	6×10^7
$\lvert\sum h_t\rvert/\sqrt{N}$:	0.67	0.48	0.69	2.08	2.29

a fitted exponent of 0.800 *over that range, with $|\sum h_t|/(N/\log N)$ flat near* 0.005*. The individual multiplicative sums thus grow strictly faster than $\sqrt{N}$* (`code/sqsieve_route.rs`)*.*

Remark 14.11 (Where the cancellation actually lives)**.** Since $T_r(N)$ itself measures at $\sqrt{N}$ across ten orders of magnitude in r (Remark 14.9) while each $\sum_m h_t(m)$ measured here grows like $N^{0.8}$, the cancellation in T_r is not present frequency by frequency: it lies in the sum over t. The magnitude is not marginal. At $r = 1009$, $N = 4 \times 10^7$, bounding each frequency separately and summing trivially gives $\sqrt{r}\,N/\log N \approx 7 \times 10^7$, against a measured $|T_r| = 1050$: a factor of about 7×10^4 is lost the moment the Gauss expansion is taken.

The deficit behind Hypothesis 14.7 is therefore not a missing Weil bound for a single character sum, which is how Remark 14.9 first framed it. It is that the only available decomposition of T_r scatters the cancellation across frequencies, and no current technique grips cancellation in that position. Stating it this way is more useful than the Weil framing, because it says what would have to be new: an estimate for T_r that never passes through the multiplicative decomposition at all.

Remark 14.11 asked for an estimate for T_r that never passes through the multiplicative decomposition. The following is the mechanism such an estimate would use, isolated and proved; the reduction of T_r to it is *not* established here, and we say precisely what is missing.

Proposition 14.12 (Kernel norm). *Since σ_r is additive, for coprime d, e Lemma 14.1 gives*

$$\Big(\frac{(de)' - 2de}{r}\Big) = \Big(\frac{d}{r}\Big)\Big(\frac{e}{r}\Big)\, K\big(\sigma_r(d), \sigma_r(e)\big), \qquad K(u,v) = \Big(\frac{u+v-2}{r}\Big).$$

The matrix K is Toeplitz in $u+v$, so $K(u,v) = \sum_\xi \widehat{f}(\xi) e_r(\xi u) e_r(\xi v)$ diagonalises it against the orthogonal system $(e_r(\xi u))_u$ of norm $\sqrt{r}$. As $\widehat{f}(\xi)$ is a normalised Gauss sum, $|\widehat{f}(\xi)| = r^{-1/2}$ for $\xi \neq 0$ and $\widehat{f}(0) = 0$, so every singular value of K equals $\sqrt{r}$ and

$$\|K\|_{\mathrm{op}} = \sqrt{r}.$$

Consequently any *bilinear form $\sum_{d\sim D}\sum_{e\sim E} \alpha_d \beta_e\, K(\sigma_r(d), \sigma_r(e))$ is at most $\sqrt{r}\,\|A\|_2\|B\|_2$, where $A(u) = \sum_{d\sim D,\ \sigma_r(d)\equiv u} \alpha_d$ and B is defined likewise. No multiplicative decomposition is used; the only input is the modulus of a Gauss sum.*

Remark 14.13 (What is missing, stated exactly). Proposition 14.12 bounds bilinear forms, not T_r. Passing from one to the other requires a decomposition of $\sum_{m\le N}$ into bilinear pieces, and two obstacles are real rather than cosmetic.

- (i) Summing over coprime pairs (d, e) with $de = m$ weights each m by $2^{\omega(m)}$: at $N = 2\times 10^6$ the coprime-pair count is 10,014,190 against 1,215,876 squarefree m, a factor 8.24. Since the summand is *signed*, no inequality relates $\sum_m F(m)$ to $\sum_m 2^{\omega(m)} F(m)$ in either direction.
- (ii) A single dyadic block $d \sim D$ sees only a minority of m: for $D = \sqrt{N}$ at $N = 2\times 10^6$, 24.7% (`code/sqsieve_bilinear.rs`).

The natural repair is the largest-prime-factor split $m = pn$ with $p = P^+(m)$, which is unique and so avoids (i); the A-side is then supported on primes, where $\|A\|_2^2 \ll \pi(D)^2/\varphi(r) + \pi(D)$ follows from Brun–Titchmarsh uniformly in r, and the B-side is governed by Theorem 14.16. What remains is the coupling $P^+(n) < p$ and the dyadic bookkeeping, a Type I/II analysis that we have not carried out. Until it is, the power saving below is **CONJECTURAL**, and Corollary 14.5 remains the best unconditional rate proved here.

Conjecture 14.14 (Power saving, pending the decomposition). *If the decomposition of Remark* 14.13 *is carried out with the stated bounds, then combining Proposition 14.12 with Theorem 14.16 and Theorem 14.8 at $P = N^{1/4}$ gives*

$$\#\{m \le N :\ m' - 2m \text{ is a positive perfect square}\} \;\ll\; N^{3/4+o(1)}.$$

The two ingredients that would feed it are themselves unconditional, and are recorded separately because they do not depend on the missing step.

Lemma 14.15 (σ is injective on squarefree integers). *For squarefree d, d', $\sigma(d) = \sigma(d')$ forces $d = d'$. Consequently $F(d,d') := D(d)\,d' - D(d')\,d = dd'(\sigma(d) - \sigma(d'))$ vanishes only on the diagonal.*

Proof. By Theorem 2.3, $\sigma(d) = D(d)/d$ is already in lowest terms, since $\gcd(D(d), d) = 1$ for squarefree d; so d is recoverable from $\sigma(d)$ as its denominator. □

Verified over all 182,377 squarefree $d \le 3\times 10^5$: 0 collisions (`code/sqsieve_pairavg.rs`).

Theorem 14.16 (The average pair bound). *Let $\mathcal{R} = \{pq : p \neq q \in \mathcal{P}\}$ and $P(D,r) = \#\{(d,d') \sim D : \sigma_r(d) \equiv \sigma_r(d')\}$. Then*

$$\sum_{r\in\mathcal{R}} P(D,r) \;\ll\; |\mathcal{R}|\, D + D^{2+o(1)}.$$

Proof. As $\gcd(d,r) = \gcd(d',r) = 1$, the condition $\sigma_r(d) \equiv \sigma_r(d')$ is exactly $r \mid F(d,d')$, so $\sum_{r\in\mathcal{R}} P(D,r) = \sum_{(d,d')} \#\{r \in \mathcal{R} : r \mid F(d,d')\}$. On the diagonal $F = 0$ is divisible by every r, contributing $|\mathcal{R}| \cdot \#\{d \le D : \mu^2(d) = 1\}$. Off it $F \ne 0$ by Lemma 14.15, and $|F| \le 2D^2\sigma_{\max}$, so $\#\{r \in \mathcal{R} : r \mid F\} \le \tau(F) \ll D^{o(1)}$, contributing $D^{2+o(1)}$. □

Measured against $D^2/r+D$ the average has ratio 1.011 at $|\mathcal{R}| = 78$ (`code/sqsieve_pairavg.rs`). Note the proof uses no equidistribution of σ_r and no uniformity in r: only the divisor bound and Lemma 14.15, the latter being the same rigidity that makes #307 a two-cycle problem at all.

Remark 14.17 (The hypothesis measures, and its constant identifies itself)**.** $P(D,r)$ against $D^2/r + D$ has ratio 0.3688, 0.3688, 0.3689 at $r = 1009$; 0.3705, 0.3696, 0.3695 at $r = 10007$; and 0.3867, 0.3727, 0.3702 at $r = 100003$, for $D = 10^6, 5 \times 10^6, 2 \times 10^7$ (`code/sqsieve_bilinear.rs`). The ratio is not merely bounded but constant, and the constant is $(6/\pi^2)^2 = 0.3696$, the square of the squarefree density, which is exactly what equidistribution of σ_r predicts once both d and d' are restricted to squarefree integers.

Compare Hypothesis 14.7. That one asks for signed square-root cancellation, and Proposition 14.10 showed the only available route to it fails. Theorem 14.16 instead bounds an *unsigned* second moment, and does so unconditionally; what it does not do is bound T_r, since the passage from bilinear forms to T_r is exactly the gap recorded in Remark 14.13.

Remark 14.18 (What made this work)**.** Every previous attack on A9 counted members of the minus layer directly, and each ran into the same obstruction: the layer is a union of lines $m' = 2m-k$, each thin by Proposition 12.24 but not summably so, and the peeling reorganisations that would combine them are all conjugate under the groupoid of Proposition 12.26. The square sieve never counts a member. It asks only whether $m' - 2m$ is a square, and answers by averaging a Jacobi symbol, so the groupoid has nothing to act on. The one obstacle, that m' is not a polynomial, is removed by Lemma 14.1: the symbol splits into a character in m times a character in $\sigma_r(m)$, and σ_r is additive, which is exactly the hypothesis Halász needs. The analytic inputs are all standard: Halász's theorem, the Salié and Weil bounds, and Mertens in progressions. Note also that the theorem does not decide #307, and cannot: it is a statement about density, and a single two-cycle is a density-zero event.

Proposition 14.19 (Slot stratification of the divisibility set)**.** *Fix c and let $D(c) = \{m$ squarefree : $(2m - m') \mid (m - c),\ m \ne c\}$. For $m \in D(c)$ put $p = (m - c)/(2m - m')$, the* slot. *Then*

$$p\,m' \;=\; (2p-1)\,m + c, \qquad \textit{equivalently} \qquad \sigma(m) = 2 - \frac{m-c}{p\,m},$$

so $D(c) = \bigsqcup_{p\ne0} L(p, 2p-1, c)$ is a disjoint union of generalized lines in the sense of Proposition 12.26*, the stratum $p = 1$ being the slope*-one *line $m' = m + c$. The identity was verified with* 0 *violations on all* 2,294 *members with $m \le 2 \times 10^7$, $|c| \le 400$* (`code/slot_stratify.rs`)*.*

Proof. $m - c = p(2m - m')$ rearranges to $pm' = (2p - 1)m + c$; dividing by pm gives the mass form. Distinct p give disjoint strata since p is a function of m. □

Theorem 14.20 (The residual difficulty sits entirely at the mass-two wall)**.** *For every $\varepsilon > 0$ and every c,*

$$\#\big(D(c) \cap [1, Z]\big) \;\le\; \big(\delta(2-\varepsilon) + o_\varepsilon(1)\big)\, Z + O(|c|),$$

where $\delta(x)$ is the density of $\{m : \sigma(m) \ge x\}$. Moreover

$$\delta(x) \;\le\; \inf_{t>0}\; e^{-tx} \prod_p \Big(1 + \frac{e^{t/p} - 1}{p}\Big),$$

which evaluates to $\delta(1) \le 10^{-0.8}$*,* $\delta(1.5) \le 10^{-7.2}$ *and* $\delta(2) \le 10^{-105.6}$ (`code/chernoff_mass.rs`)*.*

Proof. Split $D(c)$ at $|p| \le P$ and $|p| > P$. The first part is a union of at most $2P$ generalized lines, each of size $\ll Z/(\log Z)^{(1+o(1))\log_3 Z}$ by Proposition 12.24; taking $P = P(Z) \to \infty$ slowly makes the total $o(Z)$. For the second, $|p| > P$ forces $|2 - \sigma(m)| < (1 + |c|/m)/P$ by the mass form, so for $m > |c|$ we get $\sigma(m) > 2 - 2/P$, and the count is at most $\delta(2 - 2/P)Z(1 + o(1))$. For the Chernoff bound, $f(m) = \prod_{p|m} e^{t/p}$ is multiplicative with $f(p) \ge 1$, so $\sum_{m\le Z} f(m) = \sum_{d\le Z} \mu^2(d) \prod_{p|d}(f(p) - 1)\lfloor Z/d\rfloor \le Z \prod_p(1 + (f(p) - 1)/p)$, and $\#\{\sigma \ge x\} \le e^{-tx} \sum_{m\le Z} f(m)$. The product converges because $(e^{t/p} - 1)/p \sim t/p^2$. $\square$

Remark 14.21 (What this settles and what it does not)**.** Theorem 14.20 replaces the constant $\delta = 0.0420$ of Proposition 14.22 (ii) by one below 10^{-105}, an unconditional gain of more than a hundred orders of magnitude, and it does so by *using* the divisibility rather than merely the mass condition it forces. It also localises the residue exactly: away from the mass-two wall the set is a bounded union of thin lines, and *all* the remaining difficulty lies in $\{\sigma(m) \ge 2 - \varepsilon\}$. It does not prove A9. The set $\{\sigma \ge 2\}$ has *positive* density, at least $1/\Pi_{59} = 10^{-112.9}$ since every multiple of Π_{59} lies in it, so the method cannot reach $o(Z)$: the limiting constant is positive, not zero. The bound $10^{-105.6}$ sits within seven orders of that truth, so little is lost in the Chernoff step, and the obstruction is genuine rather than an artefact of the estimate. The residual question is therefore sharp: is $D(c)$ thin *within* the mass-two wall, the one place where the barrier (Theorem 4.3) also lives?

Proposition 14.22 (The divisibility set: slope one is dense, slope two is not)**.** *Let* $e \ge 1$ *and* c *be integers and let* m *be squarefree with* $(em - m') \mid (m - c)$ *and* $m \ne c$*. Then*

$$\sigma(m) \;\ge\; e - 1 - \frac{|c|}{m}.$$

Consequently the two slopes behave oppositely.

(i) *For* $e = 1$ *the condition is vacuous and the set has positive density: every prime* m *satisfies it when* $c = 1$*, since then* $m' = 1$ *and* $em - m' = m - 1 = m - c$*. Over* $m \le 2 \times 10^7$ *this accounts for* 1,270,651 *of the members, exactly* $\pi(2 \times 10^7)$*.*

(ii) *For* $e = 2$ *the condition reads* $\sigma(m) \ge 1 - |c|/m$*, so by Proposition 5.6*

$$\#\{m \le Z :\; (2m - m') \mid (m - c),\; m \ne c\} \;\le\; (\delta + o(1))\, Z + O(|c|),$$

with $\delta = 0.0420\ldots$ *the abundance density, certified in Proposition 5.6 to lie in* $[0.04202, 0.04223]$*.*

Proof. If $|em - m'| > |m - c|$ the divisibility forces $m = c$. So $|em - m'| \le |m - c| \le m + |c|$; below the barrier $\sigma(m) < e$ for $e \ge 2$, so $em - m' = m(e - \sigma(m)) > 0$ and $m(e - \sigma(m)) \le m + |c|$, which is the display. Part (i) is the computation just given, and (ii) is Proposition 5.6 applied to $\{\sigma \ge 1 - \varepsilon\}$. The mass condition was checked with 0 violations over all squarefree $m \le 2 \times 10^7$ and all $|c| \le 400$ at $e = 1, 2, 3$, where the nontrivial member counts are 2,197,230, 28 and 8 respectively (`code/a9prime_falsify.rs`). $\square$

Remark 14.23 (Consequence for the residual question)**.** Proposition 14.22 settles the shape of the residual bound. A uniform statement over all slopes is false, and false for a trivial reason: at $e = 1$ the primes sit on the set by an identity. The correct target is confined to $e \ge 2$, where a genuine mass condition appears, and there Proposition 14.22(ii) gives the first unconditional bound, δZ with $\delta < 0.043$. The gap to the $Z^{1/2}$ needed for Remark 12.33 is therefore now a gap between two *proved* quantities rather than between a proof and a wish, and the observed counts at $e = 2$ (28 over the whole range $|c| \le 400$, against $Z^{1/2} = 4472$) sit far below both.

Remark 14.24 (The two open routes are one question, asked in opposite directions)**.** The existence route of Remark 4.16 and the counting route of Remark 12.33 look unrelated (one is a search for a single solution, the other a density estimate) but they are the same question about the same kind of object. In each case an index determines a single integer, and everything turns on whether that integer is prime: in Remark 4.16 the Pell index n determines $q_n = (x_n^2 - D_S)/A_S$, a term of a ternary linear recurrence, and #307 needs it prime *once*; in Remark 12.33 the cofactor m determines the slot $(c - am)/(m' - em)$ of Proposition 12.26, and the density needs it prime *rarely*. Existence asks for a lower bound on the number of prime values, density for an upper bound, on determined values along a thin deterministic sequence, and neither estimate is available: no unconditional infinitude of prime terms is known for any non–degenerate linear recurrence of order ≥ 2, and no worst–case upper bound is known for single–class prime counts summed over all moduli. #307 sits exactly between two missing estimates on one object, which is why the heuristics can be read either way and why progress on either side would be informative about the other. The remaining walls are, in this sense, one wall seen from two directions.

Corollary 14.25 (Quadratic–residue filter for #307 solutions)**.** *Let (P, Q) solve Erdős #307 and set $a = D_P$, $b = D_Q$, $N = ab$. Since $N' = a^2 + b^2$ and hence $N' + 2N = (a+b)^2$ automatically, every prime $\ell \in P \cup Q$ satisfies $(N/\ell \mid \ell) = +1$: the cofactor N/ℓ is a nonzero quadratic residue modulo ℓ. Indeed ℓ divides exactly one of a, b (by $P \cap Q = \varnothing$), so $\ell \nmid a + b$ and $(a+b)^2 \not\equiv 0 \pmod{\ell}$; combined with the identity $N' + 2N \equiv N/\ell \pmod{\ell}$ of* (v)*, this forces N/ℓ to be a nonzero square mod ℓ. As a search filter the condition prunes candidate unions by a factor $\leq 2^{-|P\cup Q|} \leq 2^{-59}$ (heuristically each of the ≥ 59 primes imposes an independent QR constraint); for genuine cycles it is automatic and does not alter the existence heuristics. We note that Corollary 14.25 is* not *independent of Bado: it is the squared form of his congruence $D_Q \equiv D_P/\ell \pmod{\ell}$ (itself N_P reduced mod ℓ by cofactor survival), giving $N/\ell \equiv D_Q^2 \pmod{\ell}$. Squaring is lossy ($r$ and $-r$ share a square) so Bado's linear congruence is strictly stronger, pinning $D_Q \bmod \ell$ exactly where the residue filter determines it only up to sign; the filter's one added value is being* split–free *(the residue $N/\ell \bmod \ell$ is computable from the candidate union alone, with no knowledge of the $P \mid Q$ partition). The general law of Proposition 12.19(v), by contrast, holds for* any *squarefree plus–hit with no $P \mid Q$ split and is not subsumed by Bado.*

The plus– and minus–hits both organise into one–prime "slot" families, which makes the structure of Proposition 12.17 explicit at every ω.

Proposition 14.26 (Slot calculus for the discriminant layers)**.** *Let N_0 be squarefree, $r \nmid N_0$ a prime, $N = N_0 r$, and $E_0 := N_0' + 2N_0$. By Leibniz ($N' = N_0' r + N_0$),*

$$N' + 2N = rE_0 + N_0, \qquad N' - 2N = r(N_0' - 2N_0) + N_0,$$

so N is a plus–hit iff $r = (s^2 - N_0)/E_0$ for some integer s (equivalently $s^2 \equiv N_0 \pmod{E_0}$).

(a) Every ω–stratum. *A plus–hit N ($\omega \geq 2$) factors, through any prime divisor r, as $N = N_0 r$ with $N_0 = N/r$ squarefree, and lies in the quadratic family $\{N_0(s^2 - N_0)/E_0\}_s$ of its base; generalising the $p = 5$ family of Proposition 12.19(iii) beyond semiprimes (each step adds one prime, $\omega(N) = \omega(N_0) + 1$). The congruence $s^2 \equiv N_0 \pmod{E_0}$ is solvable for* 669 *of the* 3041 *squarefree $N_0 \in [2, 5000]$ (*22.0%*); solvability is necessary but not sufficient for the family to contain a hit.*

(b) Recursive towers: a trichotomy. *When the base is itself a plus–hit, $E_0 = s_0^2$ is a square; below 2×10^5 the* 114 *plus–hit bases split into three kinds.* No residue *(*84*): $s^2 \equiv N_0 \pmod{E_0}$ is* *unsolvable and the family is empty; a quadratic–residue obstruction, occurring for both even* E_0 *(*$N_0 = 305$*, with* 305 *a non–residue mod* 13*) and odd* E_0 *(*$N_0 = 1870$*,* $E_0 = 73^2$*).* Fixed divisor *(*13*, each with fixed divisor* 2*): the family is nonempty but every slot is even, so only* $r = 2$ *could serve and never gives a square; sterile (e.g.* $N_0 = 145$*,* $E_0 = 18^2$*).* Live *(*17*): fixed divisor* 1 *and a prime slot is exhibited (e.g.* $N_0 = 130$*, first prime slot* $r = 83399$*, giving the plus–hit* $N = 10{,}841{,}870$ *with* $N' + 2N = 5487^2$*).*

(c) Minus layer; the barrier in slot form. *For* $\sigma(N_0) < 2$ *the minus identity gives* $N' - 2N \geq 0$ *iff* $r \leq 1/(2 - \sigma(N_0))$*, i.e. iff* $\sigma(N) \geq 2$*: this is the slot–variable form of Corollary 12.3, not an independent proof of it. The bound* $1/(2 - \sigma(N_0))$ *is unbounded as* $\sigma(N_0) \to 2^-$*; over squarefree* $N_0 < 10^5$ *its maximum is only* 1.524 *(at* $N_0 = 30030$*), so no such base admits a prime minus–slot, yet for the* 58*–prime primorial the bound exceeds* 793 *and the next prime* 277 *fits, producing the* 59*–prime primorial, the smallest minus–hit base. A prime minus–slot thus first becomes feasible* exactly *at the* 59*–prime barrier; the minus–empty–below–barrier statement rests on the primorial growth of Corollary 12.3, not on the figure* 1.524.

(The two identities are checked symbolically and on 2×10^4 random pairs; the base trichotomy, slot lists, and the 1.524 maximum by exhaustive enumeration.)

The two square conditions over a base combine into a single binary–form representation, which places the obstruction inside Gauss's genus theory.

Proposition 14.27 (The Pythagorean condition as a binary–form representation)**.** *For squarefree* N_0 *and prime* $r \nmid N_0$*,* $N = N_0 r$*, the identity*

$$(2N_0 - N_0')(N' + 2N) + (2N_0 + N_0')(N' - 2N) = 4N_0^2$$

holds unconditionally (Leibniz; checked on 2×10^4 *random pairs). Hence if* N *is a Pythagorean pair product over* N_0 *(both* $N' + 2N = s^2$ *and* $N' - 2N = d^2$ *squares) then* (s, d) *represents* $4N_0^2$ *by the binary form* $Q_{N_0}(s, d) = A\,s^2 + B\,d^2$ *with* $A = 2N_0 - N_0' = N_0(2 - \sigma(N_0))$*,* $B = 2N_0 + N_0' = N_0(2 + \sigma(N_0))$*, of discriminant* $-4AB = 4(N_0'^2 - 4N_0^2)$*; conversely such a representation yields a Pythagorean pair over* N_0 *exactly when the recovered slot* $r = (s^2 - N_0)/E_0$ *is a positive integer, prime, and coprime to* N_0 *(the integrality* $s^2 \equiv N_0 \pmod{E_0}$ *holds on every representation we found,* $N_0 < 2 \times 10^4$*). For* $\sigma(N_0) < 2$ *the form is positive–definite (*$A > 0$*), and below the barrier representations are rare and all "ghosts": the only squarefree* $N_0 \leq 3000$ *representing* $4N_0^2$ *are* $N_0 = 30$ *(with* $(s, d) = (11, 1)$*) and* $N_0 = 1722$ *(*$(83, 1)$*), each carrying the non–admissible slot* $r = 1$*, and no live prime slot occurs. As* $\sigma(N_0) \to 2^-$ *the leading coefficient* $A = N_0(2 - \sigma(N_0)) \to 0$ *and the ellipse* $A\,s^2 + B\,d^2 = 4N_0^2$ *degenerates to an arbitrarily thin sliver: the geometric face of the mass–2 barrier. Consequently #307 is the existence of a base with* $\sigma(N_0) \geq 2 - 1/r$ *whose form* Q_{N_0} *represents* $4N_0^2$ *with an admissible, integral,* prime *slot and deviation* $k = 0$.

(We make no class–number claim: each base yields a form of a *different* discriminant, and the question is single–value representation, not a class count; only the degeneration $A \to 0$ is used.)

Proposition 14.28 (An explicit Pythagorean family, and why it stops at rung -1)**.** *Let* N_0 *be squarefree with* $N_0' = 2N_0 - 1$ *(the slope–two line* $c = -1$*, i.e.* $k_0 = 1$*) and suppose* $N_0 - 1$ *is prime. Then* $(a, b) = (N_0 - 1,\ N_0)$ *is a Pythagorean pair:*

$$a^2 + b^2 \;=\; 2N_0^2 - 2N_0 + 1 \;=\; (ab)'.$$

It arises from the form equation of Proposition 14.27 at its degenerate end: with $k_0 = 1$*,* $E_0 = 4N_0 - 1$*, the pair* $(s, d) = (2N_0 - 1,\ 1)$ *represents* $4N_0^2$ *exactly, and the slot it yields is* $r = (s^2 - N_0)/E_0 = N_0 - 1$*. Every member of the family sits on rung* $k = -1$*, and no member can sit on rung* 0*:* a *is prime, so* $a' = 1$*, and* $a' = b$ *would force* $N_0 = 1$*.*

Proof. $(ab)' = N_0'(N_0-1)+N_0(N_0-1)' = (2N_0-1)(N_0-1)+N_0 = 2N_0^2-2N_0+1 = (N_0-1)^2+N_0^2$, using $(N_0-1)'=1$ for N_0-1 prime. The form identity is $(2N_0-1)^2+(4N_0-1) = 4N_0^2$, and $(s^2-N_0)/E_0 = (4N_0^2-5N_0+1)/(4N_0-1) = N_0-1$. The rung is $k = (a'-b)/a = (1-N_0)/(N_0-1) = -1$. □

Remark 14.29 (What the family shows)**.** This is the first explicit construction of Pythagorean pairs in this note: everything earlier established that they are rare (Corollary 12.3: none below 10^{112}) without exhibiting a mechanism. The mechanism is a clean reduction (the pair exists as soon as the line $n' = 2n-1$ carries a member whose predecessor is prime) and it is Bunyakovsky–shaped, matching the observation in Remark 4.16 that a single square condition is of that class. It is also conditional in a way the barrier controls: $\sigma(N_0) = 2-1/N_0$ forces $\omega(N_0) \geq 58$, so the family is empty below the scale where the critical set of Remark 12.33 switches on.

Two further consequences. First, combining with Corollary 12.9: for odd N_0 the pair has mixed parity with odd member N_0, so $k \equiv \omega(N_0) \pmod 2$; since $k=-1$ here, *any odd member of the line* $n' = 2n-1$ *whose predecessor is prime has* $\omega(N_0)$ *odd*; a necessary condition on that line, derived rather than observed. Second, and this is the honest limitation, the construction is *permanently* off the cycle rung: primality of a is exactly what makes it work and exactly what forces $a' = 1 \neq b$. A $k=0$ analogue would need a composite with $a' = b$, which is the original problem. So the family exhibits the Pythagorean locus without approaching #307; useful as the first worked mechanism, not as a route.

Proposition 14.30 (The composite mechanism: $|P|=2$ is one line and one primality)**.** *Write* $a = 2m$ *with* m *odd squarefree, so that* $b = a' = m+2m'$ *(Theorem 4.17). The two–cycle condition is then the single equation*

$$b' \;=\; 2m, \qquad b = m+2m',$$

in the one unknown m*. For* $m=q$ *prime it reads* $b = q+2$ *and*

$$\boxed{b' = 2b-4} \qquad \textit{with } b-2 \textit{ prime},$$

and any such b *is a solution:* $a = 2q$*,* $b = q+2$ *satisfy* $a' = b$*,* $b' = a$ *and* $\sigma(a)\sigma(b) = \left(\frac{1}{2}+\frac{1}{q}\right)\left(2-\frac{4}{q+2}\right) = 1$ *identically. Hence*

$$\#307 \textit{ with } |P|=2 \iff \exists\, b \textit{ squarefree}:\; b' = 2b-4 \textit{ and } b-2 \textit{ prime}.$$

Proof. $(2m)' = m+2m'$ by Leibniz, so $a' = b$ by definition of b, and the cycle needs only $b' = a = 2m$. For $m=q$ prime, $b = q+2$ and $b' = 2q = 2(b-2) = 2b-4$; the mass identity is the displayed product. □

Remark 14.31 (Two lines, two rungs)**.** Comparing with Proposition 14.28 isolates exactly what separates the cycle from its neighbour. Both are "a member of a slope–two line whose shifted predecessor is prime":

$$\begin{array}{lll} \text{line } n' = 2n-1, & N_0-1 \text{ prime} & \Longrightarrow \text{ Pythagorean pair, rung } k=-1; \\ \text{line } n' = 2n-4, & b-2 \text{ prime} & \Longrightarrow \text{ genuine two–cycle, rung } k=0. \end{array}$$

The difference between the open problem and its solved neighbour is the constant in the line. And the obstruction changes character with it: Proposition 14.28 is barred from rung 0 *structurally* (its a is prime, so $a' = 1 \neq b$) whereas Proposition 14.30 is barred only by *scale*, since $\sigma(b) = 2-4/b$ forces $\omega(b) \geq 58$ and $b \geq \Pi_{58}$, putting the whole family above 10^{108} and the resulting $N = 2qb$ at the barrier. A direct search of the reduction over odd squarefree $m \leq 2\times10^6$ (810,590 admissible m, of which 111,464 prime) finds none, with nearest miss $|b' - 2m| = 5$ at $m = 3$ (`code/composite_a.rs`); exactly as Corollary 12.3 demands. This is the composite mechanism the rung–0 case requires; what it does not do is evade the barrier, which reappears as the mass of the line.

Proposition 14.32 (The barrier is a function of the split)**.** *For a solution with $|P| = j$ write $a = \prod P$, $b = \prod Q$. Since $\sigma(a)\sigma(b) = 1$ and $P \cap Q = \varnothing$, fixing j forces $\sigma(b) = 1/\sigma(a)$ and leaves Q only the odd primes outside P, which bounds N from below split by split. Taking $a = 2 \cdot ($the $j - 1$ smallest odd primes$)$, the most favourable configuration at that size:*

$\lvert P\rvert$	$\sigma(a)$	$\sigma(b)$	$\lvert Q\rvert \geq$	$\log_{10} N \geq$
2	0.8333	1.2000	67	137.9
3	1.0333	0.9677	56	112.9
4	1.1762	0.8502	63	132.8
5	1.2671	0.7892	73	161.0
6	1.3440	0.7440	85	195.3
12	1.5927	0.6279	181	491.1

The minimum over splits is $10^{112.9}$ at $|P| = 3$, recovering the constant of Theorem 4.3 by an independent route, and every other split is strictly dearer.

Proof. $\sigma(b) = 1/\sigma(a)$ is the cross–match. Q is disjoint from P, so its mass must be assembled from odd primes outside P; the least number of them reaching $\sigma(b)$, and the product of that many, are finite computations (`code/sector_barrier.rs`). Enlarging any prime of a lowers $\sigma(a)$, raises $\sigma(b)$ and hence $|Q|$, so the displayed configuration is extremal for each j. ☐

Remark 14.33 (Why the composite mechanism is out of reach)**.** Proposition 14.32 settles the standing of the clean reduction of Proposition 14.30. There $|P| = 2$, and the mechanism is forced through the dearest small sector: b is odd, so Q must assemble mass $\sigma(b) = 2q/(q + 2)$ from odd primes alone, and the sector cannot begin below $10^{137.9}$; some twenty–five orders of magnitude *above* the general barrier $10^{112.9}$. The line $n' = 2n - 4$ is therefore a correct and complete description of the $|P| = 2$ case and simultaneously the most expensive place to look.

This also explains the balance the barrier strikes. The cheapest configuration is the near–balanced one, $\sigma(a) \approx \sigma(b) \approx 1$ at $|P| = 3$; pushing the split either way (towards a small P with $\sigma(b) \to 2$, or towards a large P with $\sigma(b) \to 0$) costs orders rapidly, because mass near 2 needs the odd primes to $\approx 10^4$ and mass near 0 needs many large ones. The 59–prime barrier is thus not merely a statement about $|P \cup Q|$ but the *minimum of a convex profile over splits*, and Theorem 4.3's constant is the value at that minimum. (Rows with $|P| + |Q| < 60$ are further excluded by Proposition 4.5, which sharpens the $|P| = 3$ entry without moving the minimum's location.)

Proposition 14.34 (The cycle condition does not improve the barrier)**.** *A two–cycle satisfies more than the mass inequality: it forces $b = a'$, that is $b = a\,\sigma(a)$ exactly. Since $\sigma(b) = 1/\sigma(a)$ and $P \cap Q = \varnothing$, writing S for the small primes of a and $\Pi_{\min}(S)$ for the least product of primes outside S with reciprocal sum $\geq 1/\sigma(S)$, one gets*

$$a \;\geq\; \max\Bigl(\prod S,\; \Pi_{\min}(S)/\sigma(S)\Bigr), \qquad N \;=\; a^2\sigma(a) \;\geq\; \max\Bigl(\prod S,\; \Pi_{\min}(S)/\sigma(S)\Bigr)^2 \sigma(S).$$

Minimised over S, this yields $\log_{10} N \geq 113.2$, attained at $S = \{$odd $p \leq 151\} \setminus \{3, 97\}$ with $\sigma(S) = 1.0495$ and $|Q| \geq 26$. The barrier of Theorem 4.3 is $\log_{10} N \geq 112.9$: the cycle condition buys essentially nothing beyond the mass condition*, and Theorem 4.3 is already close to optimal for arguments of this class.*

Remark 14.35 (Why restricted families overstated the gain)**.** Compare the conclusion of Proposition 14.34 with what restricted families suggest, because the discrepancy is large and instructive. Confining S to an initial segment $\{2, 3, 5, \dots\}$ gives a minimum of $10^{222.9}$; enumerating all subsets of the first 14, 16, 18 primes gives $10^{195.9}$, $10^{188.9}$, $10^{181.7}$; still falling, hence not converged. Each of these looks like a large improvement of the barrier and none of them is one. The optimum sits outside all of them: it hands 2 to Q (half of Q's mass budget in a single prime), keeps the odd primes to 151 in S, and drops two of them, a configuration no initial segment and no small subset can express. Only after searching that structure does the bound settle at 113.2.

The lesson is about the shape of the search rather than the arithmetic. A bound minimised over a family that cannot express the optimum reports the family's best, not the problem's, and here that error was worth eighty orders of magnitude. We record the negative result as the substantive one: imposing $b = a'$, the strongest structural constraint available beyond the masses, does not move the barrier, which is evidence that $10^{112.9}$ is not an artefact of the AM–GM route but the true cost of the mass condition (`code/full_minimisation.rs`, `code/minimise_structured.rs`).

Remark 14.36 (Decoupling the two squares: what it buys, and what it costs)**.** Remark 4.16 locates the exponential difficulty in the *conjunction* of the two squares, so it is worth asking for a family on which one of them is free. The slot calculus supplies one: on

$$\mathcal{F}(N_0) \;=\; \left\{\, N_0 r \;:\; r = (s^2 - N_0)/E_0 \,\right\}$$

the plus–square $N' + 2N = s^2$ holds *identically*, by construction. The minus–square then costs exactly the form equation of Proposition 14.27, $k_0 s^2 + E_0 d^2 = 4N_0^2$, and the class of the residual question genuinely changes. On a coupled tail family the solutions run along a Pell orbit and the candidate slots grow like ε^n; here, for $\sigma(N_0) < 2$ the form is positive definite, so $4N_0^2$ has only *finitely* many representations and each base carries finitely many candidate slots. Decoupling therefore trades a Lehmer sequence for a finite per–base check.

What it does not do is help. Over the 30,397 squarefree bases $N_0 \le 5 \times 10^4$ with $\sigma(N_0) < 2$ the form represents $4N_0^2$ exactly *twice* (at most once per base, largest $s = 83$) and both representations are the ghost slot $r = 1$; there is no live prime slot at all (`code/plus_automatic.rs`, reproducing the sub–barrier census of Proposition 14.27 from an independent enumeration). The reason is the degeneration already noted: the barrier forces $\sigma(N_0) \uparrow 2$, so $k_0 = N_0(2 - \sigma)$ collapses and the form becomes ever more lopsided, while the target stays at $4N_0^2$. So freeing one square does not free the problem; it relocates the difficulty from the primality of an exponential sequence to the solubility of a degenerating quadratic form. The pattern is the circularity of Section 7 once more, but the relocation is informative, since a finite, degenerating representation problem is a different kind of object from a Lehmer sequence, and no no–go here forbids attacking it.

Proposition 14.37 (A principal–genus necessary condition, and a cascade on solutions)**.** *As* $\gcd(N_0, N_0') = 1$*, the value* $4N_0^2$ *is a square coprime to the odd part of* $\operatorname{disc} Q_{N_0}$*, so every genus character is* $+1$ *on it and a primitive form represents it only from the principal genus [28]. Reading the odd assigned characters off* $Q_{N_0}(1,1) = 4N_0$ *gives the computable necessary condition*

$$(N_0 \mid \ell) = +1 \quad \textit{for every odd prime } \ell \mid (N_0'^2 - 4N_0^2) \textit{ to odd multiplicity}$$

(a 2–adic condition accompanies it; for odd N_0 *of even* ω *the form is imprimitive and must be primitivised first; so we state the criterion on the odd part, with a 2–adic caveat). Both ghost bases* 30, 1722 *satisfy it, and about* 16% *of squarefree bases pass in the ranges tested (an* upper bound, *using only the odd characters, and mildly range–dependent).* Cascade. *In a solution of #307 with* $N = D_P D_Q$*, each* $\ell \in P \cup Q$ *gives a cofactor* N/ℓ *realising the Pythagorean structure of* N *over base*

N/ℓ; every cofactor with $\sigma(N/\ell) < 2$ (all of them for a minimal first–59–primes solution) must then satisfy the principal–genus criterion, a cascade of ≥ 59 computable genus conditions complementing the quadratic–residue filter of Corollary 14.25. (The further class–group and slot–admissibility stages of the fertility gap we observe only numerically at toy scale.)

Below the barrier the form has exactly one source of solutions, and it is entirely classical.

Proposition 14.38 (The sub–barrier ghosts are pronic Giuga numbers)**.** *Every representation of $4N_0^2$ by Q_{N_0} over squarefree $N_0 \leq 3 \times 10^4$ is a ghost $(s, d) = (\sqrt{4N_0 + 1}, 1)$, occurring exactly when $N_0' = N_0 + 1$ with N_0 pronic; in range only $30 = 5 \cdot 6$ and $1722 = 41 \cdot 42$. The two lines through such bases are classical: since $n' = n\,\sigma(n)$ for squarefree n,*

$$\{n : n' = n{-}1\} = \textit{primary pseudoperfect numbers } \left(\textstyle\sum_{p|n} \frac{1}{p} {+} \frac{1}{n} = 1\right), \qquad \{n : n' = n{+}1\} = \textit{Giuga numbers (quotien}$$

The derivative characterisations are due to Lava (OEIS [29, 30], 2009; the Giuga form $n' = n + 1$ being Lava's conjecture, *with the generalisation $n' = an + 1$, $a \in \mathbb{N}$, the theorem of Grau–Oller-Marcén [25]); the underlying number classes are Butske–Jaje–Mayernik [24] (exactly one primary pseudoperfect number per $\omega \leq 8$, eight known) and Giuga [21, 22], and the oblong chain (N with $N + 1$ prime breeds the next term $N(N + 1)$) is classical in both directions (Forgues and Sondow, OEIS [29]). To 2×10^6 the lines are $\{n' = n - 1\} = \{2, 6, 42, 1806, 47058\}$ and $\{n' = n + 1\} = \{30, 858, 1722, 66198\}$, and the chain correctly predicts the next terms: 47059 is prime, so $47058 \cdot 47059 = 2{,}214{,}502{,}422$ is the sixth primary pseudoperfect number, and 47057 is prime, so the pronic $47057 \cdot 47058 = 2{,}214{,}408{,}306$ is the fifth Giuga number; both already known, recovered here by structure. Our contribution is the reframing: the chain step is the one–line Leibniz identity $N' = N - 1 \Rightarrow (N(N + 1))' = N'(N + 1) + N^2 = N(N + 1) - 1$, exhibiting the chain as orbits of the Artin–Schreier map $x \mapsto x(x + 1) = x^2 + x$; the same map, with the same halt ($2 \to 6 \to 42 \to 1806$, stopping at $1807 = 13 \cdot 139$), as the $\mathbb{F}_2[t]$ tower of Proposition 12.12: characteristic 2 and $\mathbb{Z}$ run identical breeders on the two lines.*

Remark 14.39 (Converse, and the bridge)**.** Every pronic solution of $n' = n + 1$ is $m(m + 1)$ with m prime and $m + 1$ primary pseudoperfect: Leibniz gives $m'(m + 1) + m(m + 1)' = m^2 + m + 1$, and writing $m' = km + 1$ forces $(m + 1)' = (1 - k)m - k$, which is ≥ 0 only for $k = 0$, whence $m' = 1$ (m prime) and $(m + 1)' = m$. (The congruence $m' \equiv 1 \pmod{m}$ alone does *not* force m prime; its composite solutions are precisely the Giuga numbers $30, 858, 1722, 66198, \ldots$, so primality comes from the sign elimination $k = 0$.) Thus the only sub–barrier life in the form equation is the classical Giuga/primary–pseudoperfect world, known to sit in a web with Sylvester's sequence, Znám's problem and the Erdős–Moser equation [26] and with the arithmetic derivative [25, 27]; we claim none of this web, only that #307 meets it through $n' = n\,\sigma(n)$, and that its $a \geq 2$ (equivalently $\sigma > 2$) branch lands on the same ≥ 59–prime barrier [23] that bars genuine cycles (Corollary 12.3).

Writing both members of a cycle as members of these lines stratifies #307 by the gap.

Proposition 14.40 (Gap stratification: #307 as adjacency of μ–Sondow lines)**.** *For a squarefree two–cycle $a' = b$, $b' = a$ with $a > b$ and gap $d = a - b$, the smaller member lies on the line $n' = n + d$ and the larger on $n' = n - d$ (as $b' - b = d$ and $a' - a = -d$), with $\gcd(d, ab) = 1$ (a prime dividing d and a would divide $b = a - d$); conversely $x' = x + d$ with $(x + d)' = x$ is a two–cycle. These opposite–sign lines are the quotient–one μ–Sondow numbers of parameter $\mp d$ [27]. Hence #307 holds iff for some $d \geq 1$ the lines $n' = n + d$ and $n' = n - d$ carry members exactly d apart; for $d = 1$ this is* a Giuga number adjacent to a primary pseudoperfect number. *By the parity dichotomy (Theorem 4.17) the mixed–parity case forces d odd and the both–odd case (d even) carries* *the* 10^{2519} *bound, so every cycle below that bound has odd gap and* $d = 1$ *is the lead stratum. A* $d = 1$ *pair* $(k, k+1)$ *has exactly one odd member, hence requires either an odd Giuga number (known to have at least* 9 *prime factors [22, 23]) or an odd primary pseudoperfect number (none is known): the lead stratum inherits two classical open problems. Tabulating cycle–eligible members (squarefree, coprime to d) to* 2×10^6 *for all* $d \leq 50$ *gives* 21 *on the plus lines (none of even gap, consistent with the parity remark) and* 81 *on the minus lines, with* no *adjacent pair at any* $d \leq 50$*; the barrier in gap–stratified form.* (*For coprime* M *on* $n' = n - \alpha$ *and* N *on* $n' = n + \beta$*, Leibniz gives* $(MN)' - 2MN = \beta M - \alpha N$*, so* $\beta M - \alpha N = \square$ *would manufacture a minus–side square in the sense of Proposition 12.17; the case* $(\alpha, \beta) = (1, 1)$*,* $M - N = 1$ *is the* $d = 1$ *cycle itself, and the known finite Giuga/primary–pseudoperfect lists contain no coprime pair, all members being even.*) *To our knowledge this gap–stratified equivalence and the pairing of opposite–sign lines at fixed additive distance are new; the lines themselves are [27].*

The line parameter $\delta(n) = n' - n$ obeys an affine recursion that organises the classical members into a genealogy.

Proposition 14.41 (The δ–recursion and the genealogy of the classical lines)**.** *With* $\delta(n) = n' - n$*, for squarefree* M *and a prime* $p \nmid M$*,*

$$\delta(Mp) = p\,\delta(M) + M \qquad (\textit{Leibniz; checked on } 10^3 \textit{ random pairs}).$$

Solving $p\,\delta(M) + M = \pm d$ *gives the child rule* $p = (M \mp d)/e$ *from a parent* M *on the minus line* $\delta(M) = -e$*. Every known Giuga number arises this way from a small–parameter parent:* $30 = 6 \cdot 5$ *(*$e = 1$*),* $858 = 66 \cdot 13$ *(*$e = 5$*),* $1722 = 42 \cdot 41$ *(*$e = 1$*),* $66198 = 1122 \cdot 59$ *(*$e = 19$*),* $2{,}214{,}408{,}306 = 47058 \cdot 47057$ *(*$e = 1$*); the seed* 47058 *itself resolves as* $6 \to 66 \to 1518 \to 47058$ *(with* $\delta = -1, -5, -49, -1$*), so the previously unexplained members* 858 *and* 66198 *acquire genealogies. Two members of a cycle can never share a parent: siblings* $M(M \mp d)/e$ *differ by* $2Md/e > d$*, since* $e = M - M' < 2M$*. The oblong case* $e = 1$ *is the classical chain of Proposition 14.38 (its* μ*–Sondow form is [27]); the recursion, the general–*e *genealogy, and the resulting small–*$|\delta|$ *tree do not appear there.*

Corollary 14.42 (Nonemptiness families for the μ–Sondow conjecture)**.** *In the notation of [27] (their* S_μ*; the quotient–one slice is the line* $n' = n - \mu$*), the recursion produces explicit members exceeding* $|\mu|$ *in three infinite parameter families:* $2p \in S_{p-2}$ *for every odd prime* p*;* $Mp \in S_{p-M}$ *for every primary pseudoperfect number* M *and prime* $p \nmid M$*, and* $Gp \in S_{-(p+G)}$ *for every Giuga number* G *and prime* $p \nmid G$*; in each case* $\mu/n + \sum_{q|n} 1/q = 1$ *exactly (checked symbolically and on* 151 *instances). This settles their nonemptiness conjecture (*$S_\mu \cap (|\mu|, \infty) \neq \varnothing$*, their Conjecture 1(ii)) for infinitely many parameters, including infinitely many outside their verified range* $-145 < \mu < 673$*; these families do not appear among their constructions, which scale by* $\mathrm{rad}(\mu)$ *rather than extend by a new prime. Their two open instances resist precisely because every quotient–one parent route hits a composite: for* $\mu = -145$*,* $M - 145$ *is composite for all eight known primary pseudoperfect numbers (*$1661 = 11 \cdot 151$*,* $46913 = 43 \cdot 1091$*, and the three larger cases, including the recently found eighth member) while* $p + G = 145$ *has no prime solution; for* $\mu = 673$*,* $M + 673$ *is composite for all eight (*$679 = 7 \cdot 97$*,* $2479 = 37 \cdot 67$*,* $47731 = 59 \cdot 809$*, the rest ending in* 5*) and* $p = 675$ *fails. A prime in either family at the next generation would settle a published open instance; an exhaustive search found none. Concretely: no quotient–one member of* S_{-145} *or* S_{673} *exists with* any *prime–factor cofactor* $\leq 10^{11}$ *(every squarefree parent* $M \leq 10^{11}$ *was scanned against the child rules, a region whose members reach* $\sim 10^{22}$*, together with a depth–two descent), the quotient–zero slice of* S_{-145} *is empty outright (*$n' = 145$ *forces* $n \leq 5256$ *by* $n' \geq 2\sqrt{n}$*, and the finite check is empty), and quotients* ≥ 2 *force* $\sigma \gtrsim 2$*, hence at least* 59 *prime factors. Both instances therefore remain open, with the quotient–one frontier moved from the* 10^{10} *of [27] to cofactor scale* 10^{11}*.*

Remark 14.43 (The look–back)**.** Along small–$|\delta|$ paths the forced prime is $p \approx M/e$, so members grow doubly exponentially and the 10^{56}–10^{113} wall zone lies only a few generations from the seeds: a sparse, recursively enumerable tree (a child needs $(M-j)/e$ prime for small j). The adjacency of two such trees at a fixed additive distance places the final condition in the Pillai/S–unit genre (two multiplicatively generated sequences meeting at a fixed additive distance) where fixed–support finiteness is classical but growing–support statements are conjectural; a lens, not a transfer. Re–expressed in δ–coordinates the obstruction is once more $\sigma > 1$ (positivity of δ) plus integer reachability $\delta = \pm d$ under forced primes: the same additive–multiplicative exactness reappears in every coordinate tried (mass, the deviation k, the form Q_{N_0}, genus theory, μ–Sondow lines, δ–dynamics). The reframings buy structure, not a reduction.

Proposition 14.44 (Even–gap plus lines are empty below 3.23×10^9)**.** *An even squarefree $x = 2u$ has $\delta(x) = 2u' - u$ odd, so every line member of even δ is odd (with ω odd). An odd member of a plus line has $\sigma > 1$ over odd primes, and the least odd squarefree integer of mass exceeding* 1 *is* $3 \cdot 5 \cdots 29 = 3{,}234{,}846{,}615$ *(the* 8*–prime product* $3 \cdots 23 = 111{,}546{,}435$ *has mass* 0.99896*). Hence every plus line of even gap d is empty below* $3{,}234{,}846{,}615$*, proving the census observation (no even–gap plus member to* 2×10^7*) and placing an explicit floor under the smaller member of any both–odd configuration.*

The lines also admit parity, infinitude, and congruence refinements.

Proposition 14.45 (Parity floor for odd line members)**.** *For odd squarefree n every cofactor n/p is odd, so* $n' \equiv \omega(n) \pmod 2$*. Hence an odd primary pseudoperfect number ($n' = n - 1$, even) and an odd Giuga number ($n' = n + 1$, even) both have $\omega(n)$ even; combined with the classical bound $\omega \geq 9$ [22, 23] this forces $\omega \geq 10$, a parity sharpening (for the primary pseudoperfect side, the eight smallest odd primes give $\sum 1/p = 0.99896 < 1$, so $\omega \geq 9$ directly). The floor cascades when small primes are avoided: for an odd squarefree member of any plus line, or of a quotient–one line (where $\sigma(n) \geq 1 - 1/n$), exact greedy/exchange computation gives $\omega \geq 27$ if $3 \nmid n$; $\omega \geq 66$ if $3, 5 \nmid n$; $\omega \geq 144$ if $3, 5, 7 \nmid n$; and $\omega \geq 16$ if $3 \mid n$, $5 \nmid n$; each avoided small prime multiplies the required complexity, and for odd Giuga or primary pseudoperfect numbers the parity constraint rounds odd floors up (e.g. $3 \nmid n$ forces $\omega \geq 28$). (The restriction to plus or quotient–one lines is essential: on a general minus line σ can be small (every prime p trivially satisfies $p' = p - (p - 1)$) so no such floor exists there.)*

Proposition 14.46 (Infinitely many gaps carry a nonempty plus line)**.** *By the recursion of Proposition 14.41, $\delta(30p) = p\,\delta(30) + 30 = p + 30$ for every prime $p > 5$ (as $\delta(30) = 1$), so the line $n' = n + (p + 30)$ is nonempty; by Euclid infinitely many distinct gaps d carry a nonempty plus line, and the same runs from any Giuga number G ($\delta(Gp) = p + G$). This fixes the census tail:* $330, 390, 510, 570 = 30 \cdot \{11, 13, 17, 19\}$ *lie on the gaps* $d = 41, 43, 47, 49$*. (Per–gap infinitude remains open, of Bunyakovsky type.)*

Proposition 14.47 (A difference–36 congruence modulo 72)**.** *Let $n = 6m$ be a Giuga or primary pseudoperfect number with* $\gcd(m, 6) = 1$*, $n' = n + \varepsilon$ ($\varepsilon = +1$ Giuga, -1 primary pseudoperfect). Then $6m' = m + \varepsilon$, and with* $m' \equiv \omega(m) \pmod 2$ *this yields* $n \equiv 36\,\omega(n) - 78 \pmod{72}$ *on the Giuga side and* $n \equiv 36\,\omega(n) - 66 \pmod{72}$ *on the primary pseudoperfect side; in particular equal ω forces equal residue. The congruence is verified on all twelve Giuga numbers and all seven* 6*–divisible primary pseudoperfect numbers listed in the OEIS (we recomputed the factorisations of the five largest Giuga members; all are squarefree, with $\omega = 7, 7, 8, 8, 8$), and gives, at the modulus* 72*, the difference–*36 *arithmetic progression observed by Sondow–MacMillan [26]; their finer modulus–*288 *pattern remains empirical.*

The modulus–72 argument is the $D = 6$ case of a general transport of the line equation across a fixed divisor.

Proposition 14.48 (Divisor transport)**.** *Let $n = Dm$ with D, m coprime squarefree and $n' = n + \varepsilon$ ($\varepsilon = \pm 1$) a quotient–one member. Leibniz transports the line equation to*

$$D\,m' = (D - D')\,m + \varepsilon.$$

(a) *If $\sigma(D) > 1$ (so $D' > D$; least case $D = 30$), the right side is negative for $m > 1$ while the left is nonnegative; the only escape is $m = 1$, $\varepsilon = +1$. Hence* no primary pseudoperfect number is divisible by 30, and the only Giuga number divisible by 30 is 30 itself *(elementary; we have not located it in the literature), consistent with the full known corpus, of which only* 30 *carries the factor* 5*.*

(b) *At a primary pseudoperfect block D (where $D - D' = 1$) the slice $m' = 1$ reads $m = D - \varepsilon$ prime: the two classical chain rules ($D(D + 1)$ primary pseudoperfect when $D + 1$ is prime, $D(D - 1)$ Giuga when $D - 1$ is prime [27, 26]) are the two signs of one transported equation (*$1806 = 42 \cdot 43$ *and* $1722 = 42 \cdot 41$ *sit on the same slice $D = 42$).*

(c) *For any gap d the same transport stratifies the line $n' = n + d$ by divisor blocks D into the steeper lines $D\,m' = (D - D')m + d$, a further enumeration coordinate for the genealogy of Proposition 14.41.*

Finally, the form obstruction can be averaged: it eliminates almost every integer, though not the thin stratum where a solution could live.

Proposition 14.49 (Average rarity of form–barrier survivors)**.** *Call a squarefree N* blocked *at an odd prime ℓ if $\ell \nmid N$, $N' \equiv \pm 2N \pmod{\ell}$, and $(N \mid \ell) = -1$, and a* survivor *if it is blocked at no odd prime. A blocked N admits no representation $E\,s^2 + D\,d^2 = 4N^2$ (reducing mod ℓ forces N to be a quadratic residue, against $(N \mid \ell) = -1$; this is the genus fragment of Proposition 14.37), so every union of a #307 solution is a survivor. Now $N' \equiv N\,S_\ell(N) \pmod{\ell}$ with $S_\ell(N) = \sum_{p \mid N} p^{-1}$ an* additive *function, so blocking at ℓ is an additive–function event ($S_\ell \equiv \pm 2$) twisted by the quadratic character, of limiting density $1/(\ell + 1)$. As $\sum_\ell 1/(\ell + 1)$ diverges (Mertens), the Selberg–Delange/Halász theory of additive functions in residue classes [17] gives that the survivors have natural density zero among the squarefree integers (unconditional), with $\#\{\text{survivors} \le X\} = O\big(X/\log\log\log X\big)$ under a standard, but here unproved, uniformity estimate. Thus the single–prime form obstruction removes* almost every *squarefree integer. We stress the scope: survivors remain* infinite*, and a #307 solution; one integer beyond 2.09×10^{56}, in the density–zero stratum $\sigma(N) \ge 2$; is untouched by such an average statement, which speaks only to the typical integer, not to existence.*

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